FADN2Footprint — Validation Report

Comparison of package estimates with literature values

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Setup

```
library(dplyr)
library(tidyr)
library(stringr)
library(gt)

library(ggplot2)

devtools::load_all("../../")
#library(FADN2Footprint)
```

Purpose

This report compares the outputs of `FADN2Footprint` (v1.1.0) against published literature ranges. An estimate is flagged **in range** if it falls within the literature min–max, **within tolerance** if the relative deviation from the literature midpoint is below a per-indicator threshold, or **out of range** otherwise.

Produce package estimates

```
FADN dataset loaded: 26682 farm-year observations
```

```
Countries covered: FRA
```

```
Years covered: 2018, 2019, 2020, 2021
```

```
object = my_object_GHGE

# Crops
summary(object@crop)
```

```
      ID                YEAR              COUNTRY          FADN_code_letter
 Length:113744      Length:113744      Length:113744      Length:113744
 Class :character   Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character   Mode  :character



      ENA                IRA                SV                TO
 Min.   : 0.00000   Min.   :  0.000   Min.   :      0   Min.   :-194600
 1st Qu.: 0.00000   1st Qu.:  0.000   1st Qu.:      0   1st Qu.:      0
 Median : 0.00000   Median :  0.000   Median :   4168   Median :   6647
 Mean   : 0.02121   Mean   :  1.458   Mean   :  40850   Mean   :  42906
 3rd Qu.: 0.00000   3rd Qu.:  0.000   3rd Qu.:  27651   3rd Qu.:  30240
 Max.   :92.50000   Max.   :607.280   Max.   :4862533   Max.   :4924968
 NA's   :242        NA's   :242       NA's   :242       NA's   :242
       CV             area_ha           prod_t            sales_t
 Min.   :   0.0   Min.   :  0.00   Min.   :    0.00   Min.   :    0.000
 1st Qu.:   0.0   1st Qu.:  4.34   1st Qu.:   24.49   1st Qu.:    0.000
 Median :   0.0   Median : 11.28   Median :   78.85   Median :    7.681
 Mean   : 154.6   Mean   : 21.93   Mean   :  225.27   Mean   :  150.888
 3rd Qu.:   0.0   3rd Qu.: 26.82   3rd Qu.:  233.51   3rd Qu.:  101.820
 Max.   :3555.0   Max.   :950.00   Max.   :16413.80   Max.   :16413.800
 NA's   :113721                                      NA's   :1856
 land_use_type        species              yield              NUTS0
 Length:113744      Length:113744      Min.   :    0.000   Length:113744
 Class :character   Class :character   1st Qu.:    4.643   Class :character
 Mode  :character   Mode  :character   Median :    6.639   Mode  :character
                                       Mean   :   14.358
                                       3rd Qu.:    9.424
                                       Max.   :10133.333

    NUTS1              NUTS2          default_forage_yield default_forage_prod
 Length:113744      Length:113744      Mode :logical        Mode :logical
 Class :character   Class :character   FALSE:77429          FALSE:77429
```

```
 Mode  :character   Mode  :character   TRUE :36315          TRUE :36315


 default_crop_yield default_crop_prod default_sales_to_prod
 Mode :logical      Mode :logical     Mode :logical
 FALSE:106991       FALSE:106991      FALSE:113709
 TRUE :6753         TRUE :6753        TRUE :35


 default_avrg_yield_crop  sum_area_ha       sum_prod_t      keep_sup5pc
 Mode :logical            Min.   :  0.01   Min.   :    0.0   Mode:logical
 FALSE:110791             1st Qu.: 52.90   1st Qu.:  420.6   TRUE:113744
 TRUE :2953               Median :100.88   Median :  749.5
                          Mean   :118.84   Mean   : 1174.8
                          3rd Qu.:161.56   3rd Qu.: 1346.5
                          Max.   :950.00   Max.   :26566.9
```

```
(object@crop |>
    dplyr::filter(is.na(prod_t)))
```

```
# A tibble: 0 x 27
# i 27 variables: ID <chr>, YEAR <chr>, COUNTRY <chr>, FADN_code_letter <chr>,
#   ENA <dbl>, IRA <dbl>, SV <dbl>, TO <dbl>, CV <dbl>, area_ha <dbl>,
#   prod_t <dbl>, sales_t <dbl>, land_use_type <chr>, species <chr>,
#   yield <dbl>, NUTS0 <chr>, NUTS1 <chr>, NUTS2 <chr>,
#   default_forage_yield <lgl>, default_forage_prod <lgl>,
#   default_crop_yield <lgl>, default_crop_prod <lgl>,
#   default_sales_to_prod <lgl>, default_avrg_yield_crop <lgl>, ...
```

```
#sort(table(object@crop |> filter(is.na(prod_t)) |> pull(FADN_code_letter)), decreasing = TRU
#table(object@crop |> filter(is.na(prod_t)) |> pull(keep_sup5pc))
#
#
## GHGE crops
#
summary(object@footprints$GHGE$GHGE_crops)
```

```
     ID                 YEAR               COUNTRY           FADN_code_letter
Length:113744      Length:113744      Length:113744      Length:113744
Class :character   Class :character   Class :character   Class :character
Mode  :character   Mode  :character   Mode  :character   Mode  :character
```

```
   area_ha            prod_t           N2O_d_kgCO2e    N2O_ATD_kgCO2e
Min.   :  0.00   Min.   :    0.00   Min.   :     0   Min.   :    0.0
1st Qu.:  4.34   1st Qu.:   24.49   1st Qu.:  2367   1st Qu.:  171.7
Median : 11.28   Median :   78.85   Median :  8342   Median :  679.9
Mean   : 21.93   Mean   :  225.27   Mean   : 18204   Mean   : 1599.1
3rd Qu.: 26.82   3rd Qu.:  233.51   3rd Qu.: 23028   3rd Qu.: 2031.9
Max.   :950.00   Max.   :16413.80   Max.   :650359   Max.   :48278.0

 N2O_L_kgCO2e      ghg_ferti_prod_kgCO2e ghg_heat_fuel_kgCO2e
Min.   :     0.0   Min.   :     0.0      Min.   :      0
1st Qu.:   931.6   1st Qu.:   242.7      1st Qu.:      0
Median :  3160.3   Median :  2383.4      Median :      0
Mean   :  7035.3   Mean   :  8220.8      Mean   :   1289
3rd Qu.:  8840.7   3rd Qu.:  8590.6      3rd Qu.:      0
Max.   :284489.0   Max.   :322770.8      Max.   :4270760
                                         NA's   :25
ghg_diesel_tillage_kgCO2e ghg_diesel_remain_kgCO2e ghg_elec_kgCO2e
Min.   :     0.0          Min.   :    0.0          Min.   :    0.00
1st Qu.:   656.4          1st Qu.:    0.0          1st Qu.:    0.00
Median :  3402.6          Median :  163.7          Median :   13.12
Mean   :  7526.2          Mean   : 1373.5          Mean   :  242.05
3rd Qu.:  9374.5          3rd Qu.: 1427.0          3rd Qu.:  140.37
Max.   :254249.4          Max.   :52535.9          Max.   :42073.40
NA's   :25                NA's   :25               NA's   :25
total_ghg_crop_kgCO2e N2O_d_kgCO2e_per_ha N2O_d_kgCO2e_per_t
Min.   :      0       Min.   :7.064e+00   Min.   :     0.06
1st Qu.:   7933       1st Qu.:4.021e+02   1st Qu.:    54.51
Median :  23255       Median :7.656e+02   Median :    99.13
Mean   :  45488       Mean   :9.405e+02   Mean   :   143.55
3rd Qu.:  56175       3rd Qu.:1.109e+03   3rd Qu.:   150.78
Max.   :4431669       Max.   :2.793e+05   Max.   :515127.08
                                          NA's   :1517
N2O_ATD_kgCO2e_per_ha N2O_ATD_kgCO2e_per_t N2O_L_kgCO2e_per_ha
Min.   :   0.2824     Min.   :    0.002    Min.   :3.319e+00
1st Qu.:  33.7373     1st Qu.:    4.152    1st Qu.:1.744e+02
```

```
Median :  64.6598     Median :   11.271    Median :2.823e+02
Mean   :  71.5064     Mean   :   15.973    Mean   :3.676e+02
3rd Qu.:  97.4251     3rd Qu.:   15.766    3rd Qu.:3.867e+02
Max.   :5532.7632     Max.   :48874.098    Max.   :1.334e+05
                      NA's   :1517
N2O_L_kgCO2e_per_t  ghg_ferti_prod_kgCO2e_per_ha ghg_ferti_prod_kgCO2e_per_t
Min.   :2.200e-02   Min.   :    0.00             Min.   :     0.00
1st Qu.:2.301e+01   1st Qu.:   60.75             1st Qu.:     5.71
Median :4.011e+01   Median :  224.78             Median :    31.00
Mean   :5.086e+01   Mean   :  388.25             Mean   :    84.41
3rd Qu.:5.348e+01   3rd Qu.:  616.19             3rd Qu.:   100.28
Max.   :1.370e+05   Max.   :70242.67             Max.   :511093.65
NA's   :1517                                     NA's   :1517
ghg_heat_fuel_kgCO2e_per_ha ghg_heat_fuel_kgCO2e_per_t
Min.   :     0.0            Min.   :   0.000
1st Qu.:     0.0            1st Qu.:   0.000
Median :     0.0            Median :   0.000
Mean   :   649.2            Mean   :   1.833
3rd Qu.:     0.0            3rd Qu.:   0.000
Max.   :887891.9            Max.   :8028.013
NA's   :25                  NA's   :1517
ghg_diesel_tillage_kgCO2e_per_ha ghg_diesel_tillage_kgCO2e_per_t
Min.   :      0.0                Min.   :0.000e+00
1st Qu.:    143.7                1st Qu.:1.105e+01
Median :    380.1                Median :4.882e+01
Mean   :    896.2                Mean   :2.025e+02
3rd Qu.:    762.0                3rd Qu.:1.135e+02
Max.   :1069588.3                Max.   :2.004e+06
NA's   :25                       NA's   :1517
ghg_diesel_remain_kgCO2e_per_ha ghg_diesel_remain_kgCO2e_per_t
Min.   :    0.00                Min.   :     0.00
1st Qu.:    0.00                1st Qu.:     0.00
Median :   30.56                Median :     2.66
Mean   :  100.38                Mean   :    20.69
3rd Qu.:  133.93                3rd Qu.:    14.52
Max.   :72025.98                Max.   :203938.77
NA's   :25                      NA's   :1517
ghg_elec_kgCO2e_per_ha ghg_elec_kgCO2e_per_t total_ghg_crop_kgCO2e_per_ha
Min.   :    0.000      Min.   :    0.000     Min.   :2.546e+01
1st Qu.:    0.000      1st Qu.:    0.000     1st Qu.:1.117e+03
Median :    1.349      Median :    0.160     Median :2.169e+03
Mean   :   44.060      Mean   :    4.373     Mean   :3.457e+03
3rd Qu.:   11.139      3rd Qu.:    1.463     3rd Qu.:3.211e+03
```

```
 Max.   :28675.051      Max.   :48042.746     Max.   :1.071e+06
 NA's   :25             NA's   :1517
 total_ghg_crop_kgCO2e_per_t
 Min.   :6.070e+00
 1st Qu.:1.490e+02
 Median :2.682e+02
 Mean   :5.242e+02
 3rd Qu.:4.589e+02
 Max.   :2.452e+06
 NA's   :1517
```

```
#
object@footprints$GHGE$GHGE_crops |>
  dplyr::filter(!is.finite(N2O_d_kgCO2e_per_t)) #|> View()
```

```
# A tibble: 1,517 x 33
   ID            YEAR  COUNTRY FADN_code_letter area_ha prod_t N2O_d_kgCO2e
   <chr>         <chr> <chr>   <chr>               <dbl>  <dbl>        <dbl>
 1 515000000728  2018  FRA     CNURS                3.23      0        546.
 2 515000000340  2018  FRA     CFLWUG               1         0         80.2
 3 515000000340  2018  FRA     CFLW                 1         0         80.2
 4 515000000247  2018  FRA     CFLWCUT              6         0        481.
 5 515000000247  2018  FRA     CFLWUG               6         0        481.
 6 515000000853  2018  FRA     CFLWUG               2.02      0        162.
 7 515000000853  2018  FRA     CFLW                 2.02      0        162.
 8 515000000929  2018  FRA     CNURS                9.62      0       1334.
 9 515000000909  2018  FRA     CNURS                6.85      0        549.
10 5650000006622 2018  FRA     CNURS                0.5       0         40.1
# i 1,507 more rows
# i 26 more variables: N2O_ATD_kgCO2e <dbl>, N2O_L_kgCO2e <dbl>,
#   ghg_ferti_prod_kgCO2e <dbl>, ghg_heat_fuel_kgCO2e <dbl>,
#   ghg_diesel_tillage_kgCO2e <dbl>, ghg_diesel_remain_kgCO2e <dbl>,
#   ghg_elec_kgCO2e <dbl>, total_ghg_crop_kgCO2e <dbl>,
#   N2O_d_kgCO2e_per_ha <dbl>, N2O_d_kgCO2e_per_t <dbl>,
#   N2O_ATD_kgCO2e_per_ha <dbl>, N2O_ATD_kgCO2e_per_t <dbl>, ...
```

```
#sort(table(object@footprints$GHGE$GHGE_crops |> filter(is.na(prod_t)) |> pull(FADN_code_let
```

Literature references

```
# TOTAL GHGE ----

lit_GHGE_total <- readxl::read_excel(
  "../../inst/validation/validation_references.xlsx",
  sheet = "summary_total_GHGE",
  col_types = c("text", "numeric", "numeric", "text", "text")
) |>
  # filter for references with both per ha and per t
  dplyr::filter(GHGE_kgCO2e_kg >0
                | GHGE_kgCO2e_ha >0) |>
  # filter for crops, meat, milk, and eggs
  dplyr::filter(grepl("wheat|maize_grain|rapeseed|milk|meat|eggs", product)) |>
  # filter values for France if present or EU otherwise for each product
  dplyr::group_by(product) |>
  dplyr::filter(
    if (any(grepl("France", area))) grepl("France", area) else TRUE
  ) |>
  dplyr::ungroup() |>
  # add set
  dplyr::mutate(set = "GHG Emissions")

# COMPONENTS OF GHGE ----

lit_GHGE_components <- readxl::read_excel(
  "../../inst/validation/validation_references.xlsx",
  sheet = "summary_share_components"
) |>
  # filter for crops, meat and milk
  dplyr::filter(grepl("wheat|maize_grain|rapeseed|milk|meat|eggs", product)) |>
  # filter values for France if present or EU otherwise for each product and component
  dplyr::group_by(product, Harmonized_Category) |>
  dplyr::filter(
    if (any(grepl("France", area))) grepl("France", area) else TRUE
  ) |>
  dplyr::ungroup() |>
  # remove unclassified components
  dplyr::filter(!is.na(Harmonized_Category)) |>
  # sum up by product,Harmonized_Category,sheet,area
  dplyr::summarise(
```

```
    dplyr::across(
      c(GHGE_kgCO2e_ha,GHGE_kgCO2e_kg),
      ~ sum(.x)
    ),
    .by = c('product','Harmonized_Category','sheet','area')
  ) |>
  # add set
  dplyr::mutate(set = "GHG Emissions")

cat("Literature (total GHGE) loaded:", nrow(lit_GHGE_total), "rows\n")
```

```
Literature (total GHGE) loaded: 102 rows
```

```
cat("Literature (components of GHGE) loaded:", nrow(lit_GHGE_components), "rows\n")
```

```
Literature (components of GHGE) loaded: 49 rows
```

```
lit_agri_pract <- readxl::read_excel(
  "../../inst/validation/validation_references.xlsx",
  sheet = "summary_agri_practices"
)

# add EPK averages for wheat fertilization ----
tmp = data_extra$PKGC_N_ferti |>
  dplyr::filter(EPK_ferti_crop_name == "Ble_tendre" & ORGANIC == 0) |>
  dplyr::select(N_min,Nmin_sd_w,N_org,Norg_sd_w)

lit_agri_pract <- lit_agri_pract |>
  dplyr::bind_rows(
    tibble::tibble(
      indicator = c("mineral fertilization", "organic fertilizer"),
      unit = "kg N / ha",
      product = "wheat",
      area = "France",
      lit_midpoint = c(tmp$N_min, tmp$N_org),
      lit_min = 0,
      lit_max = c((tmp$N_min + tmp$Nmin_sd_w), (tmp$N_org + tmp$Norg_sd_w)),
      #tolerance_pct = "More than half\n&\nLess than twice",
      lit_source = "EPK 2017"
    )
```

```
  ) |>
  # add set
  dplyr::mutate(set = "Agricultural practices")


cat("Literature (agricultural practices) loaded:", nrow(lit_agri_pract), "rows\n")
```

```
Literature (agricultural practices) loaded: 16 rows
```

```
lit_GHGE_total |> arrange(set,product,sheet) |> gt()
```

```
lit_GHGE_components |> arrange(set,product,Harmonized_Category,sheet) |> gt()
```

```
lit_agri_pract |> arrange(set,indicator) |> gt()
```

```
lit_table = dplyr::bind_rows(
  lit_agri_pract,
  lit_GHGE_total |>
    dplyr::mutate(GHGE_kgCO2e_t = GHGE_kgCO2e_kg*10^3) |>
    tidyr::pivot_longer(cols = c(GHGE_kgCO2e_ha,GHGE_kgCO2e_t),
                        names_to = "indicator") |>
    dplyr::filter(!is.na(value)) |>
    dplyr::summarise(lit_midpoint = mean(value),
                     lit_min = min(value),
                     lit_max = max(value),
                     lit_source = paste0(unique(sheet),collapse = "; "),
                     area = paste0(unique(area),collapse = "; "),
                     .by = c('set', 'indicator', 'product')) |>
    dplyr::mutate(
      unit = ifelse(grepl("_ha",indicator),"kgCO2e_per_ha","kgCO2e_per_t")
    )
) |>
  dplyr::bind_rows(
    lit_GHGE_components |>
      dplyr::mutate(GHGE_kgCO2e_t = GHGE_kgCO2e_kg*10^3) |>
      tidyr::pivot_longer(cols = c(GHGE_kgCO2e_ha,GHGE_kgCO2e_t),
                          names_to = "indicator") |>
      dplyr::filter(!is.na(value)) |>
```

product	GHGE_kgCO2e_kg	GHGE_kgCO2e_ha	area	sheet
cattle_meat	12.0000000	9931.000	France	CAP2ER_2025_BV_E_JB_vi
cattle_meat	18.1000000	4496.000	France	CAP2ER_2025_BV_NE_boeu
cattle_meat	14.5000000	6460.620	France	Extract_Agribalyse
eggs	2.8800000	NA	France	Clune_2017
eggs	3.1900000	NA	France	Clune_2017
eggs	1.1800000	5063.667	France	Extract_Agribalyse
maize_grain	0.3560000	3684.997	France	Extract_Agribalyse
meat_chicken	2.7090909	NA	France	Clune_2017
meat_chicken	2.9731169	NA	France	Clune_2017
meat_chicken	3.5964935	NA	France	Clune_2017
meat_chicken	4.2698701	NA	France	Clune_2017
meat_chicken	4.6166667	NA	France	Clune_2017
meat_chicken	5.3511111	NA	France	Clune_2017
meat_chicken	5.3607792	NA	France	Clune_2017
meat_chicken	5.3696296	NA	France	Clune_2017
meat_chicken	5.9251852	NA	France	Clune_2017
meat_chicken	6.1103704	NA	France	Clune_2017
meat_chicken	1.9300000	6426.770	France	Extract_Agribalyse
meat_pork	2.9740000	4310.000	France	Bellassen_2021
meat_pork	2.3000000	4240.000	France	Chiriaco_2022
meat_pork	3.7865116	NA	France	Clune_2017
meat_pork	3.8330233	NA	France	Clune_2017
meat_pork	4.0190698	NA	France	Clune_2017
meat_pork	4.2000000	NA	France	Clune_2017
meat_pork	4.2516279	NA	France	Clune_2017
meat_pork	4.3213953	NA	France	Clune_2017
meat_pork	4.6702326	NA	France	Clune_2017
meat_pork	5.0655814	NA	France	Clune_2017
meat_pork	5.1000000	NA	France	Clune_2017
meat_pork	5.1120930	NA	France	Clune_2017
meat_pork	5.1586047	NA	France	Clune_2017
meat_pork	5.6884536	NA	France	Clune_2017
meat_pork	5.7167442	NA	France	Clune_2017
meat_pork	5.7400000	NA	France	Clune_2017
meat_pork	5.8946392	NA	France	Clune_2017
meat_pork	6.1586047	NA	France	Clune_2017
meat_pork	6.2039175	NA	France	Clune_2017
meat_pork	6.2283721	NA	France	Clune_2017
meat_pork	6.3213953	NA	France	Clune_2017
meat_pork	6.3679070	NA	France	Clune_2017
meat_pork	6.3894845	NA	France	Clune_2017
meat_pork	6.5306977	NA	France	Clune_2017
meat_pork	6.5772093	NA	France	Clune_2017
meat_pork	7.3213953	NA	France	Clune_2017
meat_pork	7.7000000	NA	France	Clune_2017
meat_pork	8.8000000	NA	France	Clune_2017
meat_pork	8.8097674	NA	France	Clune_2017
meat_pork	2.7900000	9068.672	France	Extract_Agribalyse
meat_pork	2.3000000	4234.300	France	Meier_2015
milk	1.1061238	NA	France	Agribalyse_2025

product	Harmonized_Category	sheet	area	GH
cattle_meat	Energy Use (Diesel/Elec)	CAP2ER_2025_BV_E_JB_viande	France	
cattle_meat	Energy Use (Diesel/Elec)	CAP2ER_2025_BV_NE_boeufs	France	
cattle_meat	Enteric Fermentation	CAP2ER_2025_BV_E_JB_viande	France	
cattle_meat	Enteric Fermentation	CAP2ER_2025_BV_NE_boeufs	France	
cattle_meat	Fertilizer Production	CAP2ER_2025_BV_E_JB_viande	France	
cattle_meat	Fertilizer Production	CAP2ER_2025_BV_NE_boeufs	France	
cattle_meat	Field N2O Emissions	CAP2ER_2025_BV_E_JB_viande	France	
cattle_meat	Field N2O Emissions	CAP2ER_2025_BV_NE_boeufs	France	
cattle_meat	Manure Management	CAP2ER_2025_BV_E_JB_viande	France	
cattle_meat	Manure Management	CAP2ER_2025_BV_NE_boeufs	France	
eggs	Energy Use (Diesel/Elec)	Lesschen_2011_compo	Europe	
eggs	Enteric Fermentation	Lesschen_2011_compo	Europe	
eggs	Fertilizer Production	Lesschen_2011_compo	Europe	
eggs	Field N2O Emissions	Lesschen_2011_compo	Europe	
eggs	Manure Management	Lesschen_2011_compo	Europe	
eggs	Other Inputs & Processes	Lesschen_2011_compo	Europe	
meat_chicken	Energy Use (Diesel/Elec)	Lesschen_2011_compo	Europe	
meat_chicken	Enteric Fermentation	Lesschen_2011_compo	Europe	
meat_chicken	Fertilizer Production	Lesschen_2011_compo	Europe	
meat_chicken	Field N2O Emissions	Lesschen_2011_compo	Europe	
meat_chicken	Manure Management	Lesschen_2011_compo	Europe	
meat_chicken	Other Inputs & Processes	Lesschen_2011_compo	Europe	
meat_pork	Energy Use (Diesel/Elec)	Bellassen_2021	France	
meat_pork	Enteric Fermentation	Bellassen_2021	France	
meat_pork	Fertilizer Production	Bellassen_2021	France	
meat_pork	Field N2O Emissions	Bellassen_2021	France	
meat_pork	Manure Management	Bellassen_2021	France	
meat_pork	Other Inputs & Processes	Bellassen_2021	France	
milk	Energy Use (Diesel/Elec)	Bellassen_2021	France	
milk	Energy Use (Diesel/Elec)	CAP2ER_2025_bovin_laitier	France	
milk	Enteric Fermentation	Bellassen_2021	France	
milk	Enteric Fermentation	CAP2ER_2025_bovin_laitier	France	
milk	Fertilizer Production	Bellassen_2021	France	
milk	Fertilizer Production	CAP2ER_2025_bovin_laitier	France	
milk	Field N2O Emissions	Bellassen_2021	France	
milk	Field N2O Emissions	CAP2ER_2025_bovin_laitier	France	
milk	Land Use Change / Soil Carbon	Agribalyse_2025	France	
milk	Manure Management	Bellassen_2021	France	
milk	Manure Management	CAP2ER_2025_bovin_laitier	France	
milk	Other Inputs & Processes	Bellassen_2021	France	
wheat	Energy Use (Diesel/Elec)	Bellassen_2021	France	
wheat	Fertilizer Production	Bamber_2025_fig1	France	
wheat	Fertilizer Production	Bellassen_2021	France	
wheat	Field N2O Emissions	Bamber_2025_fig1	France	
wheat	Field N2O Emissions	Bamber_2025_sup_tab5	France	
wheat	Field N2O Emissions	Bellassen_2021	France	
wheat	Land Use Change / Soil Carbon	Agribalyse_2025	France	
wheat	Land Use Change / Soil Carbon	Bamber_2025_sup_tab5	France	
wheat	Other Inputs & Processes	Bellassen_2021	France	

indicator	unit	product	area	lit_midpoint	lit_min	lit_
DMI	kg / head / day	dairy_cows	Germany	NA	8	24.0
GE	MJ / head / day	dairy_cows	France	3.080000e+02	NA	
Nex	kg N / head / year	dairy_cows	France	1.159100e+02	NA	
feed_Energy rich feed	percent	dairy_cows	Europe	5.524862e-03	NA	
feed_Feed cereals	percent	dairy_cows	Europe	2.569061e-01	NA	
feed_Fodder maize	percent	dairy_cows	Europe	7.458564e-02	NA	
feed_Fodder root crops	percent	dairy_cows	Europe	2.762431e-03	NA	
feed_Grass	percent	dairy_cows	Europe	2.900552e-01	NA	
feed_Other feed	percent	dairy_cows	Europe	1.104972e-02	NA	
feed_Other fodder	percent	dairy_cows	Europe	1.602210e-01	NA	
feed_Protein rich feed	percent	dairy_cows	Europe	1.988950e-01	NA	
first_calving_age	month	dairy_cows	France	3.000000e+01	27	32.0
mineral fertilization	kg N / ha	wheat	France	1.657839e+02	0	209.9
organic fertilizer	kg N / ha	wheat	France	1.058427e+01	0	48.5
renewal_rate	percent	dairy_cows	France	3.600000e+01	32	36.0
share_concent	percent	dairy_cows	Germany	NA	0	40.0

```
      dplyr::mutate(indicator = paste(indicator,Harmonized_Category,sep = "__")) |>
      dplyr::summarise(lit_midpoint = mean(value, na.rm = T),
                       lit_min = min(value, na.rm = T),
                       lit_max = max(value, na.rm = T),
                       lit_source = paste0(unique(sheet),collapse = "; "),
                       area = paste0(unique(area),collapse = "; "),
                       .by = c('set', 'indicator', 'product')) |>
      dplyr::mutate(
        unit = ifelse(grepl("_ha",indicator),"kgCO2e_per_ha","kgCO2e_per_t")
      )

  ) |>
  # arrange columns
  dplyr::select(set, indicator, unit, product, area, lit_midpoint, lit_min, lit_max, lit_sou
```

```
# add global GHGE at farm scale from ADEME data base of carbon diagnosis
ADEME_desc_stat_TF <- read.csv("ADEME_desc_stat_TF.csv")

lit_table <- dplyr::bind_rows(
  lit_table,
  ADEME_desc_stat_TF |>
    dplyr::filter(exploitation_otex %in% c(1500, 4500, 4600)) |>
```

```
    dplyr::mutate(
      set = "GHG Emissions",
      indicator = dplyr::case_when(
        variable == "emissions_ges_totales_exploitation" ~ "pseudofarm_total_ghge_kgCO2e",
        variable == "emissions_ges_co2" ~ "pseudofarm_total_CO2_kgCO2e",
        variable == "emissions_ges_ch4" ~ "pseudofarm_total_CH4_kgCO2e",
        variable == "emissions_ges_n2o" ~ "pseudofarm_total_N2O_kgCO2e",

        variable == "emissions_ges_totales_exploitation_per_ha" ~ "pseudofarm_total_ghge_kgC
        variable == "emissions_ges_co2_per_ha" ~ "pseudofarm_total_CO2_kgCO2e_per_ha",
        variable == "emissions_ges_ch4_per_ha" ~ "pseudofarm_total_CH4_kgCO2e_per_ha",
        variable == "emissions_ges_n2o_per_ha" ~ "pseudofarm_total_N2O_kgCO2e_per_ha",

        variable == "emissions_ges_totales_exploitation_per_LU" ~ "pseudofarm_total_ghge_kgC
        variable == "emissions_ges_co2_per_LU" ~ "pseudofarm_total_CO2_kgCO2e_per_LU",
        variable == "emissions_ges_ch4_per_LU" ~ "pseudofarm_total_CH4_kgCO2e_per_LU",
        variable == "emissions_ges_n2o_per_LU" ~ "pseudofarm_total_N2O_kgCO2e_per_LU"
      ),
      unit = dplyr::case_when(
        grepl("_per_ha$",variable) ~ "kgCO2e_per_ha",
        grepl("_per_LU$",variable) ~ "kgCO2e_per_LU",
        .default = "kgCO2e"
      ),
      product = dplyr::case_when(
        exploitation_otex == 1500 ~ "Field crop farms",
        exploitation_otex == 4500 ~ "Dairy farms",
        exploitation_otex == 4600 ~ "Beef farms"
      ),
      area = "France",
      lit_midpoint = mean,
      lit_min = pmax(mean - sd, 0, na.rm = T),
      lit_max = mean + sd,
      lit_source = "ADEME data base"
    ) |>
    dplyr::filter(!is.na(indicator)) |>
    # arrange columns
    dplyr::select(set, indicator, unit, product, area, lit_midpoint, lit_min, lit_max, lit_s
)
```

```
# Display

lit_table |> arrange(set, product, indicator) |> gt()
```

set	indicator	unit
Agricultural practices	DMI	kg / head / day
Agricultural practices	GE	MJ / head / day
Agricultural practices	Nex	kg N / head / year
Agricultural practices	feed_Energy rich feed	percent
Agricultural practices	feed_Feed cereals	percent
Agricultural practices	feed_Fodder maize	percent
Agricultural practices	feed_Fodder root crops	percent
Agricultural practices	feed_Grass	percent
Agricultural practices	feed_Other feed	percent
Agricultural practices	feed_Other fodder	percent
Agricultural practices	feed_Protein rich feed	percent
Agricultural practices	first_calving_age	month
Agricultural practices	renewal_rate	percent
Agricultural practices	share_concent	percent
Agricultural practices	mineral fertilization	kg N / ha
Agricultural practices	organic fertilizer	kg N / ha
GHG Emissions	pseudofarm_total_CH4_kgCO2e	kgCO2e
GHG Emissions	pseudofarm_total_CH4_kgCO2e_per_LU	kgCO2e_per_LU
GHG Emissions	pseudofarm_total_CH4_kgCO2e_per_ha	kgCO2e_per_ha
GHG Emissions	pseudofarm_total_CO2_kgCO2e	kgCO2e
GHG Emissions	pseudofarm_total_CO2_kgCO2e_per_LU	kgCO2e_per_LU
GHG Emissions	pseudofarm_total_CO2_kgCO2e_per_ha	kgCO2e_per_ha
GHG Emissions	pseudofarm_total_N2O_kgCO2e	kgCO2e
GHG Emissions	pseudofarm_total_N2O_kgCO2e_per_LU	kgCO2e_per_LU
GHG Emissions	pseudofarm_total_N2O_kgCO2e_per_ha	kgCO2e_per_ha
GHG Emissions	pseudofarm_total_ghge_kgCO2e	kgCO2e
GHG Emissions	pseudofarm_total_ghge_kgCO2e_per_LU	kgCO2e_per_LU
GHG Emissions	pseudofarm_total_ghge_kgCO2e_per_ha	kgCO2e_per_ha
GHG Emissions	pseudofarm_total_CH4_kgCO2e	kgCO2e
GHG Emissions	pseudofarm_total_CH4_kgCO2e_per_LU	kgCO2e_per_LU
GHG Emissions	pseudofarm_total_CH4_kgCO2e_per_ha	kgCO2e_per_ha
GHG Emissions	pseudofarm_total_CO2_kgCO2e	kgCO2e
GHG Emissions	pseudofarm_total_CO2_kgCO2e_per_LU	kgCO2e_per_LU
GHG Emissions	pseudofarm_total_CO2_kgCO2e_per_ha	kgCO2e_per_ha
GHG Emissions	pseudofarm_total_N2O_kgCO2e	kgCO2e
GHG Emissions	pseudofarm_total_N2O_kgCO2e_per_LU	kgCO2e_per_LU
GHG Emissions	pseudofarm_total_N2O_kgCO2e_per_ha	kgCO2e_per_ha
GHG Emissions	pseudofarm_total_ghge_kgCO2e	kgCO2e
GHG Emissions	pseudofarm_total_ghge_kgCO2e_per_LU	kgCO2e_per_LU
GHG Emissions	pseudofarm_total_ghge_kgCO2e_per_ha	kgCO2e_per_ha
GHG Emissions	pseudofarm_total_CH4_kgCO2e	kgCO2e
GHG Emissions	pseudofarm_total_CH4_kgCO2e_per_ha	kgCO2e_per_ha
GHG Emissions	pseudofarm_total_CO2_kgCO2e	kgCO2e
GHG Emissions	pseudofarm_total_CO2_kgCO2e_per_ha	kgCO2e_per_ha
GHG Emissions	pseudofarm_total_N2O_kgCO2e	kgCO2e
GHG Emissions	pseudofarm_total_N2O_kgCO2e_per_ha	kgCO2e_per_ha
GHG Emissions	pseudofarm_total_ghge_kgCO2e	kgCO2e
GHG Emissions	pseudofarm_total_ghge_kgCO2e_per_ha	kgCO2e_per_ha
GHG Emissions	GHGE_kgCO2e_ha	kgCO2e_per_ha
GHG Emissions	GHGE_kgCO2e_ha___Energy Use (Diesel/Elec)	kgCO2e_per_ha

Package outputs

Agricultural practices

Quantity of mineral and organic N per hectare

We retrieve the average quantity of mineral nitrogen per hectare for different crops from Farm Survey Practices, and used it as an allocation key. We can now compare if the average per crops from the package output are close to the retrieved averages from these surveys.

```
# select crop farms in France
tmp_object_crop = m_prune_farms(my_object_GHGE, TF8 == 1, COUNTRY == "FRA")

# Retrieve fertilization estimates for crops, filtering to common wheat only
tmp_nferti <- tmp_object_crop@practices$crops$N_ferti |>
  dplyr::filter(FADN_code_letter == "CWHTC")

my_ferti = tibble::tibble(
  indicator = c("mineral fertilization", "organic fertilizer"),
  unit = "kg N / ha",
  product = "wheat",
  estimate = c(mean(tmp_nferti$N_ferti_min, na.rm = T),
               mean(tmp_nferti$N_ferti_org, na.rm = T))
)
```

Quantity of animal manure per hectare

DMI, GE and Nex per head of dairy cows

The package estimates the feed Dry Matter Intake (DMI) and Gross Energy (GE) supplied to livestock, and deduce the livestock N excretion rates (Nex). We can also estimate the Roughage-to-concentrate ratio (DM basis), %.

```
# select dairy cattle farms in France
tmp_object_milk = m_prune_farms(my_object_GHGE, TF14 == "45", COUNTRY == "FRA")

# Retrieve model estimates
my_Nexcr <- tmp_object_milk@practices$herding$N_excretion$cattle |>
  # filter for dairy cows
  dplyr::filter(FADN_code_letter == "LCOWDAIR") |>
  # select variables
```

```
  dplyr::select(dplyr::all_of(tmp_object_milk@traceability$id_cols),
                DM_t_anim,
                GE_DMI_MJ_anim_day,GE_IPCC_MJ_anim_day,
                Nex_DMI_kgN_anim_y,Nex_IPCC_kgN_anim_y) |>
  # add share concent
  dplyr::left_join(
    tmp_object_milk@practices$herding$general |>
      # filter for dairy cows
      dplyr::filter(FADN_code_letter == "LCOWDAIR") |>
      dplyr::select(dplyr::all_of(tmp_object_milk@traceability$id_cols),
                    share_concent) |>
      # convert in percent
      dplyr::mutate(share_concent = share_concent*100),
    by = tmp_object_milk@traceability$id_cols
  )

# resume table
my_Nexcr_sumup <- my_Nexcr |>
  dplyr::summarise(dplyr::across(
    c(DM_t_anim,
      GE_DMI_MJ_anim_day,GE_IPCC_MJ_anim_day,
      Nex_DMI_kgN_anim_y,Nex_IPCC_kgN_anim_y,
      share_concent),
    list(
      mean = ~ round(mean(.x, na.rm = T),2),
      sd = ~ round(sd(.x, na.rm = T),2)
    ),
    .names = "{.col}_{.fn}"
  ))
```

Rearing parameters

For dairy cows, we estimate different rearing parameters such as the age at first calving, the renewal rate, or the number of heifers to renew dairy cows.

```
# Herd rearing parameters
herd_rear_param = f_herd_rearing_param_cattle(tmp_object_milk) |>
  dplyr::select(dplyr::all_of(tmp_object_milk@traceability$id_cols),
                t_1st_calve,rt_LCOWDAIR,LCOWDAIR_Qobs,LCOWDAIR_Fin) |>
  dplyr::mutate(
    # first_calving_age
```

```
    first_calving_age = t_1st_calve,
    # renewal_rate
    ## we convert first calving cow number into livestock unit using the heifer livestock un
    renewal_rate = mean(LCOWDAIR_Fin*0.8 / LCOWDAIR_Qobs * 100,na.rm = T)
  )
```

Quantity of purchased feed/head

source ?

Yields

```
mean(tmp_object_milk@output$other_herd_products |>
       dplyr::filter(output == "milk" & species == "cattle") |>
       dplyr::pull(yield))
```

```
[1] 6.840484
```

GHGE

GHGE crops

```
# Retrieve GHGE estimates for crops, filtering to common wheat only
tmp_ghge_wheat <- tmp_object_crop@footprints$GHGE$GHGE_crops |>
  dplyr::filter(FADN_code_letter == "CWHTC")

(tmp_ghge_wheat)
```

```
# A tibble: 4,484 x 33
   ID           YEAR  COUNTRY FADN_code_letter area_ha prod_t N2O_d_kgCO2e
   <chr>        <chr> <chr>   <chr>              <dbl>  <dbl>        <dbl>
 1 515000000629 2018  FRA     CWHTC              141   1107        199029.
 2 515000000488 2018  FRA     CWHTC               41    234.        38978.
 3 515000000474 2018  FRA     CWHTC               39.8  272         13950.
 4 515000000479 2018  FRA     CWHTC               24.6  173.        27600.
 5 515000000868 2018  FRA     CWHTC               23.2  181.        23904.
 6 515000000862 2018  FRA     CWHTC               41.8  260.        12109.
```

```
 7 515000000895 2018  FRA     CWHTC               45.7  418.        51457.
 8 515000000826 2018  FRA     CWHTC               82.6  682.        98628.
 9 515000000754 2018  FRA     CWHTC               30.8  250.        35500.
10 515000000785 2018  FRA     CWHTC               17.5   10.5        2716.
# i 4,474 more rows
# i 26 more variables: N2O_ATD_kgCO2e <dbl>, N2O_L_kgCO2e <dbl>,
#   ghg_ferti_prod_kgCO2e <dbl>, ghg_heat_fuel_kgCO2e <dbl>,
#   ghg_diesel_tillage_kgCO2e <dbl>, ghg_diesel_remain_kgCO2e <dbl>,
#   ghg_elec_kgCO2e <dbl>, total_ghg_crop_kgCO2e <dbl>,
#   N2O_d_kgCO2e_per_ha <dbl>, N2O_d_kgCO2e_per_t <dbl>,
#   N2O_ATD_kgCO2e_per_ha <dbl>, N2O_ATD_kgCO2e_per_t <dbl>, ...
```

```r
# Reshape to long format, keeping per-ha and per-t columns, dropping totals
tmp_ghge_wheat_compo_long <- tmp_ghge_wheat |>
  dplyr::select(
    dplyr::all_of(tmp_object_crop@traceability$id_cols),
    dplyr::matches("per_ha$|per_t$"),      # keep intensity estimates only
    -dplyr::matches("total_ghg")           # drop aggregated totals
  ) |>
  tidyr::pivot_longer(
    cols = dplyr::matches("per_ha$|per_t$"),
    names_to = "component"
  ) |>
  dplyr::mutate(
    # Map component names to standardised GHGE categories
    Harmonized_Category = case_when(
      str_detect(component, "N2O")        ~ "Field N2O Emissions",
      str_detect(component, "ferti_prod") ~ "Fertilizer Production",
      str_detect(component, "diesel|fuel|elec")~ "Energy Use (Diesel/Elec)",
      TRUE                                ~ "Other Inputs & Processes" # pesticides, seeds, e
    ),
    product = "wheat",
    # Build composite indicator label encoding unit type and GHGE category
    indicator = dplyr::case_when(
      grepl("per_ha$", component) ~ paste("GHGE_kgCO2e_ha", Harmonized_Category, sep = "__")
      grepl("per_t$",  component) ~ paste("GHGE_kgCO2e_t",  Harmonized_Category, sep = "__")
    ),
    # Assign unit label based on denominator (area vs. product mass)
    unit = dplyr::case_when(
      grepl("per_ha$", component) ~ "kgCO2e_per_ha",
      grepl("per_t$",  component) ~ "kgCO2e_per_t"
    )
```

```
  )

# Aggregate: sum GHGE components within each farm × category × unit combination
tmp_ghge_wheat_compo <- tmp_ghge_wheat_compo_long |>
  dplyr::summarise(
    value = sum(value, na.rm = TRUE),
    .by = c(tmp_object_crop@traceability$id_cols, "product", "Harmonized_Category", "indicat
  )
```

```
# Retrieve GHGE estimates for crops, filtering to common maize only
tmp_ghge_maize <- tmp_object_crop@footprints$GHGE$GHGE_crops |>
  dplyr::filter(FADN_code_letter == "CMZ")

tmp_ghge_maize
```

```
# A tibble: 2,405 x 33
   ID           YEAR  COUNTRY FADN_code_letter area_ha prod_t N2O_d_kgCO2e
   <chr>        <chr> <chr>   <chr>              <dbl>  <dbl>        <dbl>
 1 515000000488 2018  FRA     CMZ                21    111.         17730.
 2 515000000474 2018  FRA     CMZ                 8.27  71.6         3335.
 3 515000000786 2018  FRA     CMZ                43.4  307.         40862.
 4 515000000787 2018  FRA     CMZ                 0.47   3.91          490.
 5 515000000735 2018  FRA     CMZ                 7     36.7         7041.
 6 515000000733 2018  FRA     CMZ                28.5  175.         50802.
 7 515000000732 2018  FRA     CMZ                 5.9   58.4         4687.
 8 515000000728 2018  FRA     CMZ                21.0  122.         15824.
 9 515000000301 2018  FRA     CMZ                10.5   53.7        10868.
10 515000000302 2018  FRA     CMZ                39.2  429.         39871.
# i 2,395 more rows
# i 26 more variables: N2O_ATD_kgCO2e <dbl>, N2O_L_kgCO2e <dbl>,
#   ghg_ferti_prod_kgCO2e <dbl>, ghg_heat_fuel_kgCO2e <dbl>,
#   ghg_diesel_tillage_kgCO2e <dbl>, ghg_diesel_remain_kgCO2e <dbl>,
#   ghg_elec_kgCO2e <dbl>, total_ghg_crop_kgCO2e <dbl>,
#   N2O_d_kgCO2e_per_ha <dbl>, N2O_d_kgCO2e_per_t <dbl>,
#   N2O_ATD_kgCO2e_per_ha <dbl>, N2O_ATD_kgCO2e_per_t <dbl>, ...
```

```
# Reshape to long format, keeping per-ha and per-t columns, dropping totals
tmp_ghge_maize_compo_long <- tmp_ghge_maize |>
  dplyr::select(
    dplyr::all_of(tmp_object_crop@traceability$id_cols),
    dplyr::matches("per_ha$|per_t$"),      # keep intensity estimates only
```

```
    -dplyr::matches("total_ghg")           # drop aggregated totals
  ) |>
  tidyr::pivot_longer(
    cols = dplyr::matches("per_ha$|per_t$"),
    names_to = "component"
  ) |>
  dplyr::mutate(
    # Map component names to standardised GHGE categories
    Harmonized_Category = case_when(
      str_detect(component, "N2O")        ~ "Field N2O Emissions",
      str_detect(component, "ferti_prod") ~ "Fertilizer Production",
      str_detect(component, "diesel|fuel|elec")~ "Energy Use (Diesel/Elec)",
      TRUE                                ~ "Other Inputs & Processes" # pesticides, seeds, e
    ),
    product = "maize_grain",
    # Build composite indicator label encoding unit type and GHGE category
    indicator = dplyr::case_when(
      grepl("per_ha$", component) ~ paste("GHGE_kgCO2e_ha", Harmonized_Category, sep = "__")
      grepl("per_t$",  component) ~ paste("GHGE_kgCO2e_t",  Harmonized_Category, sep = "__")
    ),
    # Assign unit label based on denominator (area vs. product mass)
    unit = dplyr::case_when(
      grepl("per_ha$", component) ~ "kgCO2e_per_ha",
      grepl("per_t$",  component) ~ "kgCO2e_per_t"
    )
  )

# Aggregate: sum GHGE components within each farm × category × unit combination
tmp_ghge_maize_compo <- tmp_ghge_maize_compo_long |>
  dplyr::summarise(
    value = sum(value, na.rm = TRUE),
    .by = c(tmp_object_crop@traceability$id_cols, "product", "Harmonized_Category", "indicato
  )
```

```
# Retrieve GHGE estimates for crops, filtering to common rapeseed only
tmp_ghge_rapeseed <- tmp_object_crop@footprints$GHGE$GHGE_crops |>
  dplyr::filter(FADN_code_letter == "CRAPE")

tmp_ghge_rapeseed
```

```
# A tibble: 2,740 x 33
   ID           YEAR  COUNTRY FADN_code_letter area_ha prod_t N2O_d_kgCO2e
```

```
   <chr>        <chr> <chr>   <chr>              <dbl>  <dbl>        <dbl>
 1 515000000629 2018  FRA     CRAPE              58     212.        72823.
 2 515000000488 2018  FRA     CRAPE              17.4    50.1       15084.
 3 515000000474 2018  FRA     CRAPE               8      29.4        2231.
 4 515000000479 2018  FRA     CRAPE              23.4    62.8       23009.
 5 515000000868 2018  FRA     CRAPE               4.83   16.9        4309.
 6 515000000862 2018  FRA     CRAPE              11.8    32.7        2585.
 7 515000000895 2018  FRA     CRAPE              36.2   159.        34783.
 8 515000000826 2018  FRA     CRAPE              35.0   120.        36089.
 9 515000000754 2018  FRA     CRAPE              33.2   121.        33177.
10 515000000786 2018  FRA     CRAPE              10.8    33.9       10218.
# i 2,730 more rows
# i 26 more variables: N2O_ATD_kgCO2e <dbl>, N2O_L_kgCO2e <dbl>,
#   ghg_ferti_prod_kgCO2e <dbl>, ghg_heat_fuel_kgCO2e <dbl>,
#   ghg_diesel_tillage_kgCO2e <dbl>, ghg_diesel_remain_kgCO2e <dbl>,
#   ghg_elec_kgCO2e <dbl>, total_ghg_crop_kgCO2e <dbl>,
#   N2O_d_kgCO2e_per_ha <dbl>, N2O_d_kgCO2e_per_t <dbl>,
#   N2O_ATD_kgCO2e_per_ha <dbl>, N2O_ATD_kgCO2e_per_t <dbl>, ...
```

```r
# Reshape to long format, keeping per-ha and per-t columns, dropping totals
tmp_ghge_rapeseed_compo_long <- tmp_ghge_rapeseed |>
  dplyr::select(
    dplyr::all_of(tmp_object_crop@traceability$id_cols),
    dplyr::matches("per_ha$|per_t$"),      # keep intensity estimates only
    -dplyr::matches("total_ghg")           # drop aggregated totals
  ) |>
  tidyr::pivot_longer(
    cols = dplyr::matches("per_ha$|per_t$"),
    names_to = "component"
  ) |>
  dplyr::mutate(
    # Map component names to standardised GHGE categories
    Harmonized_Category = case_when(
      str_detect(component, "N2O")        ~ "Field N2O Emissions",
      str_detect(component, "ferti_prod") ~ "Fertilizer Production",
      str_detect(component, "diesel|fuel|elec")~ "Energy Use (Diesel/Elec)",
      TRUE                                ~ "Other Inputs & Processes" # pesticides, seeds, e
    ),
    product = "rapeseed",
    # Build composite indicator label encoding unit type and GHGE category
    indicator = dplyr::case_when(
      grepl("per_ha$", component) ~ paste("GHGE_kgCO2e_ha", Harmonized_Category, sep = "__")
```

```
      grepl("per_t$",  component) ~ paste("GHGE_kgCO2e_t",  Harmonized_Category, sep = "__")
    ),
    # Assign unit label based on denominator (area vs. product mass)
    unit = dplyr::case_when(
      grepl("per_ha$", component) ~ "kgCO2e_per_ha",
      grepl("per_t$",  component) ~ "kgCO2e_per_t"
    )
  )

# Aggregate: sum GHGE components within each farm × category × unit combination
tmp_ghge_rapeseed_compo <- tmp_ghge_rapeseed_compo_long |>
  dplyr::summarise(
    value = sum(value, na.rm = TRUE),
    .by = c(tmp_object_crop@traceability$id_cols, "product", "Harmonized_Category", "indicat
  )
```

GHGE milk

```
(tmp_object_milk@footprints$GHGE$GHGE_milk)
```

```
# A tibble: 3,453 x 109
   ID            YEAR  COUNTRY CH4_enteric_kgCO2e_livcat CH4_MM_kgCO2e_livcat
   <chr>         <chr> <chr>                       <dbl>                <dbl>
 1 515000000283  2018  FRA                       240929.              33873.
 2 515000000219  2018  FRA                       366101.              57663.
 3 565000000500  2018  FRA                       112239.              17227.
 4 5650000005628 2018  FRA                       241859.              36111.
 5 5650000006032 2018  FRA                       395246.              60111.
 6 5650000006096 2018  FRA                       441339.              67440.
 7 5650000006097 2018  FRA                       227522.              34048.
 8 5650000006598 2018  FRA                       361737.              53763.
 9 5650000006170 2018  FRA                       379586.              58368.
10 5650000006807 2018  FRA                       279940.              39952.
# i 3,443 more rows
# i 104 more variables: N2O_D_MM_kgCO2e_livcat <dbl>,
#   N2O_G_mm_kgCO2e_livcat <dbl>, N2O_L_mm_kgCO2e_livcat <dbl>,
#   feed_farm_area_ha_livcat <dbl>, feed_farm_N2O_d_kgCO2e_livcat <dbl>,
#   feed_farm_N2O_ATD_kgCO2e_livcat <dbl>, feed_farm_N2O_L_kgCO2e_livcat <dbl>,
#   feed_farm_ghg_ferti_prod_kgCO2e_livcat <dbl>,
#   feed_farm_ghg_heat_fuel_kgCO2e_livcat <dbl>, ...
```

```r
# Retrieve GHGE estimates for milk, filtering to cow milk output only
tmp_ghge_milk <- tmp_object_milk@footprints$GHGE$GHGE_milk |>
  dplyr::filter(species == "cattle")

# Reshape GHGE milk data to long format, keeping only per-ha (pseudofarm) and per-t columns
tmp_ghge_milk_compo_long <- tmp_ghge_milk |>
  dplyr::select(
    dplyr::all_of(tmp_object_milk@traceability$id_cols),
    dplyr::matches("per_ha_pseudofarm|per_t$"),   # keep pseudofarm and per-tonne estimates
    -dplyr::matches("total_ghg|ghge_herd|feed_farm") # drop aggregated/herd-level columns
  ) |>
  tidyr::pivot_longer(
    cols = dplyr::matches("per_ha_pseudofarm|per_t$"),
    names_to = "component"
  ) |>
  dplyr::mutate(
    # Map component names to standardised GHGE categories
    Harmonized_Category = case_when(
      str_detect(component, "(?i)enteric")  ~ "Enteric Fermentation",
      str_detect(component, "(?i)mm|MM")    ~ "Manure Management",
      #      str_detect(component, "(?i)feed")     ~ "Feed Production",
      str_detect(component, "(?i)N2O")        ~ "Field N2O Emissions",
      str_detect(component, "(?i)ferti_prod") ~ "Fertilizer Production",
      str_detect(component, "(?i)diesel|fuel|elec")~ "Energy Use (Diesel/Elec)",
      # TODO: energy use for livestock buildings is not yet accounted for
      TRUE ~ "Other Inputs & Processes"  # pesticides, seeds, transport, post-harvest
    ),
    product = "milk",
    # Build a composite indicator label encoding both unit type and GHGE category
    indicator = dplyr::case_when(
      grepl("per_ha_pseudofarm$", component) ~ paste("GHGE_kgCO2e_ha", Harmonized_Category, s
      grepl("per_t$",             component) ~ paste("GHGE_kgCO2e_t",  Harmonized_Category, s
    ),
    # Assign unit label based on whether the estimate is per ha or per tonne of product
    unit = dplyr::case_when(
      grepl("per_ha_pseudofarm$", component) ~ "kgCO2e_per_ha",
      grepl("per_t$",             component) ~ "kgCO2e_per_t"
    )
  )

# Aggregate: sum GHGE components within each farm × category × unit combination
tmp_ghge_milk_compo <- tmp_ghge_milk_compo_long |>
```

```
  dplyr::summarise(
    value = sum(value, na.rm = TRUE),
    .by = c(tmp_object_milk@traceability$id_cols, "product", "Harmonized_Category", "indicat
  )
```

GHGE beef

For beef, as wee compare our values to CAP2ER reports that consider only the farm area, we use farm values, not pseudofarm ones.

```
# select meat farms in France
tmp_object_beef = m_prune_farms(my_object_GHGE, TF14 == "49", COUNTRY == "FRA")

# keep farms with 100% grassland
#quantile(tmp_object_beef@practices$herding$general$ha_perm_grassland_pha)
#tmp_object_beef = m_keep_farms(
#  tmp_object_beef,
#  tmp_object_beef@practices$herding$general |>
#    dplyr::filter(ha_perm_grassland_pha == 1) |>
#    dplyr::select(ID,YEAR,COUNTRY)
#)

#(tmp_object_beef@footprints$GHGE$GHGE_meat)

# Retrieve GHGE estimates for meat, filtering to cow meat output only
tmp_ghge_beef <- tmp_object_beef@footprints$GHGE$GHGE_meat |>
  dplyr::filter(output == "meat_beef" | output == "meat_cull_cow")

# Reshape GHGE meat data to long format, keeping only per-ha (farm) and per-t columns
tmp_ghge_beef_compo_long <- tmp_ghge_beef |>
  dplyr::select(
    dplyr::all_of(tmp_object_beef@traceability$id_cols),
    output, species,
    dplyr::matches("per_ha_farm$|per_t$"),   # keep farm and per-tonne estimates
    -dplyr::matches("total_ghg|ghge_herd|^feed_pseudofarm") # drop aggregated/herd-level colu
    #dplyr::matches("per_ha_pseudofarm|per_t$"),   # keep pseudofarm and per-tonne estimates
    #-dplyr::matches("total_ghg|ghge_herd|feed_farm") # drop aggregated/herd-level columns

  ) |>
  tidyr::pivot_longer(
    cols = dplyr::matches("per_ha_farm|per_t$"),
```

```
    #cols = dplyr::matches("per_ha_pseudofarm|per_t$"),
    names_to = "component"
  ) |>
  dplyr::mutate(
    # Map component names to standardised GHGE categories
    Harmonized_Category = dplyr::case_when(
      str_detect(component, "(?i)enteric")  ~ "Enteric Fermentation",
      str_detect(component, "(?i)mm|MM")    ~ "Manure Management",
      #      str_detect(component, "(?i)feed")     ~ "Feed Production",
      str_detect(component, "(?i)N2O")        ~ "Field N2O Emissions",
      str_detect(component, "(?i)ferti_prod") ~ "Fertilizer Production",
      str_detect(component, "(?i)diesel|fuel|elec")~ "Energy Use (Diesel/Elec)",
      # TODO: energy use for livestock buildings is not yet accounted for
      TRUE ~ "Other Inputs & Processes"  # pesticides, seeds, transport, post-harvest
    ),
    product = "cattle_meat",
    # Build a composite indicator label encoding both unit type and GHGE category
    indicator = dplyr::case_when(
      grepl("per_ha_farm$", component) ~ paste("GHGE_kgCO2e_ha", Harmonized_Category, sep = "
      #grepl("per_ha_pseudofarm$", component) ~ paste("GHGE_kgCO2e_ha", Harmonized_Category,
      grepl("per_t$",           component) ~ paste("GHGE_kgCO2e_t",  Harmonized_Category, se
    ),
    # Assign unit label based on whether the estimate is per ha or per tonne of product
    unit = dplyr::case_when(
      grepl("per_ha_farm$", component) ~ "kgCO2e_per_ha",
      #grepl("per_ha_pseudofarm$", component) ~ "kgCO2e_per_ha",
      grepl("per_t$",           component) ~ "kgCO2e_per_t"
    )
  )

# Aggregate: sum GHGE components within each farm × category × unit combination
tmp_ghge_beef_compo <- tmp_ghge_beef_compo_long |>
  dplyr::summarise(
    value = sum(value, na.rm = TRUE),
    .by = c(tmp_object_beef@traceability$id_cols, "product", "Harmonized_Category", "indicato
  )

tmp_ghge_beef_compo
```

```
# A tibble: 38,710 x 8
   ID          YEAR  COUNTRY product  Harmonized_Category indicator unit  value
   <chr>       <chr> <chr>   <chr>    <chr>               <chr>     <chr> <dbl>
```

```
 1 515000000954 2018  FRA     cattle_~ Enteric Fermentati~ GHGE_kgC~ kgCO~ 7738.
 2 515000000954 2018  FRA     cattle_~ Manure Management   GHGE_kgC~ kgCO~ 1631.
 3 515000000954 2018  FRA     cattle_~ Field N2O Emissions GHGE_kgC~ kgCO~ 1026.
 4 515000000954 2018  FRA     cattle_~ Fertilizer Product~ GHGE_kgC~ kgCO~    0
 5 515000000954 2018  FRA     cattle_~ Energy Use (Diesel~ GHGE_kgC~ kgCO~  153.
 6 515000000954 2018  FRA     cattle_~ Enteric Fermentati~ GHGE_kgC~ kgCO~ 7236.
 7 515000000954 2018  FRA     cattle_~ Manure Management   GHGE_kgC~ kgCO~ 1525.
 8 515000000954 2018  FRA     cattle_~ Field N2O Emissions GHGE_kgC~ kgCO~  960.
 9 515000000954 2018  FRA     cattle_~ Fertilizer Product~ GHGE_kgC~ kgCO~    0
10 515000000954 2018  FRA     cattle_~ Energy Use (Diesel~ GHGE_kgC~ kgCO~  143.
# i 38,700 more rows
```

GHGE pork

```
# select meat farms in France
tmp_object_pork = m_prune_farms(my_object_GHGE, TF14 == "50", COUNTRY == "FRA")

# keep farms with pigs
quantile(tmp_object_pork@herd |> dplyr::filter(FADN_code_letter == "LPIGFAT") |> dplyr::pull
```

```
     0%      25%      50%      75%     100%
 25.000  674.750 1125.500 1831.305 9750.000
```

```
tmp_object_pork = m_keep_farms(
  tmp_object_pork,
  tmp_object_pork@herd |>
    dplyr::filter(FADN_code_letter == "LPIGFAT" & Qobs >0) |>
    dplyr::select(ID,YEAR,COUNTRY)
)

#(tmp_object_pork@footprints$GHGE$GHGE_meat)

# Retrieve GHGE estimates for meat, filtering to cow meat output only
tmp_ghge_pork <- tmp_object_pork@footprints$GHGE$GHGE_meat |>
  dplyr::filter(output == "meat_pork")

summary(tmp_ghge_pork)
```

```
      ID                YEAR             COUNTRY
 Length:921         Length:921         Length:921
```

```
Class :character   Class :character   Class :character
Mode  :character   Mode  :character   Mode  :character
```

```
CH4_enteric_kgCO2e_livcat CH4_MM_kgCO2e_livcat N2O_D_MM_kgCO2e_livcat
Min.   :  2525            Min.   :  18369      Min.   :   671.7
1st Qu.: 31888            1st Qu.: 242258      1st Qu.:  8809.1
Median : 52745            Median : 397078      Median : 14656.0
Mean   : 68180            Mean   : 487286      Mean   : 17846.4
3rd Qu.: 86501            3rd Qu.: 651896      3rd Qu.: 23856.2
Max.   :552971            Max.   :2902209      Max.   :106127.7

N2O_G_mm_kgCO2e_livcat N2O_L_mm_kgCO2e_livcat feed_farm_area_ha_livcat
Min.   :  1012         Min.   :  4.062        Min.   :  0.00
1st Qu.: 13275         1st Qu.: 53.276        1st Qu.:  0.00
Median : 22086         Median : 88.638        Median : 11.46
Mean   : 26894         Mean   :107.933        Mean   : 28.11
3rd Qu.: 35950         3rd Qu.:144.279        3rd Qu.: 44.48
Max.   :159930         Max.   :641.848        Max.   :178.02

feed_farm_N2O_d_kgCO2e_livcat feed_farm_N2O_ATD_kgCO2e_livcat
Min.   :     0                Min.   :    0
1st Qu.:     0                1st Qu.:    0
Median : 10338                Median : 1370
Mean   : 28476                Mean   : 4392
3rd Qu.: 44020                3rd Qu.: 7495
Max.   :207121                Max.   :29965

feed_farm_N2O_L_kgCO2e_livcat feed_farm_ghg_ferti_prod_kgCO2e_livcat
Min.   :    0                 Min.   :     0
1st Qu.:    0                 1st Qu.:     0
Median : 3872                 Median :  1744
Mean   :11069                 Mean   : 11384
3rd Qu.:18022                 3rd Qu.: 14993
Max.   :72522                 Max.   :131050

feed_farm_ghg_heat_fuel_kgCO2e_livcat
Min.   :0
1st Qu.:0
Median :0
Mean   :0
```

```
3rd Qu.:0
Max.   :0

feed_farm_ghg_diesel_tillage_kgCO2e_livcat
Min.   :     0
1st Qu.:     0
Median :  2822
Mean   : 11481
3rd Qu.: 16484
Max.   :199071

feed_farm_ghg_diesel_remain_kgCO2e_livcat feed_farm_ghg_elec_kgCO2e_livcat
Min.   :   0.000                          Min.   :   0.00
1st Qu.:   0.000                          1st Qu.:   0.00
Median :   0.000                          Median :   0.00
Mean   :  90.448                          Mean   :  53.04
3rd Qu.:   9.713                          3rd Qu.:  10.20
Max.   :7524.800                          Max.   :4579.70

feed_farm_total_ghg_crop_kgCO2e_livcat feed_pseudofarm_area_ha_livcat
Min.   :     0                         Min.   :   8.86
1st Qu.:     0                         1st Qu.: 116.86
Median : 26571                         Median : 194.78
Mean   : 66945                         Mean   : 236.44
3rd Qu.:103754                         3rd Qu.: 319.11
Max.   :538610                         Max.   :1399.91

feed_pseudofarm_N2O_d_kgCO2e_livcat feed_pseudofarm_N2O_ATD_kgCO2e_livcat
Min.   :   7374                     Min.   :   714.2
1st Qu.: 112555                     1st Qu.: 11936.9
Median : 186377                     Median : 20338.8
Mean   : 227579                     Mean   : 24657.5
3rd Qu.: 306384                     3rd Qu.: 33023.2
Max.   :1436117                     Max.   :144779.3

feed_pseudofarm_N2O_L_kgCO2e_livcat
Min.   :  2375
1st Qu.: 37075
Median : 62230
Mean   : 74820
3rd Qu.:100972
Max.   :451932
```

```
feed_pseudofarm_ghg_ferti_prod_kgCO2e_livcat
Min.   :      0
1st Qu.:  75761
Median : 128964
Mean   : 157424
3rd Qu.: 212304
Max.   :1091123

feed_pseudofarm_ghg_heat_fuel_kgCO2e_livcat
Min.   :0
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0

feed_pseudofarm_ghg_diesel_tillage_kgCO2e_livcat
Min.   :   4902
1st Qu.:  69219
Median : 118317
Mean   : 148772
3rd Qu.: 189822
Max.   :1126559

feed_pseudofarm_ghg_diesel_remain_kgCO2e_livcat
Min.   :  1005
1st Qu.: 11654
Median : 20093
Mean   : 24605
3rd Qu.: 32130
Max.   :169684

feed_pseudofarm_ghg_elec_kgCO2e_livcat
Min.   :  107
1st Qu.: 1271
Median : 2183
Mean   : 2720
3rd Qu.: 3492
Max.   :19867

feed_pseudofarm_total_ghg_crop_kgCO2e_livcat ghg_diesel_remain_kgCO2e_livcat
Min.   :  21824                              Min.   :     0
1st Qu.: 333676                              1st Qu.:  3549
```

```
Median : 540529                                Median :  6579
Mean   : 660576                                Mean   : 10383
3rd Qu.: 881841                                3rd Qu.: 10937
Max.   :4350654                                Max.   :550804

ghg_heat_fuel_kgCO2e_livcat ghg_elec_kgCO2e_livcat
Min.   :     0              Min.   :    0
1st Qu.:     0              1st Qu.: 2618
Median :     0              Median : 5470
Mean   :  4128              Mean   : 7753
3rd Qu.:   742              3rd Qu.:10295
Max.   :462251              Max.   :47121

farm_ghge_herd_kgCO2e_livcat pseudofarm_ghge_herd_kgCO2e_livcat
Min.   :  25697              Min.   :  47878
1st Qu.: 337107              1st Qu.: 644484
Median : 559152              Median :1034795
Mean   : 689523              Mean   :1283155
3rd Qu.: 911084              3rd Qu.:1707086
Max.   :4214329              Max.   :8317069

farm_ghge_herd_kgCO2e_per_anim pseudofarm_ghge_herd_kgCO2e_per_anim
Min.   : 141.3                 Min.   :  237.7
1st Qu.:1090.4                 1st Qu.: 2062.4
Median :1349.1                 Median : 2563.1
Mean   :1264.9                 Mean   : 2402.8
3rd Qu.:1506.8                 3rd Qu.: 2909.4
Max.   :4895.2                 Max.   :10752.1

   output            species          sales_e_output      sum_SV_species
Length:921         Length:921         Min.   :    288.6   Min.   :    368
Class :character   Class :character   1st Qu.: 260722.4   1st Qu.: 332463
Mode  :character   Mode  :character   Median : 416854.1   Median : 531556
                                      Mean   : 530678.7   Mean   : 676701
                                      3rd Qu.: 677707.5   3rd Qu.: 864186
                                      Max.   :3327702.7   Max.   :4243356

  econ_ratio         prod_t          CH4_enteric_kgCO2e_per_ha_farm
Min.   :0.7842   Min.   :   0.1541   Min.   : 251.4
1st Qu.:0.7842   1st Qu.: 120.5573   1st Qu.:1364.8
Median :0.7842   Median : 200.3295   Median :5963.6
Mean   :0.7842   Mean   : 251.9925   Mean   :   Inf
3rd Qu.:0.7842   3rd Qu.: 319.1403   3rd Qu.:   Inf
```

```
Max.   :0.7842   Max.   :1442.0127   Max.   :   Inf

CH4_MM_kgCO2e_per_ha_farm N2O_D_MM_kgCO2e_per_ha_farm
Min.   : 2640             Min.   :  96.65
1st Qu.: 9680             1st Qu.: 355.24
Median :41866             Median :1517.52
Mean   :  Inf             Mean   :    Inf
3rd Qu.:  Inf             3rd Qu.:    Inf
Max.   :  Inf             Max.   :    Inf

N2O_G_mm_kgCO2e_per_ha_farm N2O_L_mm_kgCO2e_per_ha_farm
Min.   : 145.6              Min.   :0.5845
1st Qu.: 535.3              1st Qu.:2.1484
Median :2286.8              Median :9.1778
Mean   :   Inf              Mean   :   Inf
3rd Qu.:   Inf              3rd Qu.:   Inf
Max.   :   Inf              Max.   :   Inf

feed_farm_N2O_d_kgCO2e_per_ha_farm feed_farm_N2O_ATD_kgCO2e_per_ha_farm
Min.   : 238.2                     Min.   : 31.85
1st Qu.: 800.7                     1st Qu.:113.32
Median : 964.7                     Median :152.37
Mean   : 963.9                     Mean   :150.38
3rd Qu.:1084.3                     3rd Qu.:179.17
Max.   :2295.6                     Max.   :297.68
NA's   :333                        NA's   :333
feed_farm_N2O_L_kgCO2e_per_ha_farm feed_farm_ghg_ferti_prod_kgCO2e_per_ha_farm
Min.   :112.2                      Min.   :   0.0
1st Qu.:323.0                      1st Qu.: 210.0
Median :380.9                      Median : 337.4
Mean   :377.6                      Mean   : 371.5
3rd Qu.:433.0                      3rd Qu.: 507.4
Max.   :674.5                      Max.   :1971.5
NA's   :333                        NA's   :333
feed_farm_ghg_heat_fuel_kgCO2e_per_ha_farm
Min.   :0
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0
NA's   :333
feed_farm_ghg_diesel_tillage_kgCO2e_per_ha_farm
```

```
Min.   :    0.0
1st Qu.:  220.5
Median :  377.0
Mean   :  599.0
3rd Qu.:  603.1
Max.   :28534.6
NA's   :333
feed_farm_ghg_diesel_remain_kgCO2e_per_ha_farm
Min.   :  0.000
1st Qu.:  0.000
Median :  0.000
Mean   :  4.607
3rd Qu.:  5.331
Max.   :166.648
NA's   :333
feed_farm_ghg_elec_kgCO2e_per_ha_farm
Min.   :  0.000
1st Qu.:  0.000
Median :  0.000
Mean   :  3.349
3rd Qu.:  2.959
Max.   :254.643
NA's   :333
feed_farm_total_ghg_crop_kgCO2e_per_ha_farm
Min.   :  604.1
1st Qu.: 1907.1
Median : 2253.5
Mean   : 2470.3
3rd Qu.: 2694.6
Max.   :32979.0
NA's   :333
ghg_diesel_remain_kgCO2e_per_ha_farm ghg_heat_fuel_kgCO2e_per_ha_farm
Min.   :  3.521                      Min.   :  0.0
1st Qu.:170.559                      1st Qu.:  0.0
Median :325.215                      Median :  0.0
Mean   :    Inf                      Mean   :  Inf
3rd Qu.:    Inf                      3rd Qu.:289.7
Max.   :    Inf                      Max.   :  Inf
NA's   :29                           NA's   :223
ghg_elec_kgCO2e_per_ha_farm farm_ghge_herd_kgCO2e_per_ha_farm
Min.   :  2.536             Min.   : 5459
1st Qu.:151.714             1st Qu.:15058
Median :545.192             Median :55030
```

```
Mean   :    Inf             Mean   :  Inf
3rd Qu.:    Inf             3rd Qu.:  Inf
Max.   :    Inf             Max.   :  Inf
NA's   :10
farm_ghge_herd_kgCO2e_per_anim_per_ha_farm
Min.   :  2.055
1st Qu.: 26.887
Median :102.645
Mean   :    Inf
3rd Qu.:    Inf
Max.   :    Inf

CH4_enteric_kgCO2e_per_ha_pseudofarm CH4_MM_kgCO2e_per_ha_pseudofarm
Min.   :  51.77                      Min.   :1603
1st Qu.: 237.22                      1st Qu.:2048
Median : 269.18                      Median :2073
Mean   : 291.60                      Mean   :2058
3rd Qu.: 304.70                      3rd Qu.:2073
Max.   :1667.06                      Max.   :2416

N2O_D_MM_kgCO2e_per_ha_pseudofarm N2O_G_mm_kgCO2e_per_ha_pseudofarm
Min.   :58.89                     Min.   : 88.74
1st Qu.:75.09                     1st Qu.:113.16
Median :75.81                     Median :114.24
Mean   :75.40                     Mean   :113.62
3rd Qu.:75.81                     3rd Qu.:114.24
Max.   :88.54                     Max.   :133.43

N2O_L_mm_kgCO2e_per_ha_pseudofarm
Min.   :0.3561
1st Qu.:0.4541
Median :0.4585
Mean   :0.4560
3rd Qu.:0.4585
Max.   :0.5355

feed_pseudofarm_N2O_d_kgCO2e_per_ha_pseudofarm
Min.   : 523.7
1st Qu.: 885.9
Median : 964.6
Mean   : 959.6
3rd Qu.:1044.2
Max.   :1283.8
```

```
feed_pseudofarm_N2O_ATD_kgCO2e_per_ha_pseudofarm
Min.   : 76.59
1st Qu.: 94.67
Median :104.15
Mean   :103.60
3rd Qu.:110.90
Max.   :146.34

feed_pseudofarm_N2O_L_kgCO2e_per_ha_pseudofarm
Min.   :246.4
1st Qu.:299.0
Median :324.1
Mean   :315.2
3rd Qu.:332.7
Max.   :416.3

feed_pseudofarm_ghg_ferti_prod_kgCO2e_per_ha_pseudofarm
Min.   :   0.0
1st Qu.: 603.5
Median : 675.6
Mean   : 665.3
3rd Qu.: 782.3
Max.   :1007.1

feed_pseudofarm_ghg_heat_fuel_kgCO2e_per_ha_pseudofarm
Min.   :0
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0

feed_pseudofarm_ghg_diesel_tillage_kgCO2e_per_ha_pseudofarm
Min.   : 308.1
1st Qu.: 551.5
Median : 571.3
Mean   : 627.9
3rd Qu.: 660.7
Max.   :2325.3

feed_pseudofarm_ghg_diesel_remain_kgCO2e_per_ha_pseudofarm
Min.   : 39.21
```

```
1st Qu.: 91.42
Median :109.25
Mean   :104.12
3rd Qu.:114.21
Max.   :207.54

feed_pseudofarm_ghg_elec_kgCO2e_per_ha_pseudofarm
Min.   : 4.158
1st Qu.:10.072
Median :11.522
Mean   :11.486
3rd Qu.:12.183
Max.   :21.210

feed_pseudofarm_total_ghg_crop_kgCO2e_per_ha_pseudofarm
Min.   :1934
1st Qu.:2485
Median :2701
Mean   :2787
3rd Qu.:3045
Max.   :3672

ghg_diesel_remain_kgCO2e_per_ha_pseudofarm
Min.   :  0.00
1st Qu.: 17.71
Median : 34.42
Mean   : 51.95
3rd Qu.: 55.77
Max.   :837.46

ghg_heat_fuel_kgCO2e_per_ha_pseudofarm ghg_elec_kgCO2e_per_ha_pseudofarm
Min.   :  0.000                        Min.   :  0.00
1st Qu.:  0.000                        1st Qu.: 20.35
Median :  0.000                        Median : 29.10
Mean   : 16.063                        Mean   : 31.08
3rd Qu.:  5.321                        3rd Qu.: 38.59
Max.   :518.667                        Max.   :271.15

pseudofarm_ghge_herd_kgCO2e_per_ha_pseudofarm
Min.   :4504
1st Qu.:5151
Median :5378
Mean   :5426
```

```
3rd Qu.:5689
Max.   :7271

pseudofarm_ghge_herd_kgCO2e_per_anim_per_ha_pseudofarm
Min.   :  0.8169
1st Qu.:  6.5521
Median : 11.1803
Mean   : 17.7633
3rd Qu.: 18.5272
Max.   :270.7063

CH4_enteric_kgCO2e_per_t CH4_MM_kgCO2e_per_t N2O_D_MM_kgCO2e_per_t
Min.   :   48.9          Min.   :   370.4    Min.   :   13.59
1st Qu.:  239.3          1st Qu.:  1845.1    1st Qu.:   67.44
Median :  275.0          Median :  2071.5    Median :   75.78
Mean   :  511.9          Mean   :  3705.9    Mean   :  134.57
3rd Qu.:  303.3          3rd Qu.:  2319.9    3rd Qu.:   84.99
Max.   :50095.0          Max.   :644038.5    Max.   :22554.06

N2O_G_mm_kgCO2e_per_t N2O_L_mm_kgCO2e_per_t feed_farm_N2O_d_kgCO2e_per_t
Min.   :   20.48      Min.   :  0.08221     Min.   :    0.00
1st Qu.:  101.63      1st Qu.:  0.40788     1st Qu.:    0.00
Median :  114.20      Median :  0.45833     Median :   42.75
Mean   :  202.79      Mean   :  0.81384     Mean   :  219.21
3rd Qu.:  128.07      3rd Qu.:  0.51398     3rd Qu.:  189.57
Max.   :33987.96      Max.   :136.40417     Max.   :93099.31

feed_farm_N2O_ATD_kgCO2e_per_t feed_farm_N2O_L_kgCO2e_per_t
Min.   :    0.000              Min.   :    0.00
1st Qu.:    0.000              1st Qu.:    0.00
Median :    6.707              Median :   17.05
Mean   :   33.972              Mean   :   89.17
3rd Qu.:   31.576              3rd Qu.:   78.50
Max.   :14292.775              Max.   :39765.07

feed_farm_ghg_ferti_prod_kgCO2e_per_t feed_farm_ghg_heat_fuel_kgCO2e_per_t
Min.   :    0.000                     Min.   :0
1st Qu.:    0.000                     1st Qu.:0
Median :    8.072                     Median :0
Mean   :   68.784                     Mean   :0
3rd Qu.:   68.841                     3rd Qu.:0
Max.   :20238.709                     Max.   :0
```

```
feed_farm_ghg_diesel_tillage_kgCO2e_per_t
Min.   :    0.00
1st Qu.:    0.00
Median :   15.59
Mean   :  149.04
3rd Qu.:   72.47
Max.   :86965.93

feed_farm_ghg_diesel_remain_kgCO2e_per_t feed_farm_ghg_elec_kgCO2e_per_t
Min.   :   0.0000                       Min.   :0.000e+00
1st Qu.:   0.0000                       1st Qu.:0.000e+00
Median :   0.0000                       Median :0.000e+00
Mean   :   4.9242                       Mean   :3.463e+00
3rd Qu.:   0.0355                       3rd Qu.:4.439e-02
Max.   :4153.9314                       Max.   :2.993e+03

feed_farm_total_ghg_crop_kgCO2e_per_t feed_pseudofarm_N2O_d_kgCO2e_per_t
Min.   :     0.0                      Min.   :   180.4
1st Qu.:     0.0                      1st Qu.:   824.9
Median :   109.7                      Median :   971.6
Mean   :   568.6                      Mean   :  1671.8
3rd Qu.:   450.1                      3rd Qu.:  1121.6
Max.   :261508.5                      Max.   :257094.6

feed_pseudofarm_N2O_ATD_kgCO2e_per_t feed_pseudofarm_N2O_L_kgCO2e_per_t
Min.   :   18.24                     Min.   :   57.29
1st Qu.:   86.83                     1st Qu.:  271.11
Median :  104.24                     Median :  318.74
Mean   :  183.36                     Mean   :  555.46
3rd Qu.:  120.41                     3rd Qu.:  364.66
Max.   :30649.91                     Max.   :91292.23

feed_pseudofarm_ghg_ferti_prod_kgCO2e_per_t
Min.   :     0.0
1st Qu.:   548.4
Median :   675.2
Mean   :  1128.4
3rd Qu.:   796.6
Max.   :145247.9

feed_pseudofarm_ghg_heat_fuel_kgCO2e_per_t
Min.   :0
1st Qu.:0
```

```
Median :0
Mean   :0
3rd Qu.:0
Max.   :0

feed_pseudofarm_ghg_diesel_tillage_kgCO2e_per_t
Min.   :   120.4
1st Qu.:   488.3
Median :   603.9
Mean   :  1185.2
3rd Qu.:   727.6
Max.   :188290.5

feed_pseudofarm_ghg_diesel_remain_kgCO2e_per_t
Min.   :   19.41
1st Qu.:   82.98
Median :  104.20
Mean   :  185.29
3rd Qu.:  123.05
Max.   :22196.88

feed_pseudofarm_ghg_elec_kgCO2e_per_t
Min.   :   2.059
1st Qu.:   9.192
Median :  11.416
Mean   :  23.158
3rd Qu.:  13.719
Max.   :4906.176

feed_pseudofarm_total_ghg_crop_kgCO2e_per_t ghg_diesel_remain_kgCO2e_per_t
Min.   :   532.2                            Min.   :    0.00
1st Qu.:  2363.6                            1st Qu.:   17.55
Median :  2791.3                            Median :   33.17
Mean   :  4932.7                            Mean   :  133.30
3rd Qu.:  3243.6                            3rd Qu.:   56.89
Max.   :739678.2                            Max.   :47160.85

ghg_heat_fuel_kgCO2e_per_t ghg_elec_kgCO2e_per_t farm_ghge_herd_kgCO2e_per_t
Min.   :    0.000          Min.   :   0.00       Min.   :    585.2
1st Qu.:    0.000          1st Qu.:  20.18       1st Qu.:   2565.2
Median :    0.000          Median :  28.79       Median :   2879.3
Mean   :   58.434          Mean   :  43.42       Mean   :   5359.7
3rd Qu.:    4.906          3rd Qu.:  37.98       3rd Qu.:   3283.3
```

```
 Max.   :25013.776          Max.   :9354.39       Max.   :1001077.8

 pseudofarm_ghge_herd_kgCO2e_per_t farm_ghge_herd_kgCO2e_per_anim_per_t
 Min.   :   1100                   Min.   :   0.1716
 1st Qu.:   4784                   1st Qu.:   3.1722
 Median :   5416                   Median :   5.5224
 Mean   :   9724                   Mean   :  42.6753
 3rd Qu.:   6136                   3rd Qu.:  10.2988
 Max.   :1479247                   Max.   :7041.1031

 pseudofarm_ghge_herd_kgCO2e_per_anim_per_t
 Min.   :2.567e-01
 1st Qu.:5.841e+00
 Median :1.043e+01
 Mean   :8.036e+01
 3rd Qu.:1.935e+01
 Max.   :1.127e+04
```

```r
# Reshape GHGE meat data to long format, keeping only per-ha (farm) and per-t columns
tmp_ghge_pork_compo_long <- tmp_ghge_pork |>
  dplyr::select(
    dplyr::all_of(tmp_object_pork@traceability$id_cols),
    output, species,
    #dplyr::matches("per_ha_farm$|per_t$"),   # keep farm and per-tonne estimates
    #-dplyr::matches("total_ghg|ghge_herd|^feed_pseudofarm") # drop aggregated/herd-level co
    dplyr::matches("per_ha_pseudofarm|per_t$"),   # keep pseudofarm and per-tonne estimates
    -dplyr::matches("total_ghg|ghge_herd|feed_farm") # drop aggregated/herd-level columns

  ) |>
  tidyr::pivot_longer(
    #cols = dplyr::matches("per_ha_farm|per_t$"),
    cols = dplyr::matches("per_ha_pseudofarm|per_t$"),
    names_to = "component"
  ) |>
  dplyr::mutate(
    # Map component names to standardised GHGE categories
    Harmonized_Category = dplyr::case_when(
      str_detect(component, "(?i)enteric")  ~ "Enteric Fermentation",
      str_detect(component, "(?i)mm|MM")    ~ "Manure Management",
      #      str_detect(component, "(?i)feed")     ~ "Feed Production",
      str_detect(component, "(?i)N2O")        ~ "Field N2O Emissions",
```

```
      str_detect(component, "(?i)ferti_prod") ~ "Fertilizer Production",
      str_detect(component, "(?i)diesel|fuel|elec")~ "Energy Use (Diesel/Elec)",
      # TODO: energy use for livestock buildings is not yet accounted for
      TRUE ~ "Other Inputs & Processes"  # pesticides, seeds, transport, post-harvest
    ),
    product = "meat_pork",
    # Build a composite indicator label encoding both unit type and GHGE category
    indicator = dplyr::case_when(
      #grepl("per_ha_farm$", component) ~ paste("GHGE_kgCO2e_ha", Harmonized_Category, sep =
      grepl("per_ha_pseudofarm$", component) ~ paste("GHGE_kgCO2e_ha", Harmonized_Category, s
      grepl("per_t$",            component) ~ paste("GHGE_kgCO2e_t",  Harmonized_Category, se
    ),
    # Assign unit label based on whether the estimate is per ha or per tonne of product
    unit = dplyr::case_when(
      #grepl("per_ha_farm$", component) ~ "kgCO2e_per_ha",
      grepl("per_ha_pseudofarm$", component) ~ "kgCO2e_per_ha",
      grepl("per_t$",            component) ~ "kgCO2e_per_t"
    )
  )

# Aggregate: sum GHGE components within each farm × category × unit combination
tmp_ghge_pork_compo <- tmp_ghge_pork_compo_long |>
  dplyr::summarise(
    value = sum(value, na.rm = TRUE),
    .by = c(tmp_object_pork@traceability$id_cols, "product", "Harmonized_Category", "indicato
  )

tmp_ghge_pork_compo
```

```
# A tibble: 9,210 x 8
   ID            YEAR  COUNTRY product Harmonized_Category indicator unit  value
   <chr>         <chr> <chr>   <chr>   <chr>               <chr>     <chr> <dbl>
 1 5660000005814 2018  FRA     meat_p~ Enteric Fermentati~ GHGE_kgC~ kgCO~  265.
 2 5660000005814 2018  FRA     meat_p~ Manure Management   GHGE_kgC~ kgCO~ 2272.
 3 5660000005814 2018  FRA     meat_p~ Field N2O Emissions GHGE_kgC~ kgCO~ 1542.
 4 5660000005814 2018  FRA     meat_p~ Fertilizer Product~ GHGE_kgC~ kgCO~  805.
 5 5660000005814 2018  FRA     meat_p~ Energy Use (Diesel~ GHGE_kgC~ kgCO~  940.
 6 5660000005814 2018  FRA     meat_p~ Enteric Fermentati~ GHGE_kgC~ kgCO~  223.
 7 5660000005814 2018  FRA     meat_p~ Manure Management   GHGE_kgC~ kgCO~ 1916.
 8 5660000005814 2018  FRA     meat_p~ Field N2O Emissions GHGE_kgC~ kgCO~ 1301.
 9 5660000005814 2018  FRA     meat_p~ Fertilizer Product~ GHGE_kgC~ kgCO~  679.
10 5660000005814 2018  FRA     meat_p~ Energy Use (Diesel~ GHGE_kgC~ kgCO~  793.
```

```
# i 9,200 more rows
```

GHGE chicken

```
# select meat farms in France
tmp_object_chicken = m_prune_farms(my_object_GHGE, TF14 == "50", COUNTRY == "FRA")

# keep farms with broilers
quantile(tmp_object_chicken@herd |> dplyr::filter(FADN_code_letter == "LPLTRBROYL") |> dplyr
```

```
      0%      25%      50%      75%     100%
  250.00  6358.75 12948.00 28726.00 633997.70
```

```
tmp_object_chicken = m_keep_farms(
  tmp_object_chicken,
  tmp_object_chicken@herd |>
    dplyr::filter(FADN_code_letter == "LPLTRBROYL" & Qobs >0) |>
    dplyr::select(ID,YEAR,COUNTRY)
)

#(tmp_object_chicken@footprints$GHGE$GHGE_meat)

# Retrieve GHGE estimates for meat, filtering to cow meat output only
tmp_ghge_chicken <- tmp_object_chicken@footprints$GHGE$GHGE_meat |>
  dplyr::filter(output == "meat_chicken")

summary(tmp_ghge_chicken)
```

```
      ID                 YEAR             COUNTRY
 Length:343         Length:343         Length:343
 Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character
```

```
 CH4_enteric_kgCO2e_livcat CH4_MM_kgCO2e_livcat N2O_D_MM_kgCO2e_livcat
 Min.   :0                 Min.   :  192.8      Min.   :   622.6
 1st Qu.:0                 1st Qu.: 3410.1      1st Qu.: 10540.5
 Median :0                 Median : 5867.8      Median : 18604.2
```

```
Mean   :0                 Mean   : 8311.8      Mean   : 26563.0
3rd Qu.:0                 3rd Qu.:10291.9      3rd Qu.: 32353.3
Max.   :0                 Max.   :60239.2      Max.   :194492.6

N2O_G_mm_kgCO2e_livcat N2O_L_mm_kgCO2e_livcat feed_farm_area_ha_livcat
Min.   :0              Min.   :0              Min.   : 0.0000
1st Qu.:0              1st Qu.:0              1st Qu.: 0.0000
Median :0              Median :0              Median : 0.0000
Mean   :0              Mean   :0              Mean   : 3.2243
3rd Qu.:0              3rd Qu.:0              3rd Qu.: 0.7075
Max.   :0              Max.   :0              Max.   :43.2392

feed_farm_N2O_d_kgCO2e_livcat feed_farm_N2O_ATD_kgCO2e_livcat
Min.   :    0.0               Min.   :   0.00
1st Qu.:    0.0               1st Qu.:   0.00
Median :    0.0               Median :   0.00
Mean   : 3912.0               Mean   : 538.80
3rd Qu.:  676.9               3rd Qu.:  70.21
Max.   :80376.6               Max.   :8158.13

feed_farm_N2O_L_kgCO2e_livcat feed_farm_ghg_ferti_prod_kgCO2e_livcat
Min.   :    0.0               Min.   :    0.0
1st Qu.:    0.0               1st Qu.:    0.0
Median :    0.0               Median :    0.0
Mean   : 1422.1               Mean   : 2042.3
3rd Qu.:  237.8               3rd Qu.:  173.1
Max.   :26211.5               Max.   :56698.3

feed_farm_ghg_heat_fuel_kgCO2e_livcat
Min.   :0
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0

feed_farm_ghg_diesel_tillage_kgCO2e_livcat
Min.   :    0.0
1st Qu.:    0.0
Median :    0.0
Mean   : 1704.3
3rd Qu.:  260.8
Max.   :64921.5
```

```
feed_farm_ghg_diesel_remain_kgCO2e_livcat feed_farm_ghg_elec_kgCO2e_livcat
Min.   :   0.00                          Min.   :   0.00
1st Qu.:   0.00                          1st Qu.:   0.00
Median :   0.00                          Median :   0.00
Mean   :  53.58                          Mean   :  16.28
3rd Qu.:   0.00                          3rd Qu.:   0.00
Max.   :2682.73                          Max.   :1614.05

feed_farm_total_ghg_crop_kgCO2e_livcat feed_pseudofarm_area_ha_livcat
Min.   :     0                         Min.   :   3.647
1st Qu.:     0                         1st Qu.:  63.878
Median :     0                         Median : 108.610
Mean   :  9689                         Mean   : 155.826
3rd Qu.:  1859                         3rd Qu.: 189.920
Max.   :205420                         Max.   :1139.417

feed_pseudofarm_N2O_d_kgCO2e_livcat feed_pseudofarm_N2O_ATD_kgCO2e_livcat
Min.   :  2738                      Min.   :  236.9
1st Qu.: 44194                      1st Qu.: 4632.4
Median : 81363                      Median : 8480.0
Mean   :118048                      Mean   :11627.7
3rd Qu.:145431                      3rd Qu.:14278.9
Max.   :953637                      Max.   :85852.5

feed_pseudofarm_N2O_L_kgCO2e_livcat
Min.   :   901
1st Qu.: 15520
Median : 28069
Mean   : 39360
3rd Qu.: 48570
Max.   :304416

feed_pseudofarm_ghg_ferti_prod_kgCO2e_livcat
Min.   :     0
1st Qu.: 27180
Median : 54733
Mean   : 79225
3rd Qu.:102049
Max.   :705625

feed_pseudofarm_ghg_heat_fuel_kgCO2e_livcat
Min.   :0
```

```
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0

feed_pseudofarm_ghg_diesel_tillage_kgCO2e_livcat
Min.   :   3249
1st Qu.:  54499
Median : 101857
Mean   : 143108
3rd Qu.: 180425
Max.   :1000069

feed_pseudofarm_ghg_diesel_remain_kgCO2e_livcat
Min.   :   592.1
1st Qu.:  9340.7
Median : 17023.3
Mean   : 24420.0
3rd Qu.: 31042.1
Max.   :176984.8

feed_pseudofarm_ghg_elec_kgCO2e_livcat
Min.   :   68.89
1st Qu.: 1048.92
Median : 1881.93
Mean   : 2750.53
3rd Qu.: 3442.71
Max.   :19672.63

feed_pseudofarm_total_ghg_crop_kgCO2e_livcat ghg_diesel_remain_kgCO2e_livcat
Min.   :   9711                              Min.   :    0
1st Qu.: 164501                              1st Qu.: 1723
Median : 288534                              Median : 5034
Mean   : 418539                              Mean   : 7156
3rd Qu.: 505832                              3rd Qu.:10289
Max.   :3246256                              Max.   :67461

ghg_heat_fuel_kgCO2e_livcat ghg_elec_kgCO2e_livcat
Min.   :     0              Min.   :    0.0
1st Qu.:  8804              1st Qu.:  502.4
Median : 23096              Median : 1167.2
Mean   : 36701              Mean   : 2112.5
```

```
3rd Qu.: 47422              3rd Qu.: 2750.6
Max.   :406774              Max.   :21466.3

farm_ghge_herd_kgCO2e_livcat pseudofarm_ghge_herd_kgCO2e_livcat
Min.   :  3502               Min.   :  14208
1st Qu.: 36214               1st Qu.: 196466
Median : 65935               Median : 348603
Mean   : 90534               Mean   : 499384
3rd Qu.:114690               3rd Qu.: 620361
Max.   :701788               Max.   :3948044

farm_ghge_herd_kgCO2e_per_anim pseudofarm_ghge_herd_kgCO2e_per_anim
Min.   : 1.186                  Min.   : 10.38
1st Qu.: 4.771                  1st Qu.: 27.02
Median : 7.439                  Median : 39.64
Mean   :11.149                  Mean   : 53.67
3rd Qu.:12.954                  3rd Qu.: 69.39
Max.   :75.917                  Max.   :226.60

   output            species          sales_e_output sum_SV_species
Length:343         Length:343         Min.   : NA    Min.   : NA
Class :character   Class :character   1st Qu.: NA    1st Qu.: NA
Mode  :character   Mode  :character   Median : NA    Median : NA
                                      Mean   :NaN    Mean   :NaN
                                      3rd Qu.: NA    3rd Qu.: NA
                                      Max.   : NA    Max.   : NA
                                      NA's   :343    NA's   :343
  econ_ratio      prod_t         CH4_enteric_kgCO2e_per_ha_farm
Min.   : NA   Min.   :   3.816   Min.   :0
1st Qu.: NA   1st Qu.:  51.521   1st Qu.:0
Median : NA   Median : 104.562   Median :0
Mean   :NaN   Mean   : 203.468   Mean   :0
3rd Qu.: NA   3rd Qu.: 197.747   3rd Qu.:0
Max.   : NA   Max.   :2509.261   Max.   :0
NA's   :343                      NA's   :240
CH4_MM_kgCO2e_per_ha_farm N2O_D_MM_kgCO2e_per_ha_farm
Min.   :   73.11          Min.   :  198.7
1st Qu.:11009.62          1st Qu.:35471.2
Median :     Inf          Median :    Inf
Mean   :     Inf          Mean   :    Inf
3rd Qu.:     Inf          3rd Qu.:    Inf
Max.   :     Inf          Max.   :    Inf
```

```
N2O_G_mm_kgCO2e_per_ha_farm N2O_L_mm_kgCO2e_per_ha_farm
Min.   :0                   Min.   :0
1st Qu.:0                   1st Qu.:0
Median :0                   Median :0
Mean   :0                   Mean   :0
3rd Qu.:0                   3rd Qu.:0
Max.   :0                   Max.   :0
NA's   :240                 NA's   :240
feed_farm_N2O_d_kgCO2e_per_ha_farm feed_farm_N2O_ATD_kgCO2e_per_ha_farm
Min.   :  65.13                    Min.   :  7.957
1st Qu.: 846.84                    1st Qu.:105.992
Median :1122.05                    Median :135.138
Mean   :1129.76                    Mean   :155.502
3rd Qu.:1457.84                    3rd Qu.:216.584
Max.   :2161.93                    Max.   :303.809
NA's   :240                        NA's   :240
feed_farm_N2O_L_kgCO2e_per_ha_farm feed_farm_ghg_ferti_prod_kgCO2e_per_ha_farm
Min.   : 30.86                     Min.   :   0.0
1st Qu.:319.76                     1st Qu.: 400.5
Median :401.40                     Median : 615.5
Mean   :409.73                     Mean   : 594.4
3rd Qu.:536.38                     3rd Qu.: 823.5
Max.   :693.01                     Max.   :1573.2
NA's   :240                        NA's   :240
feed_farm_ghg_heat_fuel_kgCO2e_per_ha_farm
Min.   :0
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0
NA's   :240
feed_farm_ghg_diesel_tillage_kgCO2e_per_ha_farm
Min.   :  23.74
1st Qu.: 278.81
Median : 432.44
Mean   : 542.50
3rd Qu.: 668.63
Max.   :3202.95
NA's   :240
feed_farm_ghg_diesel_remain_kgCO2e_per_ha_farm
Min.   :  0.000
1st Qu.:  6.704
```

```
Median : 14.574
Mean   : 18.077
3rd Qu.: 26.930
Max.   :145.305
NA's   :240
feed_farm_ghg_elec_kgCO2e_per_ha_farm
Min.   : 0.0000
1st Qu.: 0.4318
Median : 2.8935
Mean   : 3.9270
3rd Qu.: 4.7193
Max.   :38.6595
NA's   :240
feed_farm_total_ghg_crop_kgCO2e_per_ha_farm
Min.   : 465.2
1st Qu.:2098.1
Median :2847.0
Mean   :2853.9
3rd Qu.:3479.7
Max.   :5255.1
NA's   :240
ghg_diesel_remain_kgCO2e_per_ha_farm ghg_heat_fuel_kgCO2e_per_ha_farm
Min.   :  17.52                      Min.   :    0
1st Qu.:4416.33                      1st Qu.:36305
Median :    Inf                      Median :  Inf
Mean   :    Inf                      Mean   :  Inf
3rd Qu.:    Inf                      3rd Qu.:  Inf
Max.   :    Inf                      Max.   :  Inf
NA's   :25                           NA's   :10
ghg_elec_kgCO2e_per_ha_farm farm_ghge_herd_kgCO2e_per_ha_farm
Min.   :   0                Min.   :  2221
1st Qu.:1659                1st Qu.:103955
Median : Inf                Median :   Inf
Mean   : Inf                Mean   :   Inf
3rd Qu.: Inf                3rd Qu.:   Inf
Max.   : Inf                Max.   :   Inf
NA's   :12
farm_ghge_herd_kgCO2e_per_anim_per_ha_farm
Min.   : 0.1666
1st Qu.:10.2880
Median :    Inf
Mean   :    Inf
3rd Qu.:    Inf
```

```
Max.   :    Inf

CH4_enteric_kgCO2e_per_ha_pseudofarm CH4_MM_kgCO2e_per_ha_pseudofarm
Min.   :0                            Min.   :41.92
1st Qu.:0                            1st Qu.:52.87
Median :0                            Median :52.87
Mean   :0                            Mean   :53.47
3rd Qu.:0                            3rd Qu.:52.87
Max.   :0                            Max.   :78.42

N2O_D_MM_kgCO2e_per_ha_pseudofarm N2O_G_mm_kgCO2e_per_ha_pseudofarm
Min.   :113.9                     Min.   :0
1st Qu.:170.7                     1st Qu.:0
Median :170.7                     Median :0
Mean   :170.0                     Mean   :0
3rd Qu.:170.7                     3rd Qu.:0
Max.   :194.6                     Max.   :0

N2O_L_mm_kgCO2e_per_ha_pseudofarm
Min.   :0
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0

feed_pseudofarm_N2O_d_kgCO2e_per_ha_pseudofarm
Min.   : 459.4
1st Qu.: 679.2
Median : 731.1
Mean   : 747.4
3rd Qu.: 826.3
Max.   :1151.8

feed_pseudofarm_N2O_ATD_kgCO2e_per_ha_pseudofarm
Min.   : 55.44
1st Qu.: 65.38
Median : 74.68
Mean   : 76.78
3rd Qu.: 77.04
Max.   :136.37

feed_pseudofarm_N2O_L_kgCO2e_per_ha_pseudofarm
```

```
Min.   :212.5
1st Qu.:237.6
Median :243.7
Mean   :252.9
3rd Qu.:267.2
Max.   :380.5

feed_pseudofarm_ghg_ferti_prod_kgCO2e_per_ha_pseudofarm
Min.   :  0.0
1st Qu.:471.3
Median :501.2
Mean   :483.6
3rd Qu.:601.2
Max.   :819.9

feed_pseudofarm_ghg_heat_fuel_kgCO2e_per_ha_pseudofarm
Min.   :0
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0

feed_pseudofarm_ghg_diesel_tillage_kgCO2e_per_ha_pseudofarm
Min.   : 477.7
1st Qu.: 803.0
Median : 871.8
Mean   : 940.8
3rd Qu.: 962.5
Max.   :1900.8

feed_pseudofarm_ghg_diesel_remain_kgCO2e_per_ha_pseudofarm
Min.   : 75.57
1st Qu.:151.33
Median :153.04
Mean   :156.53
3rd Qu.:162.33
Max.   :220.78

feed_pseudofarm_ghg_elec_kgCO2e_per_ha_pseudofarm
Min.   : 8.40
1st Qu.:16.73
Median :17.09
```

```
Mean   :17.43
3rd Qu.:19.62
Max.   :23.43

feed_pseudofarm_total_ghg_crop_kgCO2e_per_ha_pseudofarm
Min.   :2107
1st Qu.:2497
Median :2726
Mean   :2675
3rd Qu.:2906
Max.   :3384

ghg_diesel_remain_kgCO2e_per_ha_pseudofarm
Min.   :   0.00
1st Qu.:  12.75
Median :  40.88
Mean   :  87.49
3rd Qu.:  83.51
Max.   :1065.86

ghg_heat_fuel_kgCO2e_per_ha_pseudofarm ghg_elec_kgCO2e_per_ha_pseudofarm
Min.   :   0.0                         Min.   :  0.000
1st Qu.: 117.5                         1st Qu.:  5.619
Median : 203.2                         Median : 10.722
Mean   : 246.3                         Mean   : 15.260
3rd Qu.: 296.9                         3rd Qu.: 19.117
Max.   :5076.0                         Max.   :105.173

pseudofarm_ghge_herd_kgCO2e_per_ha_pseudofarm
Min.   :2350
1st Qu.:2982
Median :3210
Mean   :3248
3rd Qu.:3424
Max.   :8034

pseudofarm_ghge_herd_kgCO2e_per_anim_per_ha_pseudofarm
Min.   :0.03027
1st Qu.:0.22497
Median :0.37012
Mean   :0.66867
3rd Qu.:0.75292
Max.   :8.03404
```

```
CH4_enteric_kgCO2e_per_t CH4_MM_kgCO2e_per_t N2O_D_MM_kgCO2e_per_t
Min.   :0                Min.   :   7.936    Min.   :  25.62
1st Qu.:0                1st Qu.:  37.567    1st Qu.: 121.29
Median :0                Median :  52.972    Median : 167.79
Mean   :0                Mean   :  64.631    Mean   : 204.10
3rd Qu.:0                3rd Qu.:  71.615    3rd Qu.: 225.91
Max.   :0                Max.   :1351.768    Max.   :3719.54

N2O_G_mm_kgCO2e_per_t N2O_L_mm_kgCO2e_per_t feed_farm_N2O_d_kgCO2e_per_t
Min.   :0             Min.   :0             Min.   :   0.000
1st Qu.:0             1st Qu.:0             1st Qu.:   0.000
Median :0             Median :0             Median :   0.000
Mean   :0             Mean   :0             Mean   :  70.307
3rd Qu.:0             3rd Qu.:0             3rd Qu.:   4.542
Max.   :0             Max.   :0             Max.   :8894.163

feed_farm_N2O_ATD_kgCO2e_per_t feed_farm_N2O_L_kgCO2e_per_t
Min.   :   0.0000              Min.   :   0.000
1st Qu.:   0.0000              1st Qu.:   0.000
Median :   0.0000              Median :   0.000
Mean   :  10.1997              Mean   :  26.012
3rd Qu.:   0.4406              3rd Qu.:   1.577
Max.   :1448.8646              Max.   :3466.943

feed_farm_ghg_ferti_prod_kgCO2e_per_t feed_farm_ghg_heat_fuel_kgCO2e_per_t
Min.   :   0.0000                     Min.   :0
1st Qu.:   0.0000                     1st Qu.:0
Median :   0.0000                     Median :0
Mean   :  34.4666                     Mean   :0
3rd Qu.:   0.9822                     3rd Qu.:0
Max.   :3498.1986                     Max.   :0

feed_farm_ghg_diesel_tillage_kgCO2e_per_t
Min.   :   0.000
1st Qu.:   0.000
Median :   0.000
Mean   :  45.711
3rd Qu.:   1.865
Max.   :8884.799

feed_farm_ghg_diesel_remain_kgCO2e_per_t feed_farm_ghg_elec_kgCO2e_per_t
Min.   : 0.0000                          Min.   : 0.0000
```

```
1st Qu.: 0.0000                          1st Qu.: 0.0000
Median : 0.0000                          Median : 0.0000
Mean   : 0.7585                          Mean   : 0.2486
3rd Qu.: 0.0000                          3rd Qu.: 0.0000
Max.   :67.4183                          Max.   :25.1561

feed_farm_total_ghg_crop_kgCO2e_per_t feed_pseudofarm_N2O_d_kgCO2e_per_t
Min.   :    0.00                      Min.   :  101.9
1st Qu.:    0.00                      1st Qu.:  544.1
Median :    0.00                      Median :  767.6
Mean   :  187.70                      Mean   :  874.8
3rd Qu.:   11.75                      3rd Qu.:  956.4
Max.   :26285.54                      Max.   :19065.3

feed_pseudofarm_N2O_ATD_kgCO2e_per_t feed_pseudofarm_N2O_L_kgCO2e_per_t
Min.   :   8.82                     Min.   :  33.54
1st Qu.:  48.58                     1st Qu.: 175.60
Median :  71.18                     Median : 255.87
Mean   :  98.36                     Mean   : 305.57
3rd Qu.:  99.41                     3rd Qu.: 338.25
Max.   :2328.86                     Max.   :6813.48

feed_pseudofarm_ghg_ferti_prod_kgCO2e_per_t
Min.   :    0.0
1st Qu.:  294.1
Median :  474.6
Mean   :  519.2
3rd Qu.:  622.0
Max.   :10555.5

feed_pseudofarm_ghg_heat_fuel_kgCO2e_per_t
Min.   :0
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0

feed_pseudofarm_ghg_diesel_tillage_kgCO2e_per_t
Min.   :  120.9
1st Qu.:  595.0
Median :  840.1
Mean   : 1208.7
```

```
3rd Qu.: 1151.8
Max.   :20951.9

feed_pseudofarm_ghg_diesel_remain_kgCO2e_per_t
Min.   :  22.97
1st Qu.: 105.08
Median : 148.33
Mean   : 189.99
3rd Qu.: 203.82
Max.   :2359.34

feed_pseudofarm_ghg_elec_kgCO2e_per_t
Min.   :  2.565
1st Qu.: 12.234
Median : 16.857
Mean   : 20.825
3rd Qu.: 23.322
Max.   :281.036

feed_pseudofarm_total_ghg_crop_kgCO2e_per_t ghg_diesel_remain_kgCO2e_per_t
Min.   :  361.5                             Min.   :   0.00
1st Qu.: 1884.5                             1st Qu.:  14.13
Median : 2688.4                             Median :  39.78
Mean   : 3217.4                             Mean   : 113.56
3rd Qu.: 3594.6                             3rd Qu.:  86.15
Max.   :62355.4                             Max.   :5181.42

ghg_heat_fuel_kgCO2e_per_t ghg_elec_kgCO2e_per_t farm_ghge_herd_kgCO2e_per_t
Min.   :   0.0             Min.   :  0.000       Min.   :  119.7
1st Qu.: 108.1             1st Qu.:  5.652       1st Qu.:  369.2
Median : 187.4             Median :  9.532       Median :  568.5
Mean   : 261.2             Mean   : 17.141       Mean   :  848.4
3rd Qu.: 321.0             3rd Qu.: 16.342       3rd Qu.:  836.9
Max.   :3660.9             Max.   :398.956       Max.   :36485.9

pseudofarm_ghge_herd_kgCO2e_per_t farm_ghge_herd_kgCO2e_per_anim_per_t
Min.   :  529                     Min.   :0.002919
1st Qu.: 2267                     1st Qu.:0.030170
Median : 3213                     Median :0.066702
Mean   : 3878                     Mean   :0.218511
3rd Qu.: 4287                     3rd Qu.:0.165996
Max.   :72556                     Max.   :6.189554
```

```
 pseudofarm_ghge_herd_kgCO2e_per_anim_per_t
 Min.   : 0.0133
 1st Qu.: 0.1778
 Median : 0.3994
 Mean   : 0.8776
 3rd Qu.: 0.8395
 Max.   :15.6692
```

```r
# Reshape GHGE meat data to long format, keeping only per-ha (farm) and per-t columns
tmp_ghge_chicken_compo_long <- tmp_ghge_chicken |>
  dplyr::select(
    dplyr::all_of(tmp_object_chicken@traceability$id_cols),
    output, species,
    #dplyr::matches("per_ha_farm$|per_t$"),   # keep farm and per-tonne estimates
    #-dplyr::matches("total_ghg|ghge_herd|^feed_pseudofarm") # drop aggregated/herd-level co
    dplyr::matches("per_ha_pseudofarm|per_t$"),   # keep pseudofarm and per-tonne estimates
    -dplyr::matches("total_ghg|ghge_herd|feed_farm") # drop aggregated/herd-level columns

  ) |>
  tidyr::pivot_longer(
    #cols = dplyr::matches("per_ha_farm|per_t$"),
    cols = dplyr::matches("per_ha_pseudofarm|per_t$"),
    names_to = "component"
  ) |>
  dplyr::mutate(
    # Map component names to standardised GHGE categories
    Harmonized_Category = dplyr::case_when(
      str_detect(component, "(?i)enteric")  ~ "Enteric Fermentation",
      str_detect(component, "(?i)mm|MM")    ~ "Manure Management",
      #      str_detect(component, "(?i)feed")     ~ "Feed Production",
      str_detect(component, "(?i)N2O")        ~ "Field N2O Emissions",
      str_detect(component, "(?i)ferti_prod") ~ "Fertilizer Production",
      str_detect(component, "(?i)diesel|fuel|elec")~ "Energy Use (Diesel/Elec)",
      # TODO: energy use for livestock buildings is not yet accounted for
      TRUE ~ "Other Inputs & Processes"  # pesticides, seeds, transport, post-harvest
    ),
    product = "meat_chicken",
    # Build a composite indicator label encoding both unit type and GHGE category
    indicator = dplyr::case_when(
      #grepl("per_ha_farm$", component) ~ paste("GHGE_kgCO2e_ha", Harmonized_Category, sep =
      grepl("per_ha_pseudofarm$", component) ~ paste("GHGE_kgCO2e_ha", Harmonized_Category, s
```

```
      grepl("per_t$",            component) ~ paste("GHGE_kgCO2e_t",  Harmonized_Category, se
    ),
    # Assign unit label based on whether the estimate is per ha or per tonne of product
    unit = dplyr::case_when(
      #grepl("per_ha_farm$", component) ~ "kgCO2e_per_ha",
      grepl("per_ha_pseudofarm$", component) ~ "kgCO2e_per_ha",
      grepl("per_t$",            component) ~ "kgCO2e_per_t"
    )
  )

# Aggregate: sum GHGE components within each farm × category × unit combination
tmp_ghge_chicken_compo <- tmp_ghge_chicken_compo_long |>
  dplyr::summarise(
    value = sum(value, na.rm = TRUE),
    .by = c(tmp_object_chicken@traceability$id_cols, "product", "Harmonized_Category", "indi
  )

tmp_ghge_chicken_compo
```

```
# A tibble: 3,430 x 8
   ID           YEAR  COUNTRY product  Harmonized_Category indicator unit  value
   <chr>        <chr> <chr>   <chr>    <chr>               <chr>     <chr> <dbl>
 1 515000000269 2018  FRA     meat_ch~ Enteric Fermentati~ GHGE_kgC~ kgCO~    0
 2 515000000269 2018  FRA     meat_ch~ Manure Management   GHGE_kgC~ kgCO~  224.
 3 515000000269 2018  FRA     meat_ch~ Field N2O Emissions GHGE_kgC~ kgCO~  845.
 4 515000000269 2018  FRA     meat_ch~ Fertilizer Product~ GHGE_kgC~ kgCO~    0
 5 515000000269 2018  FRA     meat_ch~ Energy Use (Diesel~ GHGE_kgC~ kgCO~ 1849.
 6 515000000269 2018  FRA     meat_ch~ Enteric Fermentati~ GHGE_kgC~ kgCO~    0
 7 515000000269 2018  FRA     meat_ch~ Manure Management   GHGE_kgC~ kgCO~  515.
 8 515000000269 2018  FRA     meat_ch~ Field N2O Emissions GHGE_kgC~ kgCO~ 1945.
 9 515000000269 2018  FRA     meat_ch~ Fertilizer Product~ GHGE_kgC~ kgCO~    0
10 515000000269 2018  FRA     meat_ch~ Energy Use (Diesel~ GHGE_kgC~ kgCO~ 4257.
# i 3,420 more rows
```

GHGE eggs

```
# select meat farms in France
tmp_object_eggs = m_prune_farms(my_object_GHGE, TF14 == "50", COUNTRY == "FRA")
```

```
# keep farms that produce eggs
quantile(tmp_object_eggs@output$other_herd_products |> dplyr::filter(output == "eggs") |> dpl
```

```
      0%      25%      50%      75%     100%
  0.7200  61.0848 171.8148 537.8874 3173.4000
```

```
tmp_object_eggs = m_keep_farms(
  tmp_object_eggs,
  tmp_object_eggs@output$other_herd_products |>
    dplyr::filter(output == "eggs" & prod_t >0) |>
    dplyr::select(ID,YEAR,COUNTRY)
)

#(tmp_object_eggs@footprints$GHGE$GHGE_eggs)

# Retrieve GHGE estimates for meat, filtering to cow meat output only
tmp_ghge_eggs <- tmp_object_eggs@footprints$GHGE$GHGE_eggs |>
  dplyr::filter(output == "eggs")

summary(tmp_ghge_eggs)
```

```
      ID                YEAR              COUNTRY
 Length:145         Length:145         Length:145
 Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character
```

```
 CH4_enteric_kgCO2e_livcat CH4_MM_kgCO2e_livcat N2O_D_MM_kgCO2e_livcat
 Min.   :0                 Min.   :   107.2     Min.   :   309.4
 1st Qu.:0                 1st Qu.:  3043.0     1st Qu.:  8512.6
 Median :0                 Median :  7024.0     Median : 19941.7
 Mean   :0                 Mean   : 15047.7     Mean   : 43107.3
 3rd Qu.:0                 3rd Qu.: 17055.7     3rd Qu.: 49238.5
 Max.   :0                 Max.   :127094.8     Max.   :366913.1

 N2O_G_mm_kgCO2e_livcat N2O_L_mm_kgCO2e_livcat feed_farm_area_ha_livcat
 Min.   :0              Min.   :0              Min.   :  0.0000
 1st Qu.:0              1st Qu.:0              1st Qu.:  0.0000
 Median :0              Median :0              Median :  0.0000
```

```
Mean   :0              Mean   :0              Mean   :  8.3871
3rd Qu.:0              3rd Qu.:0              3rd Qu.:  0.9291
Max.   :0              Max.   :0              Max.   :177.6800

feed_farm_N2O_d_kgCO2e_livcat feed_farm_N2O_ATD_kgCO2e_livcat
Min.   :     0.0              Min.   :    0.00
1st Qu.:     0.0              1st Qu.:    0.00
Median :     0.0              Median :    0.00
Mean   :  9232.7              Mean   : 1421.82
3rd Qu.:   585.6              3rd Qu.:   74.56
Max.   :230778.4              Max.   :34484.42

feed_farm_N2O_L_kgCO2e_livcat feed_farm_ghg_ferti_prod_kgCO2e_livcat
Min.   :    0.0               Min.   :     0
1st Qu.:    0.0               1st Qu.:     0
Median :    0.0               Median :     0
Mean   : 3541.0               Mean   :  3919
3rd Qu.:  221.2               3rd Qu.:     0
Max.   :87606.6               Max.   :102417

feed_farm_ghg_heat_fuel_kgCO2e_livcat
Min.   :0
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0

feed_farm_ghg_diesel_tillage_kgCO2e_livcat
Min.   :     0.0
1st Qu.:     0.0
Median :     0.0
Mean   :  7694.5
3rd Qu.:   147.7
Max.   :217109.0

feed_farm_ghg_diesel_remain_kgCO2e_livcat feed_farm_ghg_elec_kgCO2e_livcat
Min.   :  0.000                          Min.   : 0.000
1st Qu.:  0.000                          1st Qu.: 0.000
Median :  0.000                          Median : 0.000
Mean   :  9.403                          Mean   : 3.425
3rd Qu.:  0.000                          3rd Qu.: 0.000
Max.   :329.356                          Max.   :86.855
```

```
feed_farm_total_ghg_crop_kgCO2e_livcat feed_pseudofarm_area_ha_livcat
Min.   :     0                         Min.   :   1.821
1st Qu.:     0                         1st Qu.:  52.170
Median :     0                         Median : 119.298
Mean   : 25822                         Mean   : 254.706
3rd Qu.:  1608                         3rd Qu.: 294.010
Max.   :628241                         Max.   :2158.639

feed_pseudofarm_N2O_d_kgCO2e_livcat feed_pseudofarm_N2O_ATD_kgCO2e_livcat
Min.   :   1328                     Min.   :   119.5
1st Qu.:  31923                     1st Qu.:  5134.7
Median :  91096                     Median : 11109.6
Mean   : 203257                     Mean   : 21399.5
3rd Qu.: 239553                     3rd Qu.: 27564.4
Max.   :1936568                     Max.   :182807.6

feed_pseudofarm_N2O_L_kgCO2e_livcat
Min.   :   433.6
1st Qu.: 12662.0
Median : 33160.3
Mean   : 68347.6
3rd Qu.: 77410.6
Max.   :617372.6

feed_pseudofarm_ghg_ferti_prod_kgCO2e_livcat
Min.   :      0
1st Qu.:      0
Median :  49187
Mean   : 133454
3rd Qu.: 173871
Max.   :1435470

feed_pseudofarm_ghg_heat_fuel_kgCO2e_livcat
Min.   :0
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0

feed_pseudofarm_ghg_diesel_tillage_kgCO2e_livcat
Min.   :   1296
```

```
1st Qu.:  59993
Median : 123591
Mean   : 223192
3rd Qu.: 292885
Max.   :1897744

feed_pseudofarm_ghg_diesel_remain_kgCO2e_livcat
Min.   :   263
1st Qu.: 10039
Median : 18447
Mean   : 38331
3rd Qu.: 47407
Max.   :334986

feed_pseudofarm_ghg_elec_kgCO2e_livcat
Min.   :   28.98
1st Qu.:  973.86
Median : 1946.36
Mean   : 4213.23
3rd Qu.: 4986.99
Max.   :40315.39

feed_pseudofarm_total_ghg_crop_kgCO2e_livcat ghg_diesel_remain_kgCO2e_livcat
Min.   :   4405                              Min.   :     0
1st Qu.: 131820                              1st Qu.:   728
Median : 329366                              Median :  3448
Mean   : 692194                              Mean   : 10079
3rd Qu.: 797276                              3rd Qu.: 10811
Max.   :6445264                              Max.   :336292

ghg_heat_fuel_kgCO2e_livcat ghg_elec_kgCO2e_livcat
Min.   :     0              Min.   :    0.0
1st Qu.:     0              1st Qu.:  590.8
Median :     0              Median : 1758.0
Mean   :  8467              Mean   : 6673.3
3rd Qu.:  2098              3rd Qu.: 8004.9
Max.   :161107              Max.   :61323.4

farm_ghge_herd_kgCO2e_livcat pseudofarm_ghge_herd_kgCO2e_livcat
Min.   :    749.4            Min.   :   5154
1st Qu.:  20689.9            1st Qu.: 162811
Median :  46529.5            Median : 378033
Mean   : 109196.8            Mean   : 775568
```

```
3rd Qu.: 156070.9            3rd Qu.: 874593
Max.   :1080258.7            Max.   :7000595

farm_ghge_herd_kgCO2e_per_anim pseudofarm_ghge_herd_kgCO2e_per_anim
Min.   : 1.078                 Min.   : 10.97
1st Qu.: 2.776                 1st Qu.: 23.59
Median : 4.329                 Median : 29.64
Mean   : 5.559                 Mean   : 38.27
3rd Qu.: 6.817                 3rd Qu.: 48.60
Max.   :25.079                 Max.   :122.20

   output              prod_t        CH4_enteric_kgCO2e_per_ha_farm
Length:145         Min.   :   0.72   Min.   :0
Class :character   1st Qu.:  61.08   1st Qu.:0
Mode  :character   Median : 171.81   Median :0
                   Mean   : 453.39   Mean   :0
                   3rd Qu.: 537.89   3rd Qu.:0
                   Max.   :3173.40   Max.   :0
                                     NA's   :104
CH4_MM_kgCO2e_per_ha_farm N2O_D_MM_kgCO2e_per_ha_farm
Min.   :  90.16           Min.   :  242.2
1st Qu.:6595.00           1st Qu.:18956.9
Median :    Inf           Median :    Inf
Mean   :    Inf           Mean   :    Inf
3rd Qu.:    Inf           3rd Qu.:    Inf
Max.   :    Inf           Max.   :    Inf

N2O_G_mm_kgCO2e_per_ha_farm N2O_L_mm_kgCO2e_per_ha_farm
Min.   :0                   Min.   :0
1st Qu.:0                   1st Qu.:0
Median :0                   Median :0
Mean   :0                   Mean   :0
3rd Qu.:0                   3rd Qu.:0
Max.   :0                   Max.   :0
NA's   :104                 NA's   :104
feed_farm_N2O_d_kgCO2e_per_ha_farm feed_farm_N2O_ATD_kgCO2e_per_ha_farm
Min.   :  74.79                    Min.   :  7.957
1st Qu.: 794.54                    1st Qu.:110.859
Median :1066.65                    Median :150.680
Mean   :1003.35                    Mean   :150.463
3rd Qu.:1298.84                    3rd Qu.:190.792
Max.   :1750.32                    Max.   :263.383
NA's   :104                        NA's   :104
```

```
feed_farm_N2O_L_kgCO2e_per_ha_farm feed_farm_ghg_ferti_prod_kgCO2e_per_ha_farm
Min.   : 35.48                     Min.   :   0.0
1st Qu.:321.43                     1st Qu.: 175.0
Median :402.12                     Median : 456.0
Mean   :377.62                     Mean   : 460.7
3rd Qu.:476.32                     3rd Qu.: 665.2
Max.   :559.92                     Max.   :1287.7
NA's   :104                        NA's   :104
feed_farm_ghg_heat_fuel_kgCO2e_per_ha_farm
Min.   :0
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0
NA's   :104
feed_farm_ghg_diesel_tillage_kgCO2e_per_ha_farm
Min.   :  23.74
1st Qu.: 289.01
Median : 775.07
Mean   : 958.74
3rd Qu.:1186.64
Max.   :6957.66
NA's   :104
feed_farm_ghg_diesel_remain_kgCO2e_per_ha_farm
Min.   : 0.0000
1st Qu.: 0.0000
Median : 0.1992
Mean   : 6.0299
3rd Qu.: 4.9797
Max.   :44.8024
NA's   :104
feed_farm_ghg_elec_kgCO2e_per_ha_farm
Min.   : 0.0000
1st Qu.: 0.0000
Median : 0.1003
Mean   : 2.1549
3rd Qu.: 3.8694
Max.   :10.8425
NA's   :104
feed_farm_total_ghg_crop_kgCO2e_per_ha_farm
Min.   :1168
1st Qu.:2284
```

```
Median :2964
Mean   :2959
3rd Qu.:3391
Max.   :7550
NA's   :104
ghg_diesel_remain_kgCO2e_per_ha_farm ghg_heat_fuel_kgCO2e_per_ha_farm
Min.   : 136.6                       Min.   :    0
1st Qu.:2161.1                       1st Qu.:    0
Median :   Inf                       Median :30398
Mean   :   Inf                       Mean   :  Inf
3rd Qu.:   Inf                       3rd Qu.:  Inf
Max.   :   Inf                       Max.   :  Inf
NA's   :15                           NA's   :67
ghg_elec_kgCO2e_per_ha_farm farm_ghge_herd_kgCO2e_per_ha_farm
Min.   :  51.49             Min.   : 3129
1st Qu.:2125.43             1st Qu.:66118
Median :    Inf             Median :  Inf
Mean   :    Inf             Mean   :  Inf
3rd Qu.:    Inf             3rd Qu.:  Inf
Max.   :    Inf             Max.   :  Inf
NA's   :2
farm_ghge_herd_kgCO2e_per_anim_per_ha_farm
Min.   :0.04163
1st Qu.:4.66424
Median :    Inf
Mean   :    Inf
3rd Qu.:    Inf
Max.   :    Inf

CH4_enteric_kgCO2e_per_ha_pseudofarm CH4_MM_kgCO2e_per_ha_pseudofarm
Min.   :0                            Min.   :49.77
1st Qu.:0                            1st Qu.:58.88
Median :0                            Median :58.88
Mean   :0                            Mean   :58.77
3rd Qu.:0                            3rd Qu.:58.88
Max.   :0                            Max.   :63.41

N2O_D_MM_kgCO2e_per_ha_pseudofarm N2O_G_mm_kgCO2e_per_ha_pseudofarm
Min.   :133.7                     Min.   :0
1st Qu.:170.0                     1st Qu.:0
Median :170.0                     Median :0
Mean   :168.5                     Mean   :0
3rd Qu.:170.0                     3rd Qu.:0
```

```
Max.   :172.2                      Max.   :0

N2O_L_mm_kgCO2e_per_ha_pseudofarm
Min.   :0
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0

feed_pseudofarm_N2O_d_kgCO2e_per_ha_pseudofarm
Min.   : 498.4
1st Qu.: 528.9
Median : 795.7
Mean   : 739.8
3rd Qu.: 878.3
Max.   :1205.7

feed_pseudofarm_N2O_ATD_kgCO2e_per_ha_pseudofarm
Min.   : 65.66
1st Qu.: 75.39
Median : 84.69
Mean   : 91.33
3rd Qu.:111.93
Max.   :141.78

feed_pseudofarm_N2O_L_kgCO2e_per_ha_pseudofarm
Min.   :234.7
1st Qu.:243.1
Median :259.4
Mean   :264.6
3rd Qu.:283.3
Max.   :395.7

feed_pseudofarm_ghg_ferti_prod_kgCO2e_per_ha_pseudofarm
Min.   :  0.0
1st Qu.:  0.0
Median :515.2
Mean   :408.4
3rd Qu.:640.6
Max.   :835.3

feed_pseudofarm_ghg_heat_fuel_kgCO2e_per_ha_pseudofarm
```

```
Min.   :0
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0

feed_pseudofarm_ghg_diesel_tillage_kgCO2e_per_ha_pseudofarm
Min.   : 489.1
1st Qu.: 715.2
Median : 796.8
Mean   :1031.3
3rd Qu.:1464.1
Max.   :2012.2

feed_pseudofarm_ghg_diesel_remain_kgCO2e_per_ha_pseudofarm
Min.   : 65.63
1st Qu.:143.56
Median :146.53
Mean   :162.61
3rd Qu.:194.39
Max.   :230.45

feed_pseudofarm_ghg_elec_kgCO2e_per_ha_pseudofarm
Min.   : 7.217
1st Qu.:15.799
Median :16.113
Mean   :17.098
3rd Qu.:18.676
Max.   :21.229

feed_pseudofarm_total_ghg_crop_kgCO2e_per_ha_pseudofarm
Min.   :2368
1st Qu.:2566
Median :2677
Mean   :2715
3rd Qu.:2972
Max.   :3427

ghg_diesel_remain_kgCO2e_per_ha_pseudofarm
Min.   :  0.000
1st Qu.:  6.489
Median : 33.203
```

```
Mean   : 75.512
3rd Qu.: 78.978
Max.   :802.745

ghg_heat_fuel_kgCO2e_per_ha_pseudofarm ghg_elec_kgCO2e_per_ha_pseudofarm
Min.   :  0.00                         Min.   : 0.000
1st Qu.:  0.00                         1st Qu.: 7.741
Median :  0.00                         Median :17.639
Mean   : 53.21                         Mean   :22.979
3rd Qu.: 48.86                         3rd Qu.:30.436
Max.   :712.23                         Max.   :93.581

pseudofarm_ghge_herd_kgCO2e_per_ha_pseudofarm
Min.   :2639
1st Qu.:2839
Median :3078
Mean   :3094
3rd Qu.:3275
Max.   :4192

pseudofarm_ghge_herd_kgCO2e_per_anim_per_ha_pseudofarm
Min.   : 0.009759
1st Qu.: 0.081808
Median : 0.258164
Mean   : 1.192190
3rd Qu.: 0.726591
Max.   :15.807180

CH4_enteric_kgCO2e_per_t CH4_MM_kgCO2e_per_t N2O_D_MM_kgCO2e_per_t
Min.   :0                Min.   : 13.18      Min.   :  38.04
1st Qu.:0                1st Qu.: 28.70      1st Qu.:  81.46
Median :0                Median : 41.47      Median : 121.35
Mean   :0                Mean   : 55.41      Mean   : 159.07
3rd Qu.:0                3rd Qu.: 67.19      3rd Qu.: 193.99
Max.   :0                Max.   :479.39      Max.   :1383.96

N2O_G_mm_kgCO2e_per_t N2O_L_mm_kgCO2e_per_t feed_farm_N2O_d_kgCO2e_per_t
Min.   :0             Min.   :0             Min.   :   0.000
1st Qu.:0             1st Qu.:0             1st Qu.:   0.000
Median :0             Median :0             Median :   0.000
Mean   :0             Mean   :0             Mean   :  33.115
3rd Qu.:0             3rd Qu.:0             3rd Qu.:   2.242
Max.   :0             Max.   :0             Max.   :1347.098
```

```
feed_farm_N2O_ATD_kgCO2e_per_t feed_farm_N2O_L_kgCO2e_per_t
Min.   :  0.0000               Min.   :  0.0000
1st Qu.:  0.0000               1st Qu.:  0.0000
Median :  0.0000               Median :  0.0000
Mean   :  4.8516               Mean   : 12.1804
3rd Qu.:  0.2734               3rd Qu.:  0.7916
Max.   :173.8279               Max.   :447.6282

feed_farm_ghg_ferti_prod_kgCO2e_per_t feed_farm_ghg_heat_fuel_kgCO2e_per_t
Min.   :  0.00                        Min.   :0
1st Qu.:  0.00                        1st Qu.:0
Median :  0.00                        Median :0
Mean   : 16.57                        Mean   :0
3rd Qu.:  0.00                        3rd Qu.:0
Max.   :904.98                        Max.   :0

feed_farm_ghg_diesel_tillage_kgCO2e_per_t
Min.   :  0.0000
1st Qu.:  0.0000
Median :  0.0000
Mean   : 34.2640
3rd Qu.:  0.5222
Max.   :981.2086

feed_farm_ghg_diesel_remain_kgCO2e_per_t feed_farm_ghg_elec_kgCO2e_per_t
Min.   :0.00000                          Min.   :0.00000
1st Qu.:0.00000                          1st Qu.:0.00000
Median :0.00000                          Median :0.00000
Mean   :0.02998                          Mean   :0.01014
3rd Qu.:0.00000                          3rd Qu.:0.00000
Max.   :1.14416                          Max.   :0.45711

feed_farm_total_ghg_crop_kgCO2e_per_t feed_pseudofarm_N2O_d_kgCO2e_per_t
Min.   :   0.000                      Min.   : 163.3
1st Qu.:   0.000                      1st Qu.: 404.3
Median :   0.000                      Median : 509.0
Mean   : 101.017                      Mean   : 644.0
3rd Qu.:   4.883                      3rd Qu.: 682.1
Max.   :3581.818                      Max.   :5937.5

feed_pseudofarm_N2O_ATD_kgCO2e_per_t feed_pseudofarm_N2O_L_kgCO2e_per_t
Min.   : 14.71                       Min.   :  53.34
```

```
1st Qu.: 39.42                      1st Qu.: 133.89
Median : 58.65                      Median : 190.76
Mean   : 90.96                      Mean   : 244.28
3rd Qu.:122.68                      3rd Qu.: 289.46
Max.   :573.74                      Max.   :1939.08

feed_pseudofarm_ghg_ferti_prod_kgCO2e_per_t
Min.   :   0.0
1st Qu.:   0.0
Median : 271.1
Mean   : 287.5
3rd Qu.: 371.3
Max.   :4185.6

feed_pseudofarm_ghg_heat_fuel_kgCO2e_per_t
Min.   :0
1st Qu.:0
Median :0
Mean   :0
3rd Qu.:0
Max.   :0

feed_pseudofarm_ghg_diesel_tillage_kgCO2e_per_t
Min.   : 159.2
1st Qu.: 369.2
Median : 551.2
Mean   :1107.3
3rd Qu.:1443.2
Max.   :9323.5

feed_pseudofarm_ghg_diesel_remain_kgCO2e_per_t
Min.   :  32.27
1st Qu.:  68.97
Median : 105.56
Mean   : 164.95
3rd Qu.: 220.54
Max.   :1181.26

feed_pseudofarm_ghg_elec_kgCO2e_per_t
Min.   :  3.560
1st Qu.:  7.666
Median : 11.636
Mean   : 16.755
```

```
 3rd Qu.: 20.668
 Max.   :129.618

 feed_pseudofarm_total_ghg_crop_kgCO2e_per_t ghg_diesel_remain_kgCO2e_per_t
 Min.   :  541.4                             Min.   :   0.00
 1st Qu.: 1319.0                             1st Qu.:   7.04
 Median : 1938.0                             Median :  18.01
 Mean   : 2555.8                             Mean   :  67.48
 3rd Qu.: 3042.3                             3rd Qu.:  49.27
 Max.   :19700.7                             Max.   :1223.99

 ghg_heat_fuel_kgCO2e_per_t ghg_elec_kgCO2e_per_t farm_ghge_herd_kgCO2e_per_t
 Min.   :   0.00            Min.   : 0.000       Min.   :  98.31
 1st Qu.:   0.00            1st Qu.: 7.068       1st Qu.: 198.37
 Median :   0.00            Median :11.518       Median : 263.76
 Mean   :  53.46            Mean   :15.993       Mean   : 452.44
 3rd Qu.:  31.60            3rd Qu.:21.823       3rd Qu.: 443.38
 Max.   :2234.81            Max.   :78.904       Max.   :4844.46

 pseudofarm_ghge_herd_kgCO2e_per_t farm_ghge_herd_kgCO2e_per_anim_per_t
 Min.   :  748.8                   Min.   :0.0006274
 1st Qu.: 1493.8                   1st Qu.:0.0065465
 Median : 2200.0                   Median :0.0209249
 Mean   : 2907.2                   Mean   :0.3821310
 3rd Qu.: 3333.7                   3rd Qu.:0.0883216
 Max.   :21610.1                   Max.   :8.9853893

 pseudofarm_ghge_herd_kgCO2e_per_anim_per_t
 Min.   : 0.0075
 1st Qu.: 0.0481
 Median : 0.1870
 Mean   : 2.3026
 3rd Qu.: 0.8257
 Max.   :69.9357
```

```r
# Reshape GHGE meat data to long format, keeping only per-ha (farm) and per-t columns
tmp_ghge_eggs_compo_long <- tmp_ghge_eggs |>
  dplyr::select(
    dplyr::all_of(tmp_object_eggs@traceability$id_cols),
    output,
    #dplyr::matches("per_ha_farm$|per_t$"),   # keep farm and per-tonne estimates
```

```
    #-dplyr::matches("total_ghg|ghge_herd|^feed_pseudofarm") # drop aggregated/herd-level col
    dplyr::matches("per_ha_pseudofarm|per_t$"),   # keep pseudofarm and per-tonne estimates
    -dplyr::matches("total_ghg|ghge_herd|feed_farm") # drop aggregated/herd-level columns

  ) |>
  tidyr::pivot_longer(
    #cols = dplyr::matches("per_ha_farm|per_t$"),
    cols = dplyr::matches("per_ha_pseudofarm|per_t$"),
    names_to = "component"
  ) |>
  dplyr::mutate(
    # Map component names to standardised GHGE categories
    Harmonized_Category = dplyr::case_when(
      str_detect(component, "(?i)enteric")  ~ "Enteric Fermentation",
      str_detect(component, "(?i)mm|MM")    ~ "Manure Management",
      #      str_detect(component, "(?i)feed")     ~ "Feed Production",
      str_detect(component, "(?i)N2O")        ~ "Field N2O Emissions",
      str_detect(component, "(?i)ferti_prod") ~ "Fertilizer Production",
      str_detect(component, "(?i)diesel|fuel|elec")~ "Energy Use (Diesel/Elec)",
      # TODO: energy use for livestock buildings is not yet accounted for
      TRUE ~ "Other Inputs & Processes"  # pesticides, seeds, transport, post-harvest
    ),
    product = "eggs",
    # Build a composite indicator label encoding both unit type and GHGE category
    indicator = dplyr::case_when(
      #grepl("per_ha_farm$", component) ~ paste("GHGE_kgCO2e_ha", Harmonized_Category, sep =
      grepl("per_ha_pseudofarm$", component) ~ paste("GHGE_kgCO2e_ha", Harmonized_Category, s
      grepl("per_t$",             component) ~ paste("GHGE_kgCO2e_t",  Harmonized_Category, se
    ),
    # Assign unit label based on whether the estimate is per ha or per tonne of product
    unit = dplyr::case_when(
      #grepl("per_ha_farm$", component) ~ "kgCO2e_per_ha",
      grepl("per_ha_pseudofarm$", component) ~ "kgCO2e_per_ha",
      grepl("per_t$",             component) ~ "kgCO2e_per_t"
    )
  )

# Aggregate: sum GHGE components within each farm × category × unit combination
tmp_ghge_eggs_compo <- tmp_ghge_eggs_compo_long |>
  dplyr::summarise(
    value = sum(value, na.rm = TRUE),
    .by = c(tmp_object_eggs@traceability$id_cols, "product", "Harmonized_Category", "indicato
```

```
  )

tmp_ghge_eggs_compo
```

```
# A tibble: 1,450 x 8
   ID           YEAR  COUNTRY product Harmonized_Category  indicator unit  value
   <chr>        <chr> <chr>   <chr>   <chr>                <chr>     <chr> <dbl>
 1 515000000269 2018  FRA     eggs    Enteric Fermentation GHGE_kgC~ kgCO~     0
 2 515000000269 2018  FRA     eggs    Manure Management    GHGE_kgC~ kgCO~   229.
 3 515000000269 2018  FRA     eggs    Field N2O Emissions  GHGE_kgC~ kgCO~   881.
 4 515000000269 2018  FRA     eggs    Fertilizer Producti~ GHGE_kgC~ kgCO~     0
 5 515000000269 2018  FRA     eggs    Energy Use (Diesel/~ GHGE_kgC~ kgCO~ 1750.
 6 515000000269 2018  FRA     eggs    Enteric Fermentation GHGE_kgC~ kgCO~     0
 7 515000000269 2018  FRA     eggs    Manure Management    GHGE_kgC~ kgCO~   351.
 8 515000000269 2018  FRA     eggs    Field N2O Emissions  GHGE_kgC~ kgCO~ 1352.
 9 515000000269 2018  FRA     eggs    Fertilizer Producti~ GHGE_kgC~ kgCO~     0
10 515000000269 2018  FRA     eggs    Energy Use (Diesel/~ GHGE_kgC~ kgCO~ 2685.
# i 1,440 more rows
```

GHGE farm

```
(my_object_GHGE@footprints$GHGE$GHGE_farm)
```

```
# A tibble: 26,622 x 56
   ID           YEAR  COUNTRY crop_area_ha crop_prod_t crop_N2O_d_kgCO2e
   <chr>        <chr> <chr>          <dbl>       <dbl>              <dbl>
 1 515000000629 2018  FRA            265.        4158.            422461.
 2 515000000488 2018  FRA             79.4        396.             71792.
 3 515000000474 2018  FRA             76.1        470.             23787.
 4 515000000479 2018  FRA             76.3        438.             75189.
 5 515000000868 2018  FRA             54.1        904.             67169.
 6 515000000862 2018  FRA             78.5        437.             20925.
 7 515000000895 2018  FRA            104.        1114.            117829.
 8 515000000826 2018  FRA            306.        4113.            356210.
 9 515000000700 2018  FRA             41          570.             40302.
10 515000000754 2018  FRA             98.6        629.             98897.
# i 26,612 more rows
# i 50 more variables: crop_N2O_ATD_kgCO2e <dbl>, crop_N2O_L_kgCO2e <dbl>,
#   crop_ghg_ferti_prod_kgCO2e <dbl>, crop_ghg_heat_fuel_kgCO2e <dbl>,
```

```
#   crop_ghg_diesel_tillage_kgCO2e <dbl>, crop_ghg_diesel_remain_kgCO2e <dbl>,
#   crop_ghg_elec_kgCO2e <dbl>, crop_total_ghg_crop_kgCO2e <dbl>,
#   herd_Qobs <dbl>, herd_CH4_enteric_kgCO2e <dbl>, herd_CH4_MM_kgCO2e <dbl>,
#   herd_N2O_D_MM_kgCO2e <dbl>, herd_N2O_G_mm_kgCO2e <dbl>, ...
```

```
# Crop farms ----
tmp_ghge_crop_farm <- tmp_object_crop@footprints$GHGE$GHGE_farm |>
  dplyr::select(
    dplyr::all_of(tmp_object_crop@traceability$id_cols),
    dplyr::matches("^pseudofarm_total")   # keep pseudofarm estimates
  ) |>
  tidyr::pivot_longer(
    cols = dplyr::matches("pseudofarm_total"),
    names_to = "component"
  ) |>
  # add total area
  dplyr::left_join(
    tmp_object_crop@farm |>
      dplyr::select(dplyr::all_of(tmp_object_crop@traceability$id_cols),SE025),
    by = tmp_object_crop@traceability$id_cols
  ) |>
  # divide by total area
  dplyr::mutate(
    value_per_ha = value / dplyr::na_if(SE025,0)
  ) |>
  # estimate averages
  dplyr::summarise(
    estimate = mean(value_per_ha, na.rm = T),
    .by = 'component'
  ) |>
  dplyr::mutate(
    indicator = paste0(component,"_per_ha"),
    unit = "kgCO2e_per_ha",
    product = "Field crop farms"
  ) |>
  dplyr::select(
    indicator, unit, product, estimate
  )

# Dairy farms ----
tmp_ghge_dairy_farm <- tmp_object_milk@footprints$GHGE$GHGE_farm |>
  dplyr::select(
```

```
    dplyr::all_of(tmp_object_milk@traceability$id_cols),
    dplyr::matches("^pseudofarm_total")   # keep pseudofarm estimates
  ) |>
  tidyr::pivot_longer(
    cols = dplyr::matches("pseudofarm_total"),
    names_to = "component"
  ) |>
  # add total LU
  dplyr::left_join(
    tmp_object_milk@herd |>
      dplyr::filter(species == "cattle") |>
      dplyr::summarise(
        tot_LU = sum(ALU, na.rm = T),
        .by =  dplyr::all_of(tmp_object_milk@traceability$id_cols)
      ),
    by = tmp_object_milk@traceability$id_cols
  ) |>
  # divide by total LU
  dplyr::mutate(
    value_per_LU = value / dplyr::na_if(tot_LU,0)
  ) |>
  # estimate averages
  dplyr::summarise(
    estimate = mean(value_per_LU, na.rm = T),
    .by = 'component'
  ) |>
  dplyr::mutate(
    indicator = paste0(component,"_per_LU"),
    unit = "kgCO2e_per_LU",
    product = "Dairy farms"
  ) |>
  dplyr::select(
    indicator, unit, product, estimate
  )

# Beef farms ----
tmp_ghge_beef_farm <- tmp_object_beef@footprints$GHGE$GHGE_farm |>
  dplyr::select(
    dplyr::all_of(tmp_object_beef@traceability$id_cols),
    dplyr::matches("^pseudofarm_total")   # keep pseudofarm estimates
  ) |>
  tidyr::pivot_longer(
```

```
    cols = dplyr::matches("pseudofarm_total"),
    names_to = "component"
  ) |>
  # add total LU
  dplyr::left_join(
    tmp_object_beef@herd |>
      dplyr::filter(species == "cattle") |>
      dplyr::summarise(
        tot_LU = sum(ALU, na.rm = T),
        .by =  dplyr::all_of(tmp_object_beef@traceability$id_cols)
      ),
    by = tmp_object_beef@traceability$id_cols
  ) |>

  # divide by total LU
  dplyr::mutate(
    value_per_LU = value / dplyr::na_if(tot_LU,0)
  ) |>
  # estimate averages
  dplyr::summarise(
    estimate = mean(value_per_LU, na.rm = T),
    .by = 'component'
  ) |>
  dplyr::mutate(
    indicator = paste0(component,"_per_LU"),
    unit = "kgCO2e_per_LU",
    product = "Beef farms"
  ) |>
  dplyr::select(
    indicator, unit, product, estimate
  )
```

Combine F2F outputs

```
# Build agricultural practice indicators for dairy cows from herd nutrition & rearing parame
F2F_output_agri_pract <- tibble::tribble(
  ~indicator,          ~product,       ~estimate,
  "DMI",               "dairy_cows",   my_Nexcr_sumup$DM_t_anim_mean[[1]],           # dry ma
  "GE",                "dairy_cows",   my_Nexcr_sumup$GE_DMI_MJ_anim_day_mean[[1]],  # gross
  "Nex",               "dairy_cows",   my_Nexcr_sumup$Nex_DMI_kgN_anim_y_mean[[1]],  # nitrog
```

```
  "share_concent",     "dairy_cows",   my_Nexcr_sumup$share_concent_mean[[1]],        # conce
  "first_calving_age", "dairy_cows",   mean(herd_rear_param$first_calving_age, na.rm = TRUE)
  "renewal_rate",      "dairy_cows",   mean(herd_rear_param$renewal_rate, na.rm = TRUE)
) |>
  # Attach units from the literature reference table
  dplyr::left_join(
    lit_agri_pract |> dplyr::select(indicator, unit),
    by = "indicator"
  ) |>
  # add N ferti
  dplyr::bind_rows(
    my_ferti
  )


# Build total GHGE summary (per ha and per tonne)
F2F_output_ghge0 <- dplyr::bind_rows(
  tmp_ghge_wheat |>
    dplyr::select(ID, YEAR, COUNTRY, total_ghg_crop_kgCO2e_per_ha, total_ghg_crop_kgCO2e_per_
    dplyr::rename(GHGE_kgCO2e_ha = total_ghg_crop_kgCO2e_per_ha,
                  GHGE_kgCO2e_t = total_ghg_crop_kgCO2e_per_t) |>
    dplyr::mutate(product = "wheat"),
  tmp_ghge_maize |>
    dplyr::select(ID, YEAR, COUNTRY, total_ghg_crop_kgCO2e_per_ha, total_ghg_crop_kgCO2e_per_
    dplyr::rename(GHGE_kgCO2e_ha = total_ghg_crop_kgCO2e_per_ha,
                  GHGE_kgCO2e_t = total_ghg_crop_kgCO2e_per_t) |>
    dplyr::mutate(product = "maize_grain"),
  tmp_ghge_rapeseed |>
    dplyr::select(ID, YEAR, COUNTRY, total_ghg_crop_kgCO2e_per_ha, total_ghg_crop_kgCO2e_per_
    dplyr::rename(GHGE_kgCO2e_ha = total_ghg_crop_kgCO2e_per_ha,
                  GHGE_kgCO2e_t = total_ghg_crop_kgCO2e_per_t) |>
    dplyr::mutate(product = "rapeseed"),
  tmp_ghge_milk |>
    dplyr::select(ID, YEAR, COUNTRY, pseudofarm_ghge_herd_kgCO2e_per_ha_pseudofarm, pseudofa
    dplyr::rename(GHGE_kgCO2e_ha = pseudofarm_ghge_herd_kgCO2e_per_ha_pseudofarm,
                  GHGE_kgCO2e_t = pseudofarm_ghge_herd_kgCO2e_per_t) |>
    dplyr::mutate(product = "milk"),
  tmp_ghge_beef |>
    dplyr::select(ID, YEAR, COUNTRY, pseudofarm_ghge_herd_kgCO2e_per_ha_pseudofarm, pseudofa
    dplyr::rename(GHGE_kgCO2e_ha = pseudofarm_ghge_herd_kgCO2e_per_ha_pseudofarm,
                  GHGE_kgCO2e_t = pseudofarm_ghge_herd_kgCO2e_per_t) |>
    dplyr::mutate(product = "cattle_meat"),
```

```
  tmp_ghge_pork |>
    dplyr::select(ID, YEAR, COUNTRY, pseudofarm_ghge_herd_kgCO2e_per_ha_pseudofarm, pseudofa
    dplyr::rename(GHGE_kgCO2e_ha = pseudofarm_ghge_herd_kgCO2e_per_ha_pseudofarm,
                  GHGE_kgCO2e_t = pseudofarm_ghge_herd_kgCO2e_per_t) |>
    dplyr::mutate(product = "meat_pork"),
  tmp_ghge_chicken |>
    dplyr::select(ID, YEAR, COUNTRY, pseudofarm_ghge_herd_kgCO2e_per_ha_pseudofarm, pseudofa
    dplyr::rename(GHGE_kgCO2e_ha = pseudofarm_ghge_herd_kgCO2e_per_ha_pseudofarm,
                  GHGE_kgCO2e_t = pseudofarm_ghge_herd_kgCO2e_per_t) |>
    dplyr::mutate(product = "meat_chicken"),
  tmp_ghge_eggs |>
    dplyr::select(ID, YEAR, COUNTRY, pseudofarm_ghge_herd_kgCO2e_per_ha_pseudofarm, pseudofa
    dplyr::rename(GHGE_kgCO2e_ha = pseudofarm_ghge_herd_kgCO2e_per_ha_pseudofarm,
                  GHGE_kgCO2e_t = pseudofarm_ghge_herd_kgCO2e_per_t) |>
    dplyr::mutate(product = "eggs")
) |>
  tidyr::pivot_longer(
    cols = c(GHGE_kgCO2e_ha, GHGE_kgCO2e_t),
    names_to = "indicator",
    values_to = "estimate"
  ) |>
  dplyr::mutate(
    unit = ifelse(indicator == "GHGE_kgCO2e_ha", 'kgCO2e_per_ha', 'kgCO2e_per_t')
  )

F2F_output_ghge <- F2F_output_ghge0 |>
  dplyr::filter(is.finite(estimate)) |>
  dplyr::summarise(
    estimate = mean(estimate, na.rm = TRUE),
    .by = c(product, indicator, unit)
  )  |>
  dplyr::bind_rows(tmp_ghge_crop_farm) |>
  dplyr::bind_rows(tmp_ghge_dairy_farm) |>
  dplyr::bind_rows(tmp_ghge_beef_farm)

# Compute mean GHGE by harmonized category (component breakdown)
F2F_output_ghge_compo0 <- dplyr::bind_rows(
  tmp_ghge_wheat_compo,
  tmp_ghge_maize_compo,
  tmp_ghge_rapeseed_compo,
  tmp_ghge_milk_compo,
  tmp_ghge_beef_compo,
```

```
  tmp_ghge_pork_compo,
  tmp_ghge_chicken_compo,
  tmp_ghge_eggs_compo
)

## average each GHGE component across farms
F2F_output_ghge_compo <- F2F_output_ghge_compo0 |>
  dplyr::filter(is.finite(value)) |>
  dplyr::summarise(
    estimate = mean(value, na.rm = TRUE),
    .by = c(product, indicator, unit)
  )

# Stack all output tables (practices, total GHGE, GHGE components) into a single flat table
F2F_output <- Reduce(
  function(x, y) dplyr::bind_rows(x, y),
  list(F2F_output_agri_pract,
       F2F_output_ghge,
       F2F_output_ghge_compo)
)

# Render the final output table sorted by product and indicator
F2F_output |> arrange(product, indicator) |> gt()
```

Comparison

Total GHGE

```
library(dplyr)
library(stringr)
library(ggplot2)
library(forcats)

# 1. Prepare data for plotting
tmp_F2F_values0 <- F2F_output_ghge0 %>%
  # Extract the clean category name from the indicator string
  mutate(
    Category = str_extract(indicator, "(?<=__).*"),
    # Make unit names prettier for the plot labels
```

indicator	product	estimate	unit
pseudofarm_total_CH4_kgCO2e_per_LU	Beef farms	3406.70138	kgCO2e_per_LU
pseudofarm_total_CO2_kgCO2e_per_LU	Beef farms	702.52307	kgCO2e_per_LU
pseudofarm_total_N2O_kgCO2e_per_LU	Beef farms	1408.26719	kgCO2e_per_LU
pseudofarm_total_ghge_kgCO2e_per_LU	Beef farms	5517.49165	kgCO2e_per_LU
pseudofarm_total_CH4_kgCO2e_per_LU	Dairy farms	3971.86348	kgCO2e_per_LU
pseudofarm_total_CO2_kgCO2e_per_LU	Dairy farms	1208.54693	kgCO2e_per_LU
pseudofarm_total_N2O_kgCO2e_per_LU	Dairy farms	1826.66296	kgCO2e_per_LU
pseudofarm_total_ghge_kgCO2e_per_LU	Dairy farms	7007.07337	kgCO2e_per_LU
pseudofarm_total_CH4_kgCO2e_per_ha	Field crop farms	164.10567	kgCO2e_per_ha
pseudofarm_total_CO2_kgCO2e_per_ha	Field crop farms	1152.84176	kgCO2e_per_ha
pseudofarm_total_N2O_kgCO2e_per_ha	Field crop farms	1562.02386	kgCO2e_per_ha
pseudofarm_total_ghge_kgCO2e_per_ha	Field crop farms	2878.97129	kgCO2e_per_ha
GHGE_kgCO2e_ha	cattle_meat	5045.55072	kgCO2e_per_ha
GHGE_kgCO2e_ha___Energy Use (Diesel/Elec)	cattle_meat	442.11786	kgCO2e_per_ha
GHGE_kgCO2e_ha___Enteric Fermentation	cattle_meat	4302.21513	kgCO2e_per_ha
GHGE_kgCO2e_ha___Fertilizer Production	cattle_meat	213.47741	kgCO2e_per_ha
GHGE_kgCO2e_ha___Field N2O Emissions	cattle_meat	1135.97236	kgCO2e_per_ha
GHGE_kgCO2e_ha___Manure Management	cattle_meat	1122.59694	kgCO2e_per_ha
GHGE_kgCO2e_t	cattle_meat	22200.89893	kgCO2e_per_t
GHGE_kgCO2e_t___Energy Use (Diesel/Elec)	cattle_meat	1479.52859	kgCO2e_per_t
GHGE_kgCO2e_t___Enteric Fermentation	cattle_meat	16320.73200	kgCO2e_per_t
GHGE_kgCO2e_t___Fertilizer Production	cattle_meat	607.72344	kgCO2e_per_t
GHGE_kgCO2e_t___Field N2O Emissions	cattle_meat	4232.24572	kgCO2e_per_t
GHGE_kgCO2e_t___Manure Management	cattle_meat	3928.87982	kgCO2e_per_t
DMI	dairy_cows	8.09000	kg / head / day
GE	dairy_cows	413.64000	MJ / head / day
Nex	dairy_cows	143.47000	kg N / head / yea
first_calving_age	dairy_cows	30.35420	month
renewal_rate	dairy_cows	22.41787	percent
share_concent	dairy_cows	25.85000	percent
GHGE_kgCO2e_ha	eggs	3094.04610	kgCO2e_per_ha
GHGE_kgCO2e_ha___Energy Use (Diesel/Elec)	eggs	1362.69100	kgCO2e_per_ha
GHGE_kgCO2e_ha___Enteric Fermentation	eggs	0.00000	kgCO2e_per_ha
GHGE_kgCO2e_ha___Fertilizer Production	eggs	408.43226	kgCO2e_per_ha
GHGE_kgCO2e_ha___Field N2O Emissions	eggs	1095.67980	kgCO2e_per_ha
GHGE_kgCO2e_ha___Manure Management	eggs	227.24304	kgCO2e_per_ha
GHGE_kgCO2e_t	eggs	2907.22984	kgCO2e_per_t
GHGE_kgCO2e_t___Energy Use (Diesel/Elec)	eggs	1425.99256	kgCO2e_per_t
GHGE_kgCO2e_t___Enteric Fermentation	eggs	0.00000	kgCO2e_per_t
GHGE_kgCO2e_t___Fertilizer Production	eggs	287.52210	kgCO2e_per_t
GHGE_kgCO2e_t___Field N2O Emissions	eggs	979.23581	kgCO2e_per_t
GHGE_kgCO2e_t___Manure Management	eggs	214.47938	kgCO2e_per_t
GHGE_kgCO2e_ha	maize_grain	2558.65979	kgCO2e_per_ha
GHGE_kgCO2e_ha___Energy Use (Diesel/Elec)	maize_grain	449.95976	kgCO2e_per_ha
GHGE_kgCO2e_ha___Fertilizer Production	maize_grain	698.71586	kgCO2e_per_ha
GHGE_kgCO2e_ha___Field N2O Emissions	maize_grain	1409.98417	kgCO2e_per_ha
GHGE_kgCO2e_t	maize_grain	378.66901	kgCO2e_per_t
GHGE_kgCO2e_t___Energy Use (Diesel/Elec)	maize_grain	69.08248	kgCO2e_per_t
GHGE_kgCO2e_t___Fertilizer Production	maize_grain	103.99187	kgCO2e_per_t
GHGE_kgCO2e_t___Field N2O Emissions	maize_grain	205.59466	kgCO2e_per_t

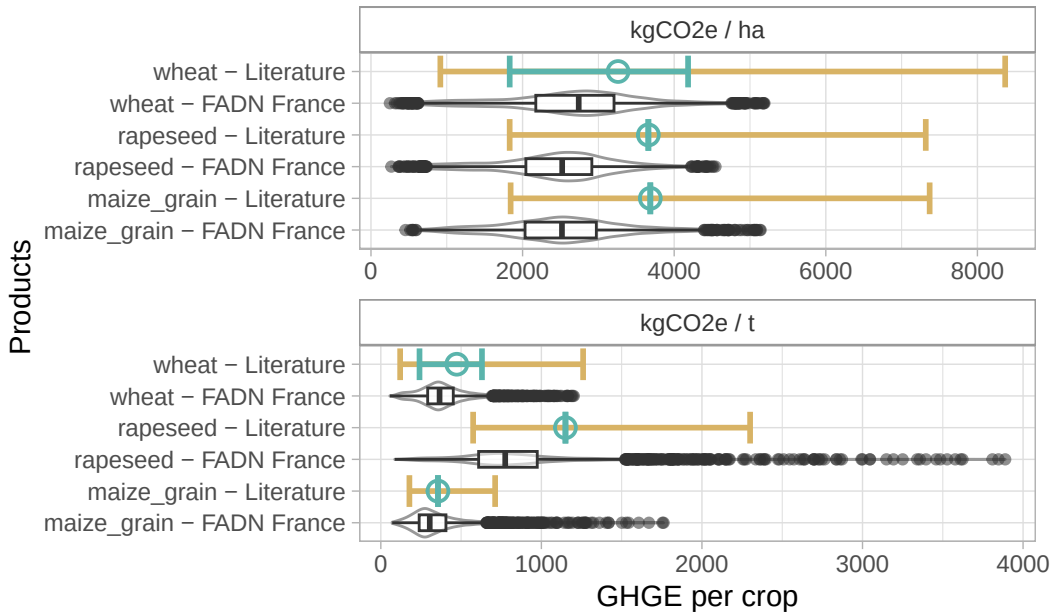
```
    unit_label = gsub("_per_"," / ", unit),
    # Facet by product_type
    product_type = as.factor(ifelse(grepl("wheat|maize|rapeseed", product), "Crops", "Animal
    product_source = paste(product, "FADN France", sep = " - ")
  ) |>
# Compute 99th percentile for visual winsorisation by unit
  mutate(
    p99 = quantile(estimate, 0.99, na.rm = TRUE),
    .by = c(Category, unit_label, product_type, product_source)
  )

tmp_lit_values0 <- lit_table |>
  dplyr::semi_join(
    tmp_F2F_values0 |>
      dplyr::select(product, indicator, unit) |>
      dplyr::distinct(),
    by = c('indicator', 'unit', 'product')
  ) |>
  # Extract the clean category name from the indicator string
  mutate(
    Category = str_extract(indicator, "(?<=__).*"),
    # Make unit names prettier for the plot labels
    unit_label = gsub("_per_"," / ", unit),
    # Facet by product_type
    product_type = as.factor(ifelse(grepl("wheat|maize|rapeseed", product), "Crops", "Animal
     product_source = paste(product, "Literature", sep = " - ")
  )
```

```
# 3. Filter product and Winsorisation
tmp_plot_data <- dplyr::bind_rows(
  tmp_F2F_values0 |>
    dplyr::filter(estimate <= p99),
  tmp_lit_values0
) |>
  dplyr::filter(product_type == "Crops")


# 2. Create the plot
ggplot(tmp_plot_data) +
  # accepted range
  geom_errorbar(aes(x = product_source,
                    ymin = pmin(lit_midpoint, lit_min, na.rm = T) / 2,
```

```
                    ymax = pmax(lit_midpoint, lit_max, na.rm = T) * 2
                ), colour = "#d8b365",
                linewidth=1,width = 1,position = position_dodge(width = 0.75)) +
# literature values
geom_point(aes(x = product_source, y = lit_midpoint), colour = "#5ab4ac",
           shape = 1, size=3, stroke = 1, position = position_dodge(width = 0.75)) +
# min and max
geom_errorbar(aes(x = product_source, ymin = lit_min, ymax = lit_max), colour = "#5ab4ac",
                linewidth=1,width = 1,position = position_dodge(width = 0.75)) +
# estimates
geom_violin(aes(x = product_source, y = estimate, alpha = 0),
            colour = "#999999"
) +
geom_boxplot(aes(x = product_source, y = estimate, alpha = 0),
             colour = "#333333", width = 0.5
) +
# theme
coord_flip() +
facet_wrap(vars(unit_label), scales = "free",ncol = 1) +
labs(y = "GHGE per crop",x= "Products",
     caption = "For visual readability, FADN France estimates are capped at the 99th percen
scale_fill_viridis_d() +
#  scale_fill_manual(values = tmp_FQS_names$colors,labels = tmp_FQS_names$labels,) +
#  scale_colour_manual(values = tmp_FQS_names$colors,labels = tmp_FQS_names$labels) +
theme_light() +
theme(
  legend.position = c(0,-1),legend.direction = "horizontal",legend.title = element_blank()
  strip.background = element_rect(fill = NA, colour = "#999999"),strip.text = element_text
```

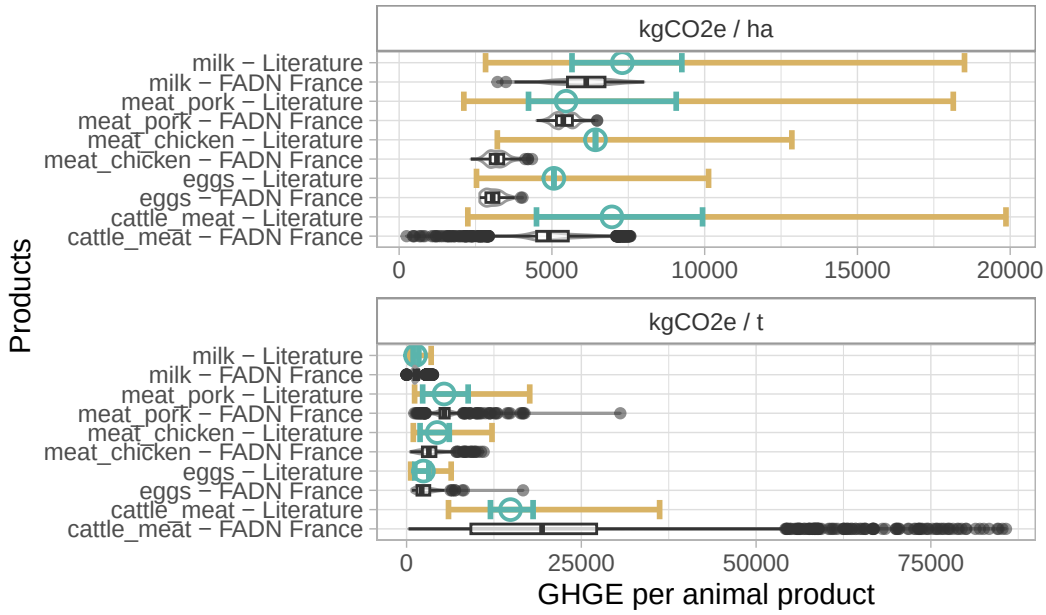


For visual readability, FADN France estimates are capped at the 99th percentile.

```
# 3. Filter product and Winsorisation
tmp_plot_data <- dplyr::bind_rows(
  tmp_F2F_values0 |>
    dplyr::filter(estimate <= p99),
  tmp_lit_values0
) |>
  dplyr::filter(product_type != "Crops")


# 2. Create the plot
ggplot(tmp_plot_data) +
  # accepted range
  geom_errorbar(aes(x = product_source,
                    ymin = pmin(lit_midpoint, lit_min, na.rm = T) / 2,
                    ymax = pmax(lit_midpoint, lit_max, na.rm = T) * 2
                ), colour = "#d8b365",
                linewidth=1,width = 1,position = position_dodge(width = 0.75)) +
  # literature values
  geom_point(aes(x = product_source, y = lit_midpoint), colour = "#5ab4ac",
             shape = 1, size=3, stroke = 1, position = position_dodge(width = 0.75)) +
  # min and max
  geom_errorbar(aes(x = product_source, ymin = lit_min, ymax = lit_max), colour = "#5ab4ac",
                linewidth=1,width = 1,position = position_dodge(width = 0.75)) +
```

```
  # estimates
  geom_violin(aes(x = product_source, y = estimate, alpha = 0),
              colour = "#999999"
  ) +
  geom_boxplot(aes(x = product_source, y = estimate, alpha = 0),
               colour = "#333333", width = 0.5
  ) +
  # theme
  coord_flip() +
  facet_wrap(vars(unit_label), scales = "free",ncol = 1) +
  labs(y = "GHGE per animal product",x= "Products",
       caption = "For visual readability, FADN France estimates are capped at the 99th percer
  scale_fill_viridis_d() +
  #  scale_fill_manual(values = tmp_FQS_names$colors,labels = tmp_FQS_names$labels,) +
  #  scale_colour_manual(values = tmp_FQS_names$colors,labels = tmp_FQS_names$labels) +
  theme_light() +
  theme(
    legend.position = c(0,-1),legend.direction = "horizontal",legend.title = element_blank()
    strip.background = element_rect(fill = NA, colour = "#999999"),strip.text = element_text
```



For visual readability, FADN France estimates are capped at the 99th percentile.

```
tmp_outliers = head(tmp_F2F_values0 |>
                      dplyr::filter(grepl("per_t", unit)) |>
```

```
                    dplyr::arrange(-estimate),
                  n = 15)

tmp_outliers
```

```
# A tibble: 15 x 12
   ID         YEAR  COUNTRY product indicator estimate unit  Category unit_label
   <chr>      <chr> <chr>   <chr>   <chr>        <dbl> <chr> <chr>    <chr>
 1 105000000~ 2018  FRA     meat_p~ GHGE_kgC~ 1479247. kgCO~ <NA>     kgCO2e / t
 2 571000000~ 2020  FRA     cattle~ GHGE_kgC~ 1459985. kgCO~ <NA>     kgCO2e / t
 3 576000000~ 2021  FRA     cattle~ GHGE_kgC~  709421. kgCO~ <NA>     kgCO2e / t
 4 521000000~ 2020  FRA     meat_p~ GHGE_kgC~  681225. kgCO~ <NA>     kgCO2e / t
 5 105000000~ 2020  FRA     cattle~ GHGE_kgC~  587175. kgCO~ <NA>     kgCO2e / t
 6 521000000~ 2021  FRA     meat_p~ GHGE_kgC~  537774. kgCO~ <NA>     kgCO2e / t
 7 594000000~ 2020  FRA     cattle~ GHGE_kgC~  520144. kgCO~ <NA>     kgCO2e / t
 8 571000000~ 2021  FRA     cattle~ GHGE_kgC~  458026. kgCO~ <NA>     kgCO2e / t
 9 105000000~ 2019  FRA     cattle~ GHGE_kgC~  454956. kgCO~ <NA>     kgCO2e / t
10 521000000~ 2021  FRA     cattle~ GHGE_kgC~  434021. kgCO~ <NA>     kgCO2e / t
11 104000000~ 2020  FRA     meat_p~ GHGE_kgC~  427740. kgCO~ <NA>     kgCO2e / t
12 567000000~ 2019  FRA     cattle~ GHGE_kgC~  258788. kgCO~ <NA>     kgCO2e / t
13 104000000~ 2020  FRA     cattle~ GHGE_kgC~  248584. kgCO~ <NA>     kgCO2e / t
14 574000000~ 2019  FRA     cattle~ GHGE_kgC~  223699. kgCO~ <NA>     kgCO2e / t
15 567000000~ 2018  FRA     cattle~ GHGE_kgC~  195560. kgCO~ <NA>     kgCO2e / t
# i 3 more variables: product_type <fct>, product_source <chr>, p99 <dbl>
```

```
tmp_outliers_obj = m_keep_farms(
  my_object_GHGE,
  tmp_outliers
)

# compare feed
dplyr::left_join(
  my_object_GHGE@practices$herding$feed$feed_intake$total |>
    dplyr::summarise(
      DM_t_anim = mean(DM_t_anim, na.rm = TRUE),
      GE_MJ_anim = mean(GE_MJ_anim, na.rm =TRUE),
      .by = FADN_code_letter
    ),
  tmp_outliers_obj@practices$herding$feed$feed_intake$total |>
    dplyr::summarise(
      DM_t_anim_outliers = mean(DM_t_anim, na.rm = TRUE),
```

```
      GE_MJ_anim_outliers = mean(GE_MJ_anim, na.rm =TRUE),
      .by = FADN_code_letter
    )
)
```

```
# A tibble: 22 x 5
   FADN_code_letter DM_t_anim GE_MJ_anim DM_t_anim_outliers GE_MJ_anim_outliers
   <chr>                <dbl>      <dbl>              <dbl>                <dbl>
 1 LPIGFAT             0.943      17746.             1.14                21321.
 2 LPIGLET             0.0985      1861.             0.102                1919.
 3 LPIGOTH             0.967      18192.             1.14                21321.
 4 LSOWBRE             1.60       30204.             1.99                37017.
 5 LHENSLAY            0.0827      1566.             0.0380                704.
 6 LPLTRBROYL          0.0376       719.            NA                     NA
 7 LBOV1               0.704      13130.             0.861               15989.
 8 LBOV1_2F            2.66       49653.             3.39                62896.
 9 LBOV2               4.45       82537.             7.43               137750.
10 LCOWOTH             7.35      136378.            11.0                203740.
# i 12 more rows
```

```
# compare GHGE components

dplyr::left_join(
  my_object_GHGE@footprints$GHGE$GHGE_meat |>
    dplyr::summarise(dplyr::across(
      .cols = dplyr::matches("kgCO2e"),
      .fns = ~sum(.x, na.rm = TRUE)
    ),
    .by = c(output, species)) |>
    tidyr::pivot_longer(
      cols = dplyr::matches("kgCO2e")
    ),

  tmp_outliers_obj@footprints$GHGE$GHGE_meat |>
    dplyr::summarise(dplyr::across(
      .cols = dplyr::matches("kgCO2e"),
      .fns = ~sum(.x, na.rm = TRUE)
    ),
    .by = c(output, species)) |>
    tidyr::pivot_longer(
      cols = dplyr::matches("kgCO2e"),
      values_to = "value_outlier"
```

```
    ),
  by = c('output', 'species', 'name')
) |>
  dplyr::filter(!is.na(value_outlier)) |>
  dplyr::mutate(
    diff_p100 = abs(value_outlier - value) / value
  ) |>
  dplyr::arrange(-diff_p100)
```

```
# A tibble: 490 x 6
   output        species name                      value value_outlier diff_p100
   <chr>         <chr>   <chr>                     <dbl>         <dbl>     <dbl>
 1 meat_cull_cow cattle  ghg_heat_fuel_kgCO2e_li~ 4.22e5          0        1
 2 meat_cull_cow cattle  ghg_heat_fuel_kgCO2e_pe~ 1.01e5          0        1
 3 meat_cull_cow cattle  ghg_heat_fuel_kgCO2e_pe~ 5.82e4          0        1
 4 meat_cull_cow cattle  ghg_heat_fuel_kgCO2e_pe~ 4.77e4          0        1
 5 meat_cull_cow cattle  ghg_diesel_remain_kgCO2~ 7.49e5          9.30     1.000
 6 meat_cull_cow cattle  ghg_elec_kgCO2e_per_ha_~ 1.95e5          2.71     1.000
 7 meat_cull_cow cattle  ghg_diesel_remain_kgCO2~ 4.77e5          7.94     1.000
 8 meat_cull_cow cattle  ghg_elec_kgCO2e_per_ha_~ 1.18e5          2.31     1.000
 9 meat_cull_cow cattle  ghg_diesel_remain_kgCO2~ 4.01e6        173.       1.000
10 meat_chicken  <NA>    ghg_heat_fuel_kgCO2e_li~ 2.19e7       1010.       1.000
# i 480 more rows
```

```
tmp_data = tmp_F2F_values0 |>
                      dplyr::filter(grepl("per_ha", unit),
                                    grepl("cattle_meat", product))

tmp_quantile = quantile(tmp_data$estimate)

tmp_outliers_obj = m_keep_farms(
  my_object_GHGE,
  tmp_data |>
  dplyr::filter(estimate < tmp_quantile['25%'])
)

tmp_OK_farms <- m_keep_farms(
  my_object_GHGE,
  tmp_data |>
  dplyr::filter(estimate > tmp_quantile['75%'])
)
```

Source of GHGE

```
# 0. Summary table
F2F_output_ghge_compo0 |>
  summarise(
    n        = n(),
    mean     = mean(value, na.rm = TRUE),
    sd       = sd(value, na.rm = TRUE),
    min      = min(value, na.rm = TRUE),
    q25      = quantile(value, 0.25, na.rm = TRUE),
    median   = median(value, na.rm = TRUE),
    q75      = quantile(value, 0.75, na.rm = TRUE),
    max      = max(value, na.rm = TRUE),
    n_NA     = sum(is.na(value)),
    .by = c(product, unit, indicator)
  )
```

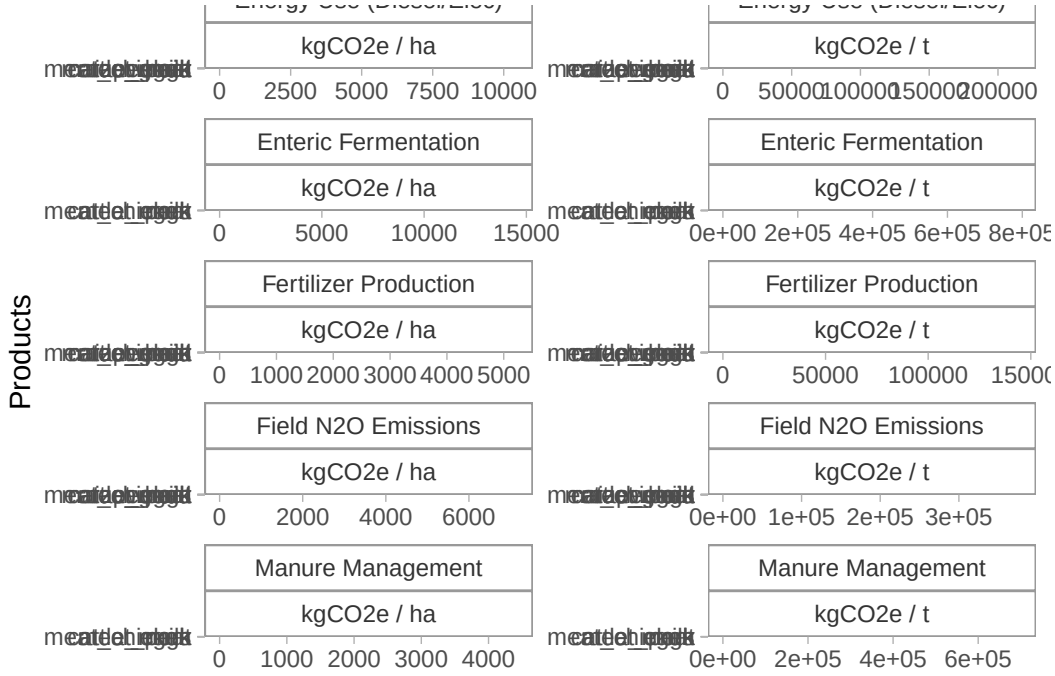
```
# A tibble: 68 x 12
   product  unit  indicator     n   mean    sd   min    q25 median    q75    max
   <chr>    <chr> <chr>     <int>  <dbl> <dbl> <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
 1 wheat    kgCO~ GHGE_kgC~  4484 1466.   501. 157.  1195.  1515.  1765.   5996.
 2 wheat    kgCO~ GHGE_kgC~  4484  215.   167.  53.5  162.   203.   245.   8728.
 3 wheat    kgCO~ GHGE_kgC~  4484  827.   381.   0    635.   864.  1042.   4423.
 4 wheat    kgCO~ GHGE_kgC~  4484  123.   127.   0     82.8  115.   148.   6467.
 5 wheat    kgCO~ GHGE_kgC~  4484  407.   301.   0    266.   367.   485.  10493.
 6 wheat    kgCO~ GHGE_kgC~  4484   65.9  129.   0     35.3   50.1   71.8  5191.
 7 maize_g~ kgCO~ GHGE_kgC~  2405 1410.   441. 279.  1143.  1413.  1663.   4874.
 8 maize_g~ kgCO~ GHGE_kgC~  2405  206.   175.  49.3  136.   170.   224.   4479.
 9 maize_g~ kgCO~ GHGE_kgC~  2405  699.   325.   0    521.   716.   865.   3554.
10 maize_g~ kgCO~ GHGE_kgC~  2405  104.   113.   0     60.9   83.9  119.   2795.
# i 58 more rows
# i 1 more variable: n_NA <int>
```

```
# 1. Prepare data for plotting
tmp_F2F_values <- F2F_output_ghge_compo0 %>%
  # Extract the clean category name from the indicator string
  mutate(
    Category = str_extract(indicator, "(?<=__).*"),
    # Make unit names prettier for the plot labels
    unit_label = gsub("_per_"," / ", unit)
  )
```

```r
tmp_lit_values <- lit_table |>
  dplyr::semi_join(
    tmp_F2F_values |>
      dplyr::select(product, unit, indicator) |>
      dplyr::distinct(),
    by = c('product', 'unit', 'indicator')
  ) |>
  # Extract the clean category name from the indicator string
  mutate(
    Category = str_extract(indicator, "(?<=__).*"),
    # Make unit names prettier for the plot labels
    unit_label = gsub("_per_"," / ", unit)
  )


# 2. Create the plot
ggplot() +
  # accepted range
  geom_errorbar(data = tmp_lit_values,
                aes(x = product,
                    ymin = pmin(lit_midpoint, lit_min, na.rm = T) / 2,
                    ymax = pmax(lit_midpoint, lit_max, na.rm = T) * 2
                ), colour = "lightpink",
                linewidth=1,width = 1,position = position_dodge(width = 0.75)) +
  # literature values
  geom_point(data = tmp_lit_values,
             aes(x = product, y = lit_midpoint), colour = "plum4",
             shape = 1, size=3, stroke = 1, position = position_dodge(width = 0.75)) +
  # min and max
  geom_errorbar(data = tmp_lit_values,
                aes(x = product, ymin = lit_min, ymax = lit_max), colour = "plum4",
                linewidth=1,width = 1,position = position_dodge(width = 0.75)) +
  # estimates
  geom_violin(data = tmp_F2F_values,
              aes(x = product, y = value, alpha = 0),
              colour = "#999999"
  ) +
  geom_boxplot(data = tmp_F2F_values,
               aes(x = product, y = value, alpha = 0),
               colour = "#333333", width = 0.25
  ) +
  # theme
```

```
  coord_flip() +
  facet_wrap(vars(Category, unit_label), scales = "free", ncol = 2) +
  labs(y = "Total GHGE",x= "Products") +
  scale_fill_viridis_d() +
  #  scale_fill_manual(values = tmp_FQS_names$colors,labels = tmp_FQS_names$labels,) +
  #  scale_colour_manual(values = tmp_FQS_names$colors,labels = tmp_FQS_names$labels) +
  theme_light() +
  theme(
    legend.position = c(0,-1),legend.direction = "horizontal",legend.title = element_blank()
    strip.background = element_rect(fill = NA, colour = "#999999"),strip.text = element_text
```




```
# 1. Prepare data for plotting

tmp_plot_share <- dplyr::bind_rows(
  lit_table %>%
  # Filter only for the rows that contain the component breakdown (identified by __)
  filter(str_detect(indicator, "__")
         & unit == "kgCO2e_per_t") %>%
  # Extract the clean category name from the indicator string
  mutate(
    Category = str_extract(indicator, "(?<=__).*"),
    # Make unit names prettier for the plot labels
```

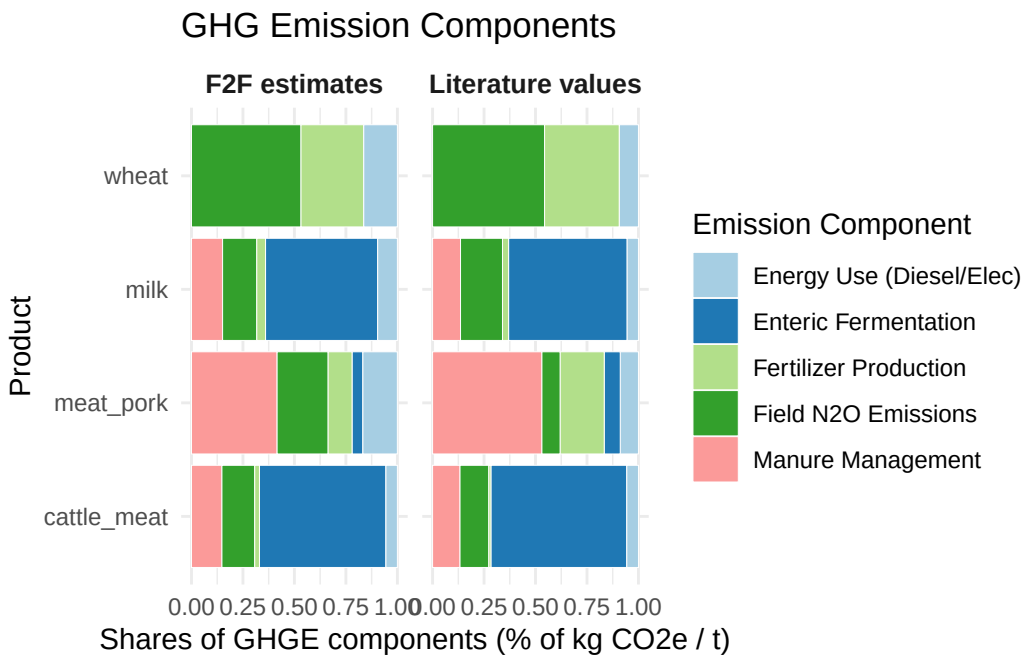
```
    unit_label = "kg CO2e / tonne"
  ) |>
    dplyr::filter(indicator %in% F2F_output_ghge_compo$indicator) |>
    dplyr::mutate(
      source = "Literature values",
      value = lit_midpoint
    ),
  F2F_output_ghge_compo %>%
  # Extract the clean category name from the indicator string
  mutate(
    Category = str_extract(indicator, "(?<=__).*"),
    # Make unit names prettier for the plot labels
    unit_label = gsub("_per_"," / ", unit),
          source = "F2F estimates",
      value = estimate
    )
) |>
  dplyr::filter(
    unit == "kgCO2e_per_t",
    grepl("wheat|milk|cattle_meat|meat_pork", product)
  )

cat(paste("Literature sources: ",paste0(unique(unlist(str_split(tmp_plot_share$lit_source, "
```

```
Literature sources:  Bamber_2025_fig1; Bellassen_2021; Bamber_2025_sup_tab5; CAP2ER_2025_bov
```

```
# 2. Create the plot
ggplot(tmp_plot_share, aes(x = product, y = value, fill = Category)) +
  # Use stat = "identity" to plot the literal values in the table
  geom_bar(stat = "identity", position = "fill", color = "white", linewidth = 0.1) +
  # Facet by Product (rows) and Unit (columns)
  # scales = "free_y" is crucial because ha and t values are not comparable
  facet_grid(~ source) +
  # Styling
  scale_fill_brewer(palette = "Paired") + # Or use scale_fill_viridis_d()
  theme_minimal() +
  labs(
    title = "GHG Emission Components",
    #subtitle = paste("Literature sources: ",paste0(unique(unlist(str_split(tmp_plot_share$li
    x = "Product",
    y = "Shares of GHGE components (% of kg CO2e / t)",
```

```
    fill = "Emission Component"
  ) +
  theme(
    strip.text = element_text(face = "bold", size = 10),
    legend.position = "right",
    #axis.text.x = element_blank(), # Remove x axis text because it's redundant with facets
    axis.ticks.x = element_blank()
  ) +
  guides(fill = guide_legend(ncol = 1, title.position = "top")) +
  coord_flip()
```



Validation

We considered a check point validated when the estimates is more than half and less than twice the literature midpoint.

```
comparison <- validate_against_literature(
  estimates = F2F_output,
  references = lit_table)
```

Validation summary

status	n
missing	39
in range	70
out of range	12

```
# TODO: check missing share concent
# TODO: check lit value for Bellassen
```

```
comparison |>
  count(status) |>
  gt() |>
  tab_header(title = "Validation summary")
```

```
comp_table = comparison |>
  dplyr::filter(!is.na(estimate)) |>
  dplyr::select(indicator, product, unit,
                lit_midpoint, lit_min, lit_max, lit_source,
                estimate, rel_deviation_pct, status) |>
  arrange(product,indicator) |>
  gt() |>
  tab_header(
    title    = "Detailed validation results",
    subtitle = paste("Estimation on France data: ",
                     paste(sort(unique(my_object_GHGE@farm$YEAR)),
                           collapse = ", "))
  ) |>
  fmt_number(columns = c(lit_midpoint, lit_min, lit_max, estimate), decimals = 1) |>
  fmt_number(columns = c(rel_deviation_pct), decimals = 1) |>
  tab_style(
    style     = cell_fill(color = "#fdd"),
    locations = cells_body(rows = status == "\U274C out of range")
  ) |>
  tab_style(
    style     = cell_fill(color = "#dfd"),
    locations = cells_body(rows = status == "\U2705 in range")
  ) |>
  tab_style(
    style     = cell_fill(color = "#ffd"),
```

```
    locations = cells_body(rows = grepl("tolerance", status))
  )

comp_table
```

```
gtsave(comp_table, filename = paste0("F2F_comp_lit_summary_",Sys.Date(),".html"))
gtsave(comp_table, filename = paste0("F2F_comp_lit_summary_",Sys.Date(),".docx"))
```

```
library(ggplot2)

comparison |>
  filter(!is.na(estimate)) |>
  ggplot(aes(x = reorder(paste(indicator, product, sep = " - "),
                         rel_deviation_pct),
             y = rel_deviation_pct,
             fill = status)) +
  geom_col() +
  geom_hline(aes(yintercept = 50), linetype = "dashed", colour = "red") +
  scale_fill_discrete(direction = -1) +
  coord_flip() +
  labs(x = NULL, y = "Relative deviation from midpoint (%)") +
  theme_minimal()
```

Detailed validation results

Estimation on France data: 2018, 2019, 2020, 2021

indicator	product	unit	lit_midpoi
pseudofarm_total_CH4_kgCO2e_per_LU	Beef farms	kgCO2e_per_LU	6,068
pseudofarm_total_CO2_kgCO2e_per_LU	Beef farms	kgCO2e_per_LU	2,754
pseudofarm_total_N2O_kgCO2e_per_LU	Beef farms	kgCO2e_per_LU	1,921
pseudofarm_total_ghge_kgCO2e_per_LU	Beef farms	kgCO2e_per_LU	10,670
pseudofarm_total_CH4_kgCO2e_per_LU	Dairy farms	kgCO2e_per_LU	4,389
pseudofarm_total_CO2_kgCO2e_per_LU	Dairy farms	kgCO2e_per_LU	1,697
pseudofarm_total_N2O_kgCO2e_per_LU	Dairy farms	kgCO2e_per_LU	1,292
pseudofarm_total_ghge_kgCO2e_per_LU	Dairy farms	kgCO2e_per_LU	7,370
pseudofarm_total_CH4_kgCO2e_per_ha	Field crop farms	kgCO2e_per_ha	43
pseudofarm_total_CO2_kgCO2e_per_ha	Field crop farms	kgCO2e_per_ha	900
pseudofarm_total_N2O_kgCO2e_per_ha	Field crop farms	kgCO2e_per_ha	1,254
pseudofarm_total_ghge_kgCO2e_per_ha	Field crop farms	kgCO2e_per_ha	2,322
GHGE_kgCO2e_ha	cattle_meat	kgCO2e_per_ha	6,962
GHGE_kgCO2e_ha___Energy Use (Diesel/Elec)	cattle_meat	kgCO2e_per_ha	355
GHGE_kgCO2e_ha___Enteric Fermentation	cattle_meat	kgCO2e_per_ha	4,573
GHGE_kgCO2e_ha___Fertilizer Production	cattle_meat	kgCO2e_per_ha	95
GHGE_kgCO2e_ha___Field N2O Emissions	cattle_meat	kgCO2e_per_ha	1,303
GHGE_kgCO2e_ha___Manure Management	cattle_meat	kgCO2e_per_ha	844
GHGE_kgCO2e_t	cattle_meat	kgCO2e_per_t	14,866
GHGE_kgCO2e_t___Energy Use (Diesel/Elec)	cattle_meat	kgCO2e_per_t	850
GHGE_kgCO2e_t___Enteric Fermentation	cattle_meat	kgCO2e_per_t	9,900
GHGE_kgCO2e_t___Fertilizer Production	cattle_meat	kgCO2e_per_t	150
GHGE_kgCO2e_t___Field N2O Emissions	cattle_meat	kgCO2e_per_t	2,100
GHGE_kgCO2e_t___Manure Management	cattle_meat	kgCO2e_per_t	2,000
DMI	dairy_cows	kg / head / day	16
GE	dairy_cows	MJ / head / day	308
Nex	dairy_cows	kg N / head / year	115
first_calving_age	dairy_cows	month	30
renewal_rate	dairy_cows	percent	36
share_concent	dairy_cows	percent	20
GHGE_kgCO2e_ha	eggs	kgCO2e_per_ha	5,063
GHGE_kgCO2e_ha___Energy Use (Diesel/Elec)	eggs	kgCO2e_per_ha	333
GHGE_kgCO2e_ha___Enteric Fermentation	eggs	kgCO2e_per_ha	0
GHGE_kgCO2e_ha___Fertilizer Production	eggs	kgCO2e_per_ha	166
GHGE_kgCO2e_ha___Field N2O Emissions	eggs	kgCO2e_per_ha	833
GHGE_kgCO2e_ha___Manure Management	eggs	kgCO2e_per_ha	333
GHGE_kgCO2e_t	eggs	kgCO2e_per_t	2,416
GHGE_kgCO2e_ha	maize_grain	kgCO2e_per_ha	3,685
GHGE_kgCO2e_t	maize_grain	kgCO2e_per_t	356
GHGE_kgCO2e_ha	meat_chicken	kgCO2e_per_ha	6,426
GHGE_kgCO2e_ha___Energy Use (Diesel/Elec)	meat_chicken	kgCO2e_per_ha	300
GHGE_kgCO2e_ha___Enteric Fermentation	meat_chicken	kgCO2e_per_ha	0
GHGE_kgCO2e_ha___Fertilizer Production	meat_chicken	kgCO2e_per_ha	300
GHGE_kgCO2e_ha___Field N2O Emissions	meat_chicken	kgCO2e_per_ha	700
GHGE_kgCO2e_ha___Manure Management	meat_chicken	kgCO2e_per_ha	300
GHGE_kgCO2e_t	meat_chicken	kgCO2e_per_t	4,382
GHGE_kgCO2e_ha	meat_pork	kgCO2e_per_ha	5,463

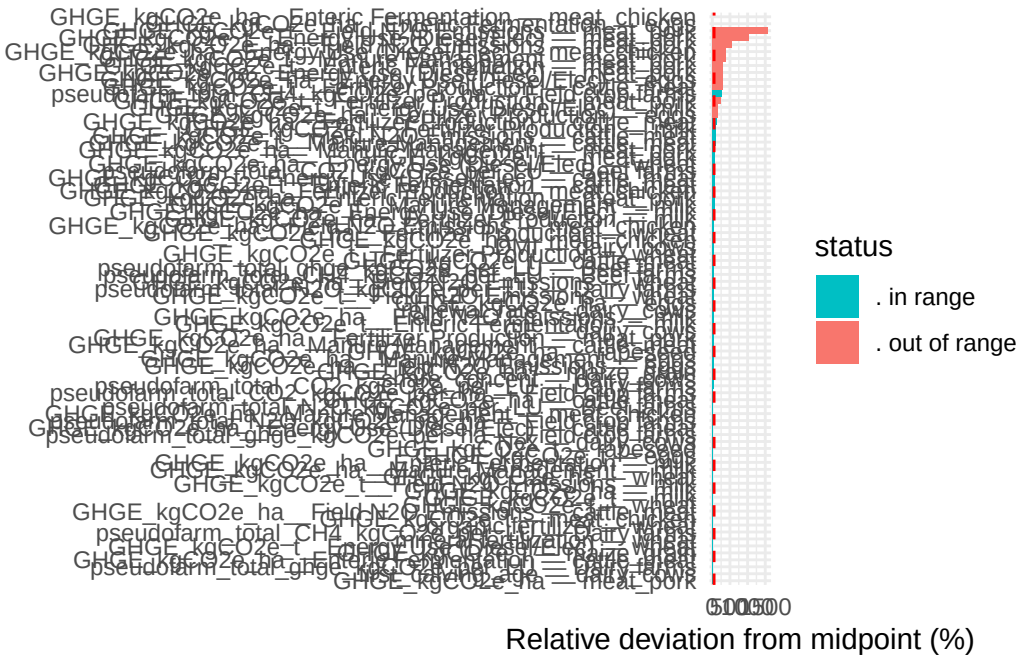


Figure 1: Relative deviation from literature midpoint by indicator

Checks

```
object = my_object_GHGE

id_cols = object@traceability$id_cols

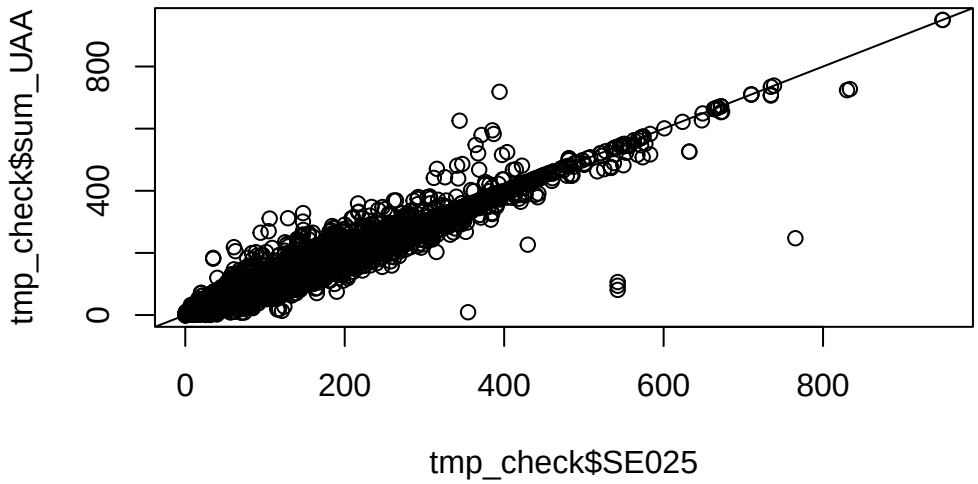
tmp_tot_UAA = object@farm |>
  dplyr::select(dplyr::all_of(id_cols),
                SE025)

tmp_sum_UAA = object@crop |>
  #dplyr::filter(!grepl("CTOT",FADN_code_letter)) |>
  dplyr::summarise(
    sum_UAA = sum(area_ha, na.rm = TRUE),
    .by = dplyr::all_of(id_cols)
  )

tmp_check = dplyr::left_join(
  tmp_tot_UAA,
```

```
  tmp_sum_UAA,
  by = id_cols
)

plot(tmp_check$SE025, tmp_check$sum_UAA)
abline(0,1)
```




```
load("C:/Users/srhuet/OneDrive/Research/GitHub/FADN2Footprint/data_raw/raw_FADN_FRA_18_21.RDa

object = my_object_GHGE

id_cols = object@traceability$id_cols

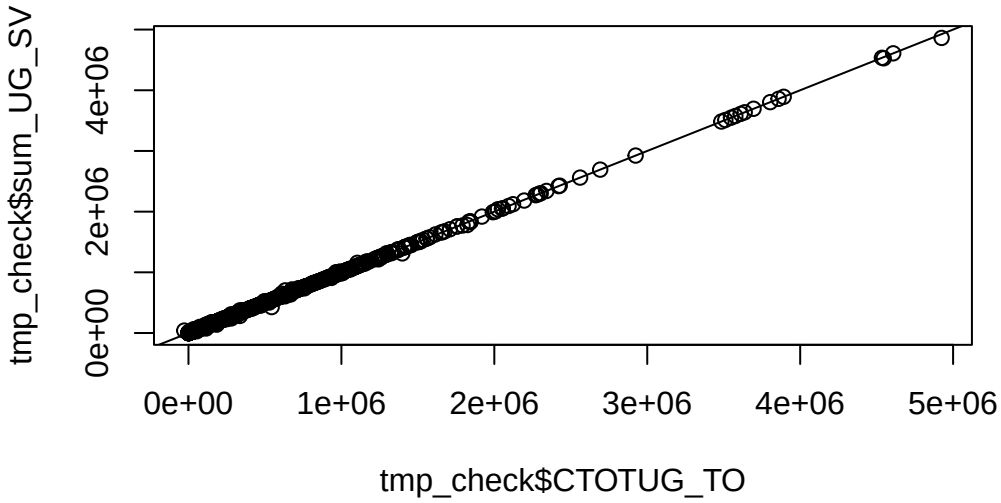
tmp_tot_UG = my_FADN_data |>
  dplyr::select(dplyr::all_of(id_cols),
                CTOTUG_TO) |>
  dplyr::mutate(ID = as.character(ID),
                YEAR = as.character(YEAR))

tmp_sum_UG = object@crop |>
  dplyr::filter(grepl("UG$",FADN_code_letter)) |>
  dplyr::summarise(
```

```
    sum_UG_TO = sum(TO, na.rm = TRUE),
    sum_UG_SV = sum(SV, na.rm = TRUE),
    .by = dplyr::all_of(id_cols)
  )

tmp_check = dplyr::left_join(
  tmp_tot_UG,
  tmp_sum_UG,
  by = id_cols
)

plot(tmp_check$CTOTUG_TO, tmp_check$sum_UG_SV)
abline(0,1)
```




```
object = my_object_GHGE

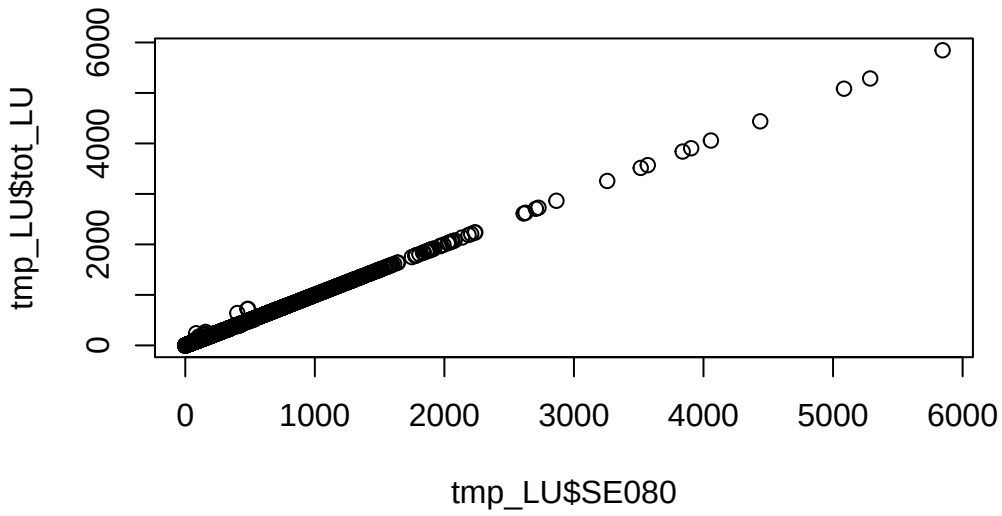
id_cols = object@traceability$id_cols


tmp_farm_LU = object@farm |>
  dplyr::select(dplyr::all_of(id_cols),
                SE080)
```

```r
tmp_herd_LU = object@herd |>
  dplyr::summarise(
    tot_LU = sum(ALU, na.rm = T),
    .by =  dplyr::all_of(id_cols)
  )

tmp_LU = dplyr::left_join(
  tmp_farm_LU,
  tmp_herd_LU,
  by = id_cols
)

plot(tmp_LU$SE080, tmp_LU$tot_LU)
```



Meat production

I want to compare our estimation for meat production to EUROSTAT price : https://ec.europa.eu/eurostat/data
Les prix pour la France sont bizarre : on est ~360€ pour 100kg et par tête pour les génisses alors qu'en Espagne par exemple, on est à 285€ pour 100 kg (donc 2850€/t) et 1200€ par tête
le rendement dans agribalyse est de 1/22e-4=450 kg/ha vs 7.2/15*10^3 = 480 kg/ha dans CAP2ER

```r
# Heiffers

tmp_heif = tmp_object_beef@output$meat |>
  dplyr::filter(FADN_code_letter == "LHEIFFAT")

cat("Average weight for heiffers: ",
    mean(tmp_heif$prod_t / tmp_heif$sales_nb, na.rm = TRUE)," t/head")
```

```
Average weight for heiffers:  0.4409379  t/head
```

```r
cat("\n")
```

```r
cat("Mean price for heiffer: ",
    mean(tmp_heif$sales_e / tmp_heif$sales_nb, na.rm = TRUE)," €/head and ",
    mean(tmp_heif$sales_e / tmp_heif$prod_t, na.rm = TRUE)," €/t")
```

```
Mean price for heiffer:  1512.5  €/head and  3430.189  €/t
```

```r
cat("\n")
```

```r
cat("\n")
```

```r
cat("\nOn average, beef farms sell ",
    (mean(tmp_object_beef@herd |>
            #dplyr::filter(grepl("LHEIF",FADN_code_letter)) |>
            dplyr::pull(share_SSN),
          na.rm = T))*100,"% of their animals for slaughter")
```

```
On average, beef farms sell  61.58834 % of their animals for slaughter
```

I want to compare our liveweight production per livestock units to those reported in CAP2ER => OK, eventhough we overestimate veal live weight

```r
object = tmp_object_beef

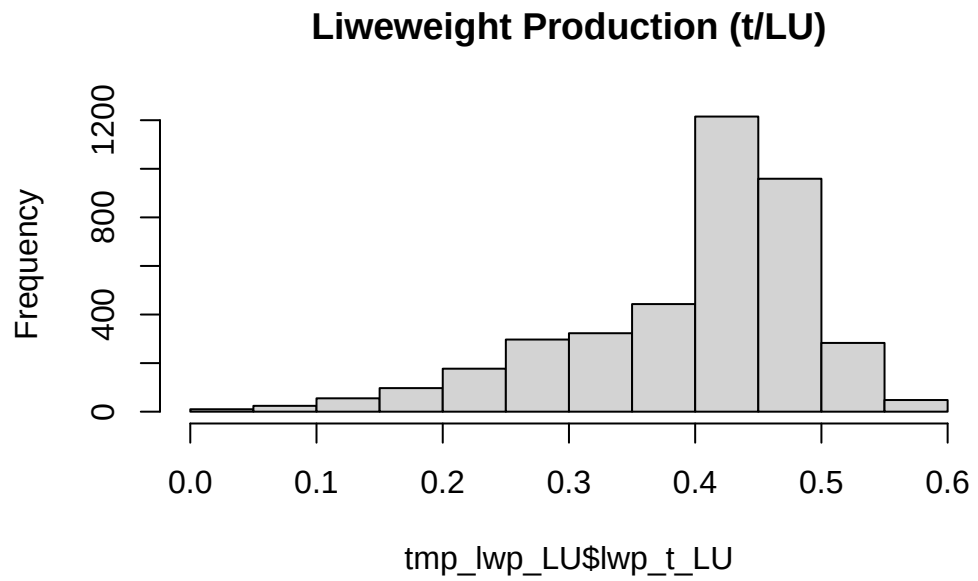
tmp_lwp <- object@output$meat |>
  dplyr::filter(grepl("meat", output) & species == "cattle") |>
  dplyr::summarise(
```

```r
    LWP_t = sum(prod_t, na.rm = TRUE),
    .by = dplyr::all_of(id_cols)
  )

tmp_sold_LU = object@herd |>
  dplyr::filter(species == "cattle") |>
  dplyr::summarise(
    tot_LU = sum(SN * livestock_unit_coef, na.rm = T),
    .by =  dplyr::all_of(id_cols)
  )

tmp_lwp_LU <- dplyr::left_join(
  tmp_lwp,
  tmp_sold_LU,
  by = id_cols
) |>
  dplyr::mutate(
    lwp_t_LU = LWP_t / tot_LU
  )

hist(tmp_lwp_LU$lwp_t_LU, main = "Liweweight Production (t/LU)")
```



```
quantile(tmp_lwp_LU$lwp_t_LU)
```

```
        0%        25%        50%        75%       100%
0.04006993 0.34997468 0.43529124 0.45889267 0.57552340
```

```
mean(tmp_lwp_LU$lwp_t_LU)
```

```
[1] 0.399804
```

```
object@herd |> filter(ID == "52100000050798")
```

```
# A tibble: 20 x 26
   ID       YEAR  COUNTRY FADN_code_letter    AN    PN    PV    SN    SV      TO
   <chr>    <chr> <chr>   <chr>            <dbl> <dbl> <dbl> <dbl> <dbl>   <dbl>
 1 5210000~ 2018  FRA     LBOV1_2F           2.5     0     0     5  5667  8203.
 2 5210000~ 2018  FRA     LBOV1            237       0     0   560 54145 54145
 3 5210000~ 2018  FRA     LBOV2              0.5     1  1350     1   650  -700
 4 5210000~ 2018  FRA     LCOWOTH           20       0     0     0     0  -115.
 5 5210000~ 2018  FRA     LHEIFFAT           3.5     0     0     0     0 -3687.
 6 5210000~ 2019  FRA     LBOV1_2F           8       0     0     0     0   -14.8
 7 5210000~ 2019  FRA     LBOV1            234.      0     0   560 54570 54148
 8 5210000~ 2019  FRA     LBOV2              0.5     1  1600     0     0  -382.
 9 5210000~ 2019  FRA     LCOWOTH           17.5     0     0     0     0 -4673.
10 5210000~ 2019  FRA     LHEIFFAT           4.5     0     0     0     0  5892.
11 5210000~ 2020  FRA     LBOV1_2F           7.5     0     0     6  6464  5484.
12 5210000~ 2020  FRA     LBOV1            232       0     0     7  4404  4826
13 5210000~ 2020  FRA     LBOV2              0.5     0     0     1  1114  -139.
14 5210000~ 2020  FRA     LCOWOTH           15       0     0     0     0  -113.
15 5210000~ 2020  FRA     LHEIFFAT           6.5     0     0     0     0 -1319.
16 5210000~ 2021  FRA     LBOV1_2F          10       0     0     0     0  5061.
17 5210000~ 2021  FRA     LBOV1            232       0     0     7  5775  4931
18 5210000~ 2021  FRA     LBOV2              0.5     1  1700     0     0  -466.
19 5210000~ 2021  FRA     LCOWOTH           14       0     0     4  6018  3975.
20 5210000~ 2021  FRA     LHEIFFAT           6.5     0     0     4  5686  6058.
# i 16 more variables: livestock_unit_coef <dbl>, species <chr>, ALU <dbl>,
#   ON <dbl>, CN <dbl>, Qobs <dbl>, SSN <dbl>, SSV <dbl>, SRN <dbl>, SRV <dbl>,
#   NUTS2 <chr>, TF14 <int>, Country_ISO_3166_1_A3 <chr>, share_SSN <dbl>,
#   share_SRN <dbl>, complete_sales <lgl>
```

```
object@output$meat |> filter(ID == "52100000050798")
```

```
# A tibble: 10 x 15
   ID      YEAR  COUNTRY FADN_code_letter output species sales_nb prod_t sales_t
   <chr>   <chr> <chr>   <chr>            <chr>  <chr>      <dbl>  <dbl>   <dbl>
 1 521000~ 2018  FRA     LBOV1_2F         meat_~ cattle     0.318  0.140   0.140
 2 521000~ 2018  FRA     LBOV1            meat_~ cattle   218.    96.2    96.2
 3 521000~ 2018  FRA     LBOV2            meat_~ cattle     1.000  0.441   0.441
 4 521000~ 2019  FRA     LBOV1            meat_~ cattle   218.    96.2    96.2
 5 521000~ 2020  FRA     LBOV1_2F         meat_~ cattle     0.382  0.168   0.168
 6 521000~ 2020  FRA     LBOV1            meat_~ cattle     2.73   1.20    1.20
 7 521000~ 2020  FRA     LBOV2            meat_~ cattle     1.000  0.441   0.441
 8 521000~ 2021  FRA     LBOV1            meat_~ cattle     2.73   1.20    1.20
 9 521000~ 2021  FRA     LCOWOTH          meat_~ cattle     3.45   1.52    1.52
10 521000~ 2021  FRA     LHEIFFAT         meat_~ cattle     4.00   1.76    1.76
# i 6 more variables: sales_e <dbl>, output_e <dbl>, yield_t_sales_nb <dbl>,
#   prop_sales_Qobs <dbl>, yield_t_Qobs <dbl>, yield_e_Qobs <dbl>
```

```
my_FADN_data |> filter(ID == "52100000050798") |> dplyr::select(dplyr::all_of(id_cols),dplyr
```

```
# A tibble: 128 x 5
        ID  YEAR COUNTRY name         value
     <dbl> <int> <chr>   <chr>        <dbl>
 1 5.21e13  2018 FRA     LBOV0_AN_X     0
 2 5.21e13  2018 FRA     LBOV0_PN_X     0
 3 5.21e13  2018 FRA     LBOV0_PV_X     0
 4 5.21e13  2018 FRA     LBOV0_SN_X     0
 5 5.21e13  2018 FRA     LBOV1_2F_AN    2.5
 6 5.21e13  2018 FRA     LBOV1_2F_PN    0
 7 5.21e13  2018 FRA     LBOV1_2F_PV    0
 8 5.21e13  2018 FRA     LBOV1_2F_SN    5
 9 5.21e13  2018 FRA     LBOV1_2F_SV 5667
10 5.21e13  2018 FRA     LBOV1_2F_TO 8203.
# i 118 more rows
```

Eggs

```
# LHENSLAY
```

```
tmp_LU <- tmp_object_eggs@herd |>
  dplyr::filter(FADN_code_letter == "LHENSLAY")


tmp_prod <- tmp_object_eggs@output$other_herd_products |>
  dplyr::filter(grepl("PEGG", FADN_code_letter_output)) |>
  dplyr::summarise(
    prod_t = sum(prod_t, na.rm = T),
    .by = c(dplyr::all_of(id_cols), output)
  )

tmp_yield <- dplyr::left_join(
  tmp_LU, tmp_prod,
  by = id_cols
) |>
  dplyr::mutate(
    yield = prod_t / Qobs
  )

quantile(tmp_yield$yield)
```

```
         0%         25%         50%         75%        100%
0.002330097 0.012885048 0.015852588 0.017494737 0.022718949
```

Feed

```
object = my_object_GHGE
id_cols = my_object_GHGE@traceability$id_cols

th_feed <- f_feed_theo_ration(object)

# create the "would-be" allocation exactly like f_feed_offfarm does (uses traceability id co
feed_purchased_qty <- object@input |>
  # Retrieve farm purchases values for concentrate and rough feed
  dplyr::select(dplyr::all_of(id_cols),
                dplyr::matches("feed_purch")) |>
  ## Purchased concentrated feedstuffs for grazing stock (equines, ruminants) Value in EUR
  ##feed_purch_concent_graz = IGRFEDCNCTRPUR_V,
  ## Purchased coarse fodder for grazing stock (equines, ruminants) Value in EUR
  ##feed_purch_rough_graz = IGRFEDCRSPUR_V,
```

```
  ## Purchased feedstuffs for pigs Value in EUR
  ##feed_purch_concent_pigs = IPIGFEDPUR_V,
  ## Purchased feedstuffs for poultry and other small animals Value in EUR
  ##feed_purch_concent_poultry = IPLTRFEDPUR_V,
  tidyr::pivot_longer(cols = tidyselect::matches("feed_purch"),
                      names_to = "feed_var",
                      values_to = "feed_purch_value") |>
  dplyr::filter(feed_purch_value >0) |>
  dplyr::mutate(
    feed_type = dplyr::case_when(
      feed_var == "feed_purch_rough_graz" ~ "feed_rough",
      .default = "feed_concent"
    ),
    species = dplyr::case_when(
      feed_var %in% c("feed_purch_concent_graz","feed_purch_rough_graz") ~ "grazing",
      feed_var == "feed_purch_concent_pigs" ~ "swine",
      feed_var == "feed_purch_concent_poultry" ~ "poultry"
    )
  ) |>
  #filter(feed_purch_value >0) |> # keep null purchases for traceability
  # Add theoretical values
  dplyr::left_join(th_feed$th_feed_value |>
                     dplyr::select(dplyr::all_of(id_cols),
                                   species, FADN_code_letter,
                                   feed_type, Sailley_feed,
                                   th_crop_value_p100_feed_type, euros_t),
                   by = c(id_cols,"feed_type","species")) |>
  # Estimate share of purchased feed value per livestock category and per crop
  dplyr::mutate(
    feed_purch_value_share = feed_purch_value * th_crop_value_p100_feed_type
  ) |>
  # Convert estimated value in ton
  dplyr::mutate(
    feed_purch_t_livcat = feed_purch_value_share / euros_t
  )

# summary per feed_var
tmp_comp <- feed_purchased_qty |>
  dplyr::summarise(
    total_input = mean(feed_purch_value, na.rm=TRUE),
    total_allocated = sum(feed_purch_value_share, na.rm=TRUE),
    n_rows = n(),
```

```
    #n_unallocated = sum(is.na(feed_purch_value_share)),
    #.by = feed_var
    .by = c(dplyr::all_of(id_cols), species, feed_type)
  ) |>
  dplyr::mutate(
    diff_p100 = abs(total_allocated - total_input) / total_input,
    diff = ifelse(round(diff_p100,5) > 0, TRUE, FALSE))

table(tmp_comp$diff)
```

```
FALSE  TRUE
21874   476
```

```
# see if livestock corresponding to feed
tmp_comp |>
  dplyr::filter(diff) |>
  dplyr::left_join(
    object@herd |>
      #dplyr::mutate(
      #    species = case_when(
      #        species %in% c("cattle","sheep","goats","horse") ~ "grazing",
      #        .default = species
      #    )
      #) |>
      dplyr::summarise(
        Qobs = sum(Qobs),
        .by = c(dplyr::all_of(id_cols), species)
      ) |>
      tidyr::pivot_wider(names_from = species, values_from = Qobs),
    #by = c(id_cols, 'species')
    by = c(id_cols)
  ) #|> View()
```

```
# A tibble: 476 x 17
   ID         YEAR  COUNTRY species feed_type total_input total_allocated n_rows
   <chr>      <chr> <chr>   <chr>   <chr>           <dbl>           <dbl>  <int>
 1 515000000~ 2018  FRA     poultry feed_con~         982               0      1
 2 515000000~ 2018  FRA     poultry feed_con~         332               0      1
 3 515000000~ 2018  FRA     poultry feed_con~        6047               0      1
 4 515000000~ 2018  FRA     poultry feed_con~       15441               0      1
```

```
 5 515000000~ 2018  FRA     poultry feed_con~         265                0      1
 6 515000000~ 2018  FRA     poultry feed_con~         174                0      1
 7 515000000~ 2018  FRA     poultry feed_con~         724                0      1
 8 515000000~ 2018  FRA     poultry feed_con~         427                0      1
 9 565000000~ 2018  FRA     poultry feed_con~          37                0      1
10 565000000~ 2018  FRA     poultry feed_con~         229                0      1
# i 466 more rows
# i 9 more variables: diff_p100 <dbl>, diff <lgl>, sheep <dbl>, cattle <dbl>,
#   goats <dbl>, poultry <dbl>, others <dbl>, swine <dbl>, horse <dbl>
```

Some farms purchase poultry feed but do not have poultry (some do not have livestock at all sometimes)

Heating fuels

```
semi_join(
  my_object_GHGE@footprints$GHGE$GHGE_herd |>
    dplyr::summarise(ghg_heat_fuel_kgCO2e_livcat = sum(ghg_heat_fuel_kgCO2e_livcat, na.rm = T
                     .by = dplyr::all_of(id_cols)) |>
    dplyr::filter(ghg_heat_fuel_kgCO2e_livcat == 0),
  my_object_GHGE@input |> dplyr::filter(heating_fuels_l >0)
) #|> View()
```

```
# A tibble: 292 x 4
   ID            YEAR  COUNTRY ghg_heat_fuel_kgCO2e_livcat
   <chr>         <chr> <chr>                         <dbl>
 1 5610000005933 2018  FRA                               0
 2 5640000001924 2018  FRA                               0
 3 5640000001928 2018  FRA                               0
 4 5640000001921 2018  FRA                               0
 5 5640000001370 2018  FRA                               0
 6 5640000001998 2018  FRA                               0
 7 5680000005992 2018  FRA                               0
 8 5670000003311 2018  FRA                               0
 9 5670000003901 2018  FRA                               0
10 5670000003962 2018  FRA                               0
# i 282 more rows
```

```r
tmp_farms = semi_join(
  my_object_GHGE@footprints$GHGE$GHGE_herd |>
    dplyr::summarise(ghg_heat_fuel_kgCO2e_livcat = sum(ghg_heat_fuel_kgCO2e_livcat, na.rm = T
                     .by = dplyr::all_of(id_cols)) |>
    dplyr::filter(ghg_heat_fuel_kgCO2e_livcat == 0),
  my_object_GHGE@input |> dplyr::filter(heating_fuels_l >0)
) |>
  dplyr::select(dplyr::all_of(my_object_GHGE@traceability$id_cols))

tmp_obbj = m_keep_farms(my_object_GHGE, tmp_farms)

tmp_econ_alloc = f_output_econ_alloc(tmp_obbj)

summary(tmp_econ_alloc$all_outputs)
```

```
      ID                YEAR              COUNTRY          FADN_code_letter
 Length:798         Length:798         Length:798         Length:798
 Class :character   Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character   Mode  :character



    output            species              prod_t            sales_t
 Length:798         Length:798         Min.   :   0.012   Min.   :   0.00
 Class :character   Class :character   1st Qu.:  13.361   1st Qu.:   0.00
 Mode  :character   Mode  :character   Median :  43.431   Median :  16.94
                                       Mean   : 104.111   Mean   :  80.44
                                       3rd Qu.: 106.694   3rd Qu.:  78.05
                                       Max.   :1912.160   Max.   :1912.16
                                                          NA's   :7
    sales_e          output_e          yield             activity
 Min.   :     0   Min.   :     0   Min.   : 0.01812   Length:798
 1st Qu.:     0   1st Qu.:     0   1st Qu.: 2.36943   Class :character
 Median :  6258   Median :  6952   Median : 4.82965   Mode  :character
 Mean   : 20248   Mean   : 20445   Mean   : 7.51871
 3rd Qu.: 20456   3rd Qu.: 20672   3rd Qu.: 7.22888
 Max.   :553150   Max.   :560377   Max.   :95.70748
                                   NA's   :7
   farm_use_e        TO_e            sales_nb     yield_t_sales_nb
 Min.   :0     Min.   :     0   Min.   :51612   Min.   :0.00191
 1st Qu.:0     1st Qu.:     0   1st Qu.:52342   1st Qu.:0.00191
```

```
 Median :0     Median :  6258   Median :53071   Median :0.00191
 Mean   :0     Mean   : 20248   Mean   :63405   Mean   :0.00191
 3rd Qu.:0     3rd Qu.: 20456   3rd Qu.:69302   3rd Qu.:0.00191
 Max.   :0     Max.   :553150   Max.   :85532   Max.   :0.00191
 NA's   :788                    NA's   :795     NA's   :795
 prop_sales_Qobs  yield_t_Qobs      yield_e_Qobs    FADN_code_letter_output
 Min.   :1.970   Min.   :0.00376   Min.   : 8.661   Length:798
 1st Qu.:1.998   1st Qu.:0.00381   1st Qu.: 9.275   Class :character
 Median :2.026   Median :0.00387   Median : 9.889   Mode  :character
 Mean   :2.332   Mean   :0.00445   Mean   :11.902
 3rd Qu.:2.513   3rd Qu.:0.00480   3rd Qu.:13.523
 Max.   :3.001   Max.   :0.00573   Max.   :17.155
 NA's   :795     NA's   :795       NA's   :795
      Qobs       sum_sales_e_farm sum_output_e_farm econ_alloc_ratio_farm
 Min.   :  5.0   Min.   :     0   Min.   :     0    Min.   :0.000000
 1st Qu.:177.2   1st Qu.: 22208   1st Qu.: 29303    1st Qu.:0.009883
 Median :200.2   Median : 69284   Median : 70450    Median :0.131234
 Mean   :184.0   Mean   :105424   Mean   :108209    Mean   :0.217087
 3rd Qu.:212.0   3rd Qu.:134005   3rd Qu.:141490    3rd Qu.:0.318758
 Max.   :313.0   Max.   :749898   Max.   :764118    Max.   :1.000000
 NA's   :792                                       NA's   :84
```

```
tmp_econ_alloc$all_outputs #|> View()
```

```
# A tibble: 798 x 24
   ID      YEAR  COUNTRY FADN_code_letter output species  prod_t sales_t sales_e
   <chr>   <chr> <chr>   <chr>            <chr>  <chr>     <dbl>   <dbl>   <dbl>
 1 561000~ 2018  FRA     CCEROTH          crop   cereal  1.62e+1    16.2    3557
 2 561000~ 2018  FRA     CRAPE            crop   oilsee~ 7.23e+1    72.3   24864
 3 561000~ 2018  FRA     CRG              crop   grassl~ 3.48e-1     0         0
 4 561000~ 2018  FRA     CSUGBT           crop   root_v~ 1.47e+3  1472.    31745
 5 561000~ 2018  FRA     CWHTC            crop   cereal  2.31e+2   231.    40123
 6 564000~ 2018  FRA     CBRL             crop   cereal  6.75e+1    67.5    9492
 7 564000~ 2018  FRA     CGRSTMP          crop   <NA>    1.73e+1     0         0
 8 564000~ 2018  FRA     CRAPE            crop   oilsee~ 2.6 e+1    26      9065
 9 564000~ 2018  FRA     CWHTC            crop   cereal  8.84e+1    88.4   14797
10 564000~ 2018  FRA     CRG              crop   grassl~ 5.50e+0     0         0
# i 788 more rows
# i 15 more variables: output_e <dbl>, yield <dbl>, activity <chr>,
#   farm_use_e <dbl>, TO_e <dbl>, sales_nb <dbl>, yield_t_sales_nb <dbl>,
#   prop_sales_Qobs <dbl>, yield_t_Qobs <dbl>, yield_e_Qobs <dbl>,
#   FADN_code_letter_output <chr>, Qobs <dbl>, sum_sales_e_farm <dbl>,
```

```
#   sum_output_e_farm <dbl>, econ_alloc_ratio_farm <dbl>
```

Farms with zero GHGE

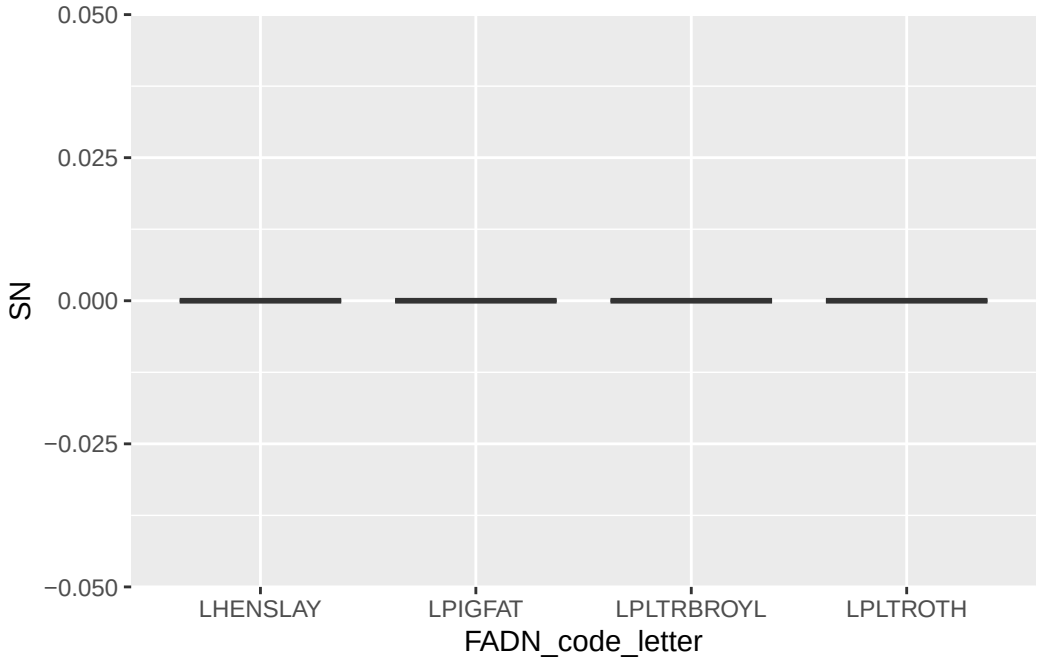
```
object = my_object_GHGE
id_cols = object@traceability$id_cols

tmp_obj_farms_zero <- m_keep_farms(
  object,
  object@footprints$GHGE$GHGE_farm |>
    dplyr::filter(is.na(farm_total_ghge_kgCO2e ) | farm_total_ghge_kgCO2e  == 0)
)

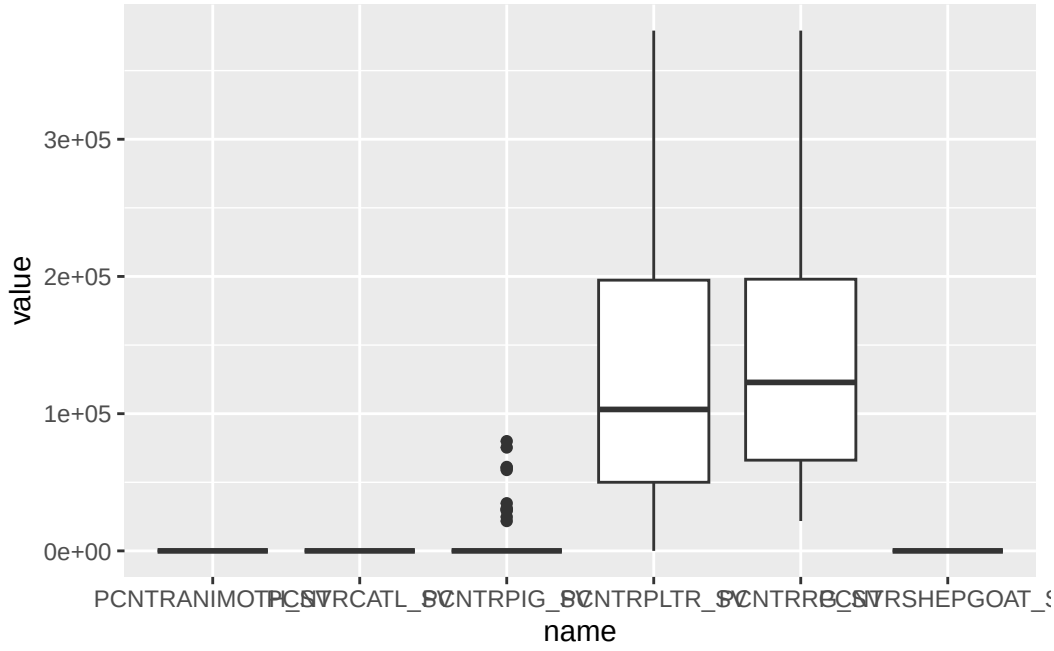
table(tmp_obj_farms_zero@farm$TF14)
```

```
50
89
```

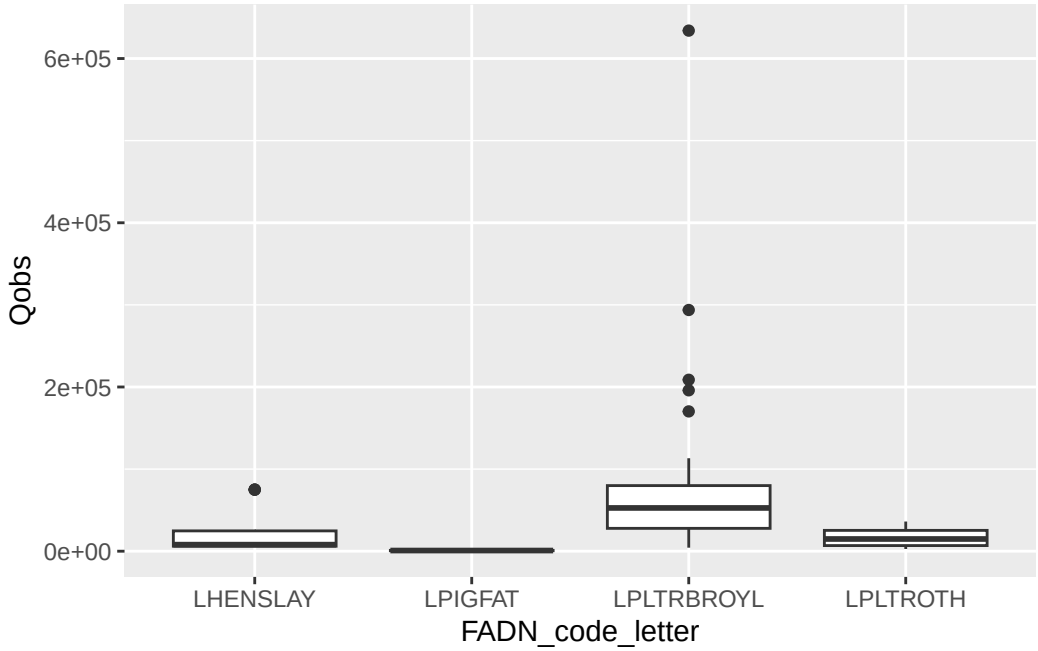
```
tmp_obj_farms_zero@herd |>
  ggplot() +
  aes(x = FADN_code_letter,
      y = SN) +
  geom_boxplot()
```




```
my_FADN_data |>
  dplyr::select(dplyr::all_of(id_cols),
                dplyr::matches("PCNT")) |>
  dplyr::mutate(ID = as.character(ID),
                YEAR = as.character(YEAR)) |>
  dplyr::semi_join(tmp_obj_farms_zero@farm,
                   by = id_cols) |>
  tidyr::pivot_longer(cols = -dplyr::all_of(id_cols)) |>
  dplyr::filter(grepl("_SV$", name))|>
  ggplot() +
  aes(x = name,
      y = value) +
  geom_boxplot()
```




```
tmp_obj_farms_zero@herd |>
  ggplot() +
  aes(x = FADN_code_letter,
      y = Qobs) +
  geom_boxplot()
```



Regressions